

TEST REPORT

Applicant/Manufacturer: Dragino Technology Co., Limited
Address: Room 202, BaoChengTai industrial park, No.8 CaiYun
LongCheng Street, LongGang District, Shenzhen China
Report Number: 2501X31094E-RF-22BA1

Test Standard (s)

ETSI EN 300 220-1 V3.1.1 (2017-02); ETSI EN 300 220-2 V3.2.1 (2018-06)

Sample Description

Product Type: LoRaWAN Sensor Node
Model No.: DDS75-LB
Multiple Model(s) No.: please refer to Product Description for Equipment under
Test (EUT)
Trade Mark: DRAGINO
Date Received: 2025/09/26
Issue Date: 2026/01/04

Test Result:	Pass▲
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▲ In the configuration tested, the EUT complied with the standards above.

Prepared and Checked By:

Ekko Wu

Ekko Wu
RF Engineer

Approved By:

Nancy Wang

Nancy Wang
RF Supervisor



Note: The information marked # is provided by the applicant, the laboratory is not responsible for its authenticity and this information can affect the validity of the result in the test report. Customer model name, addresses, names, trademarks etc. are included.
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Bay Area Compliance Laboratories Corp. (Shenzhen)

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DOCUMENT REVISION HISTORY

Revision Number	Report Number	Description of Revision	Date of Revision
0	SZ1230802-44707E-RF-22B	Original Report	2023-10-13
1	2501X31094E-RF-22BA1	Amended Report	2026-01-04

Note:

This is an amended report based on the report SZ1230802-44707E-RF-22B, the details as below:

(1) Adding the model number “LDS25-LB, LDS25-LS, LDS25-LB2, CS01-LB, CS01-LS, CS01-LB2, SE01-LS, SE01-LB2, DDS75-LS, DDS75-LB2, DDS45-LS, DDS45-LB2, DDS20-LS, DDS20-LB2, MDS120-LS, MDS120-LB2, SPH01-LS, SPH01-LB2, LMS01-LS, LMS01-LB2, WQS-LB, WQS-LS, WQS-LB2, SE0X-LB, SE0X-LS, SE0X-LB2, WSC2-LB, WSC2-LS, WSC2-LB2, MDS200-LS, MDS200-LB2, LDS40-LB, LDS40-LS, LDS40-LB2, LDS12-LS, LDS12-LB2, DDS04-LS, DDS04-LB2, RS485-LB2, RS485-LS, RS485W-LB, RS485W-LS, RS485W-LB2”, with the different power module or battery from the original model.

Based on above difference listed, the modifications will not impact any test data, we only updated the related photos in the report, all the test data and other photos please refer to the report SZ1230802-44707E-RF-22B.

GENERAL INFORMATION

Product Description for Equipment under Test (EUT)

Product	LoRaWAN Sensor Node
Tested Model	DDS75-LB
Multiple Model(s)	LDS25-LB, LDS25-LS, LDS25-LB2, CS01-LB, CS01-LS, CS01-LB2, SE01-LB, SE01-LS, SE01-LB2, DDS75-LS, DDS75-LB2, DDS45-LB, DDS45-LS, DDS45-LB2, DDS20-LB, DDS20-LS, DDS20-LB2, MDS120-LB, MDS120-LS, MDS120-LB2, SPH01-LB, SPH01-LS, SPH01-LB2, LMS01-LB, LMS01-LS, LMS01-LB2, WQS-LB, WQS-LS, WQS-LB2, SE0X-LB, SE0X-LS, SE0X-LB2, WSC2-LB, WSC2-LS, WSC2-LB2, MDS200-LB, MDS200-LS, MDS200-LB2, LDS40-LB, LDS40-LS, LDS40-LB2, LDS12-LB, LDS12-LS, LDS12-LB2, DDS04-LB, DDS04-LS, DDS04-LB2, RS485-LB, RS485-LB2, RS485-LS, RS485-LB, RS485W-LS, RS485W-LB2
Frequency Range	865-868MHz & 868-868.6MHz
Maximum Transmit Power (ERP)	13.17dBm
Modulation Technique	FSK
Antenna Specification [#]	2dBi (It is provided by the manufacturer)
Voltage Range	LB Series: DC 3.6V from battery 1/4 LB2 Series: DC 3.6V from battery 3 LS Series: DC 3.7V from battery 2
Sample/EUT Status	Good condition
Normal/Extreme Condition [#]	L.V.: Low Voltage 2.5V _{DC} ; L.T.: Low Temperature -40°C N.V.: Normal Voltage 3.6V _{DC} ; N.T.: Normal Temperature +25°C H.V.: High Voltage 3.7V _{DC} ; H.T.: High Temperature +85°C Note: the extreme condition was declared by manufacturer.
Adapter Information	N/A
Battery Information	Original: Battery 1(EVE ENERGY): DC 3.6V, 8500mAh New: Battery 2(YO): DC 3.7V, 3000mAh 11.1Wh Battery 3(RAMWAY): DC 3.6V, 19000mAh Battery 4(RAMWAY): DC 3.6V, 8500mAh
Note: The Multiple models are electrically identical with the test model except for model no., probes, PCB and battery. Please refer to the declaration letter [#] for more detail, which was provided by manufacturer.	

EXHIBIT A - EUT PHOTOGRAPHS

Please refer to the report number is 2501X31094E-EUT.

EXHIBIT B - PRODUCT SIMILARITY DECLARATION OF LETTER

Dragino Technology Co., Limited.
Address: Room 202, BaoChengTai industrial park, No.8 CaiYun LongCheng Street, LongGang District,
Shenzhen China
Tel: 86-0755-86610821; Fax: 86-0755-61673660
Email: edwin@dragino.com

Product Similarity Declaration

Date: 2025-11-09

Dear Sir/Madam,

We, **Dragino Technology Co., Limited.** hereby declare that we have a product named as LoRaWAN Sensor Node (model: DDS75-LB), were tested by Bay Area Compliance Laboratories Corp. (Shenzhen). Meanwhile, For our marketing purpose, we would like to list a series model (LDS25-LB, LDS25-LS, LDS25-LB2, CS01-LB, CS01-LS, CS01-LB2, SE01-LB, SE01-LS, SE01-LB2, DDS75-LB, DDS75-LS, DDS75-LB2, DDS45-LB, DDS45-LS, DDS45-LB2, DDS20-LB, DDS20-LS, DDS20-LB2, MDS120-LB, MDS120-LS, MDS120-LB2, SPH01-LB, SPH01-LS, SPH01-LB2, LMS01-LB, LMS01-LS, LMS01-LB2, WQS-LB, WQS-LS, WQS-LB2, SE0X-LB, SE0X-LS, SE0X-LB2, WSC2-LB, WSC2-LS, WSC2-LB2, MDS200-LB, MDS200-LS, MDS200-LB2, LDS40-LB, LDS40-LS, LDS40-LB2, LDS12-LB, LDS12-LS, LDS12-LB2, DDS04-LB, DDS04-LS, DDS04-LB2, RS485-LB, RS485-LB2, RS485-LS, RS485W-LB, RS485W-LS, RS485W-LB2) on reports and certificate. All the models are identical in electrically, the differences are shown in the table below. No other changes were made to them.

Model	Probes	PCB	Battery
DDS75-LB	Ultrasonic Probe	without charging module	8500mAh battery
RS485-LB	N/A		
SE01-LB	Soil Moisture Probe		
DDS45-LB	Ultrasonic Probe		
DDS20-LB	Ultrasonic Probe		
MDS120-LB	Microwave Probe		
MDS200-LB	Microwave Probe		
LDS12-LB	LiDar Probe		
SPH01-LB	Soil PH probe		
LMS01-LB	Leaf Moisture Probe		
DDS04-LB	Ultrasonic Probe		
LDS25-LB	LiDAR Distance Probe		
CS01-LB	Current Probe		
WQS-LB	Smoke Probe		
SE0X-LB	DR-SE-6P Soil Moisture & EC. Probe		
WSC2-LB	Alarm		
LDS40-LB	LiDAR Distance Probe		

DDS75-LS	Soil Moisture Probe	Has a charging module	3000mAh rechargeable battery
RS485-LS	without Probe		
SE01-LS	Ultrasonic Probe		
DDS45-LS	Microwave Probe		
DDS20-LS	Microwave Probe		
MDS120-LS	LiDar Probe		
MDS200-LS	Soil PH probe		
LDS12-LS	Leaf Moisture Probe		
SPH01-LS	Ultrasonic Probe		
LMS01-LS	Soil Moisture Probe		
DDS04-LS	Ultrasonic Probe		
LDS25-LS	LiDAR Distance Probe		
CS01-LS	Current Probe		
WQS-LS	Smoke Probe		
SE0X-LS	DR-SE-6P Soil Moisture & EC. Probe		
WSC2-LS	Alarm		
LDS40-LS	LiDAR Distance Probe		
DDS75-LB2	Soil Moisture Probe	Without charging module	19000mAh battery
RS485-LB2	without Probe		
SE01-LB2	Ultrasonic Probe		
DDS45-LB2	Microwave Probe		
DDS20-LB2	Microwave Probe		
MDS120-LB2	LiDar Probe		
MDS200-LB2	Soil PH probe		
LDS12-LB2	Leaf Moisture Probe		
SPH01-LB2	Ultrasonic Probe		
LMS01-LB2	Soil Moisture Probe		
DDS04-LB2	Ultrasonic Probe		
LDS25-LB2	LiDAR Distance Probe		
CS01-LB2	Current Probe		
WQS-LB2	Smoke Probe		
SE0X-LB2	DR-SE-6P Soil Moisture & EC. Probe		
WSC2-LB2	Alarm		
LDS40-LB2	LiDAR Distance Probe		
RS485W-LS	Without probe	1. Has a charging module 2. Support	3000mAh rechargeable battery

		12Vdc output	
RS485W-LB	Without probe	3. Has a charging module 4. Support 12Vdc output	8500mAh battery
RS485W-LB2	Without probe	1. Has a charging module 2. Support 12Vdc output	19000mAh battery

We confirm that all information above is true, and we'll be responsible for all the consequences. Please contact me if you have any question.

Sincerely Yours,

Signature

edwin chen

Manager



BELOW IS THE REFERENCED REPORT

TEST REPORT

Applicant Name : Dragino Technology Co., Limited
Address : Room 202,BaoChengTai industrial park,No.8 CaiYun
LongCheng Street,LongGang District,Shenzhen China
Report Number : SZ1230802-44707E-RF-22B

Test Standard (s)

ETSI EN 300 220-1 V3.1.1 (2017-02); ETSI EN 300 220-2 V3.2.1 (2018-06)

Sample Description

Product Type: LoRaWAN Sensor Node
Model No.: DDS75-LB
Multiple Model(s) No.: RS485-LB, SE01-LB, DDS45-LB, DDS20-LB, MDS120-LB,
MDS200-LB, LDS12-LB, SPH01-LB, LMS01-LB, DDS04-LB
Trade Mark: DRAGINO
Date Received: 2023/08/02
Report Date: 2023/10/13

Test Result:	Pass*
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* In the configuration tested, the EUT complied with the standards above.

Prepared and Checked By:

April Zhang

April Zhang
RF Engineer

Approved By:

Nancy Wang

Nancy Wang
RF Supervisor

Note: The information marked * is provided by the applicant, the laboratory is not responsible for its authenticity and this information can affect the validity of the result in the test report. Customer model name, addresses, names, trademarks etc. are included.

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DOCUMENT REVISION HISTORY

Revision Number	Report Number	Description of Revision	Date of Revision
0	SZ1230802-44707E-RF-22B	Original Report	2023-10-13

GENERAL INFORMATION

Product Description for Equipment under Test (EUT)

Product	LoRaWAN Sensor Node
Test Model	DDS75-LB
Multiple Model(s)	RS485-LB, SE01-LB, DDS45-LB, DDS20-LB, MDS120-LB, MDS200-LB, LDS12-LB, SPH01-LB, LMS01-LB, DDS04-LB (Please refer to the DoS* provided by the applicant)
Frequency Range	865-868MHz & 868-868.6MHz
Maximum Transmit Power (ERP)	13.17dBm
Modulation Technique	FSK
Antenna Specification*	2dBi (It is provided by the manufacturer)
Voltage Range	DC 3.6V from battery
Sample serial number	299I-1 (RF Conducted Test) 299I-2 (RF Radiated Test) (Assigned by BACL, Shenzhen)
Sample/EUT Status	Good condition
Normal/Extreme Condition*	L.V.: Low Voltage 2.5 V _{DC} ; L.T.: Low Temperature -40°C N.V.: Normal Voltage 3.6 V _{DC} ; N.T.: Normal Temperature +25°C H.V.: High Voltage 3.7 V _{DC} ; H.T.: High Temperature +85°C Note: the extreme condition was declared by manufacturer.

Objective

The test report is in accordance with ETSI EN 300 220-2 V3.2.1 (2018-06), Short Range Devices (SRD) operating in the frequency range 25 MHz to 1 000 MHz; Part 2: Harmonised Standard for access to radio spectrum for non specific radio equipment

The objective is to determine the compliance of the EUT with ETSI EN 300 220-2 V3.2.1 (2018-06).

Measurement Uncertainty

Parameter	Flab	Maximum allow uncertainty
Radio frequency	±0,5 ppm	±0,5 ppm
RF power, conducted	±0.73dB	±1.5dB
Conducted spurious emission of transmitter, valid up to 6 GHz	±1.6dB	±3dB
Conducted emission of receivers	±1.6dB	±3dB
Below 1GHz emissions, radiated	±4.75dB	±6dB
Above 1GHz emissions, radiated	±4.88dB	±6dB
RF level uncertainty for a given BER	±1.5dB	±1.5 dB
Occupied BandWidth	±5%	±5%
Temperature	±1 °C	±2,5 °C
Humidity	±1%	±10%

Note: The extended uncertainty given in this report is obtained by combining the standard uncertainty times the coverage factor K with the 95% confidence interval. Otherwise required by the applicant or Product Regulations, Decision Rule in this report did not consider the uncertainty.

Test Facility

The Test site used by Bay Area Compliance Laboratories Corp. (Shenzhen) to collect test data is located on the 5F(B-West) , 6F, 7F, the 3rd Phase of Wan Li Industrial Building D, Shihua Rd, FuTian Free Trade Zone, Shenzhen, China.

Each test item follows test standards and with no deviation.

SYSTEM TEST CONFIGURATION

Description of Test Configuration

The system was configured for testing in a test mode.

EUT Exercise Software

“SmartSnippets Toolbox V5.0.12.exe” exercise software was used and the power level is 0*. The software and power level was provided by the manufacturer.

Special Accessories

No special accessories.

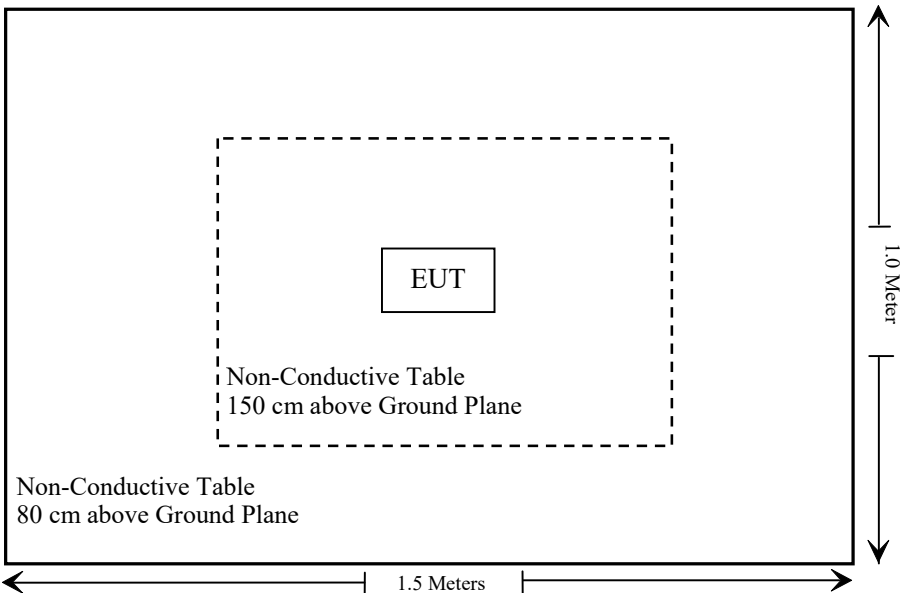
Support Equipment List and Details

Manufacturer	Description	Model	Serial Number
/	/	/	/

External I/O Cable

Cable Description	Length (m)	From/Port	To
/	/	/	/

Block Diagram of Test Setup



SUMMARY OF TEST RESULTS

ETSI EN 300 220-2 V3.2.1 (2018-06)

Rules	Description of Test	Result	Condition
§4.2.1	Operating frequency	Compliant	/
§4.2.2	Unwanted emissions in the spurious domain	Compliant	/
§4.3.1	Effective radiated power	Compliant	/
§4.3.2	Maximum e.r.p. spectral density	Not Applicable	Applies to EUT using annex B band I. Applies to EUT using DSSS or wideband techniques other than FHSS modulation, using annex C band W, AA or AC.
§4.3.3	Duty cycle	Not Applicable	Not applicable to EUT with polite spectrum access where permitted in annex B, table B.1.
§4.3.4	Occupied Bandwidth	Compliant	/
§4.3.5	Tx Out of Band Emissions	Compliant	Applies to EUT with OCW > 25 kHz.
§4.3.6	Transient Power	Compliant	/
§4.3.7	Adjacent channel power	Not Applicable	Applies to EUT with OCW ≤ 25 kHz.
§4.3.8	TX behaviour under Low Voltage Conditions	Compliant	Applies to battery powered EUT.
§4.3.9	Adaptive Power Control	Not Applicable	Applies to EUT with adaptive power control using annex C band AF.
§4.3.10	FHSS equipment	Not Applicable	Applies to FHSS EUT using the band 863 MHz to 870 MHz.
§4.3.11	Short term behaviour	Not Applicable	Applies to EUT using annex C bands AD, AE, AF, AG, AH, or AI.
§4.4.1	RX sensitivity	Compliant	Applies to EUT with polite spectrum access.
§4.4.2	Blocking	Compliant	/
§4.5.2	Clear Channel Assessment threshold	Compliant	Applies to EUT with polite spectrum access.
§4.5.3	Polite spectrum access timing parameters	Compliant	Applies to EUT with polite spectrum access.
§4.5.4	Adaptive Frequency Agility	Not Applicable	Applies to EUT with AFA.

TEST EQUIPMENT LIST

Manufacturer	Description	Model	Serial Number	Calibration Date	Calibration Due Date
Radiated Emission Test					
R&S	EMI Test Receiver	ESR3	102455	2023/02/08	2024/02/07
Sonoma instrument	Pre-amplifier	310 N	186238	2023/06/08	2024/06/07
Sunol Sciences	Broadband Antenna	JB1	A040904-1	2023/07/20	2024/07/19
Unknown	Cable	Chamber Cable 1	F-03-EM236	2023/8/3	2024/8/2
Unknown	Cable	Chamber Cable 4	EC-007	2023/8/3	2024/8/2
COM-POWER	Dipole Antenna	AD-100	721027	NCR	NCR
Rohde & Schwarz	Spectrum Analyzer	FSV40	101605	2023/04/18	2024/04/17
COM-POWER	Pre-amplifier	PA-122	181919	2023/06/29	2024/06/28
Schwarzbeck	Horn Antenna	BBHA9120D(1201)	1143	2023/07/26	2024/07/25
A.H.System	Horn Antenna	SAS-200/571	135	2021/07/14	2024/07/13
Insulted Wire Inc.	RF Cable	SPS-2503-3150	02222010	2022/11/25	2023/11/24
Unknown	RF Cable	W1101-EQ1 OUT	F-19-EM005	2022/11/25	2023/11/24
Unknown	1.3G High Pass filter	1.3GHz	101120	2023/03/30	2024/03/29
Agilent	Signal Generator	N5183A	MY51040755	2023/02/08	2024/02/07
RF Conducted Test					
Rohde & Schwarz	SPECTRUM ANALYZER	FSU26	200982	2023/02/08	2024/02/07
BACL	Temperature & Humidity Chamber	BTH-150-40	30145	2023/02/08	2024/02/07
WEINSCHL	3dB Attenuator	Unknown	F-03-EM121	2022/11/25	2023/11/24
Keysight	MXG Vector Signal Generator	N5182B	MY53051503	2023/02/08	2024/02/07
Agilent	Signal Generator	N5183A	MY51040755	2023/02/08	2024/02/07
instek	DC Power Supply	GPS-3030DD	EM832096	NCR	NCR
Fluke	Digital Multimeter	287	19000011	2023/06/08	2024/06/07

* **Statement of Traceability:** Bay Area Compliance Laboratories Corp. (Shenzhen) attests that all calibrations have been performed in accordance to requirements that traceable to National Primary Standards and International System of Units (SI).

ETSI EN 300 220-2 V3.2.1 (2018-06) §4.2.1 – OPERATING FREQUENCY**Applicable Standard**

According to ETSI EN 300 220-1 V3.1.1 (2017-02) clause 5.1.1, the nominal operating frequency is the centre of a channel of width OCW.

Limit: The manufacturer may declare either one or more operating frequencies and operating channels. Operating channel(s) shall be entirely within operational frequency bands allowed by annexes B, C or any NRI

The below information shall be recorded in the test report

Value	Note
Operational Frequency band or bands	Declared by the manufacturer
Nominal Operating Frequency or Frequencies	Declared by the manufacturer
Operating Channel width(s) - OCW	Declared by the manufacturer

Test Result

The operational frequency band or bands, nominal operating frequency or Frequencies and operating channel width(s) – OCW are declared by the manufacturer

Note: Pass, which is declared by the manufacturer.

Operating Channel width(s) – OCW	Nominal Operating Frequency (MHz)	Operational Frequency band or bands (MHz)	EN 300220-2 Annex B Table B.1
130kHz	867.1	865-868	Band L
	867.9		
	868.1	868-868.6	Band M
	868.5		
260kHz	868.3	868-868.6	Band M

ETSI EN 300 220-2 V3.2.1 (2018-06) §4.2.2 – UNWANTED EMISSIONS IN THE SPURIOUS DOMAIN

Applicable Standard

According to ETSI EN 300 220-1 V3.1.1 (2017-02) clause 5.9.1.

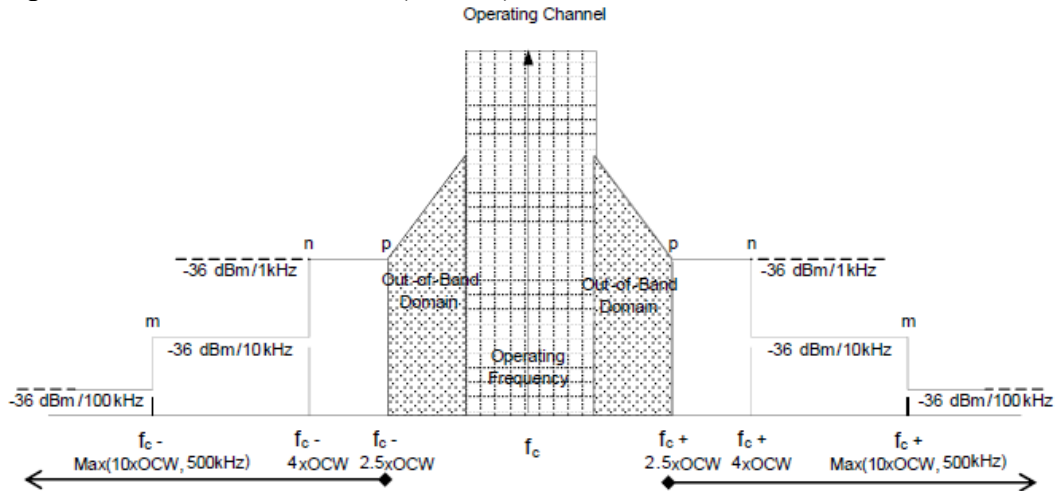


Figure 7: Spectrum Mask for Unwanted Emissions in the Spurious Domain with reference BW

Spurious emissions are unwanted emissions in the spurious domain at frequencies other than those of the Operating Channel and its Out Of Band Domain. The relevant spurious domain is shown in Figure 7.

Limit: The power of any unwanted emission in the spurious domain shall not exceed the values given in Table 19.

Table 19: Spurious domain emission limits

Frequency	47 MHz to 74 MHz 87.5 MHz to 118 MHz 174 MHz to 230 MHz 470 MHz to 790 MHz	Other frequencies below 1 000 MHz	Frequencies above 1 000 MHz
State			
TX mode	-54 dBm	-36 dBm	-30 dBm
RX and all other modes	-57 dBm	-57 dBm	-47 dBm

Method of Measurement

According to ETSI EN 300 220-1 V3.1.1 (2017-02) clause 5.9.3.

Test Data

Environmental Conditions

Temperature:	25.5~26°C
Relative Humidity:	50~55 %
ATM Pressure:	101.0 kPa

The testing was performed by Warren Huang on 2023-08-28 for below 1GHz, Dylan Yang from 2023-08-28 to 2023-10-13 for above 1GHz and Hanic Pan from 2023-09-07 to 2023-10-11 for conducted emission.

Test result: Pass

Please refer to the following tables.

Radiated spurious emissions:

Below 1GHz:

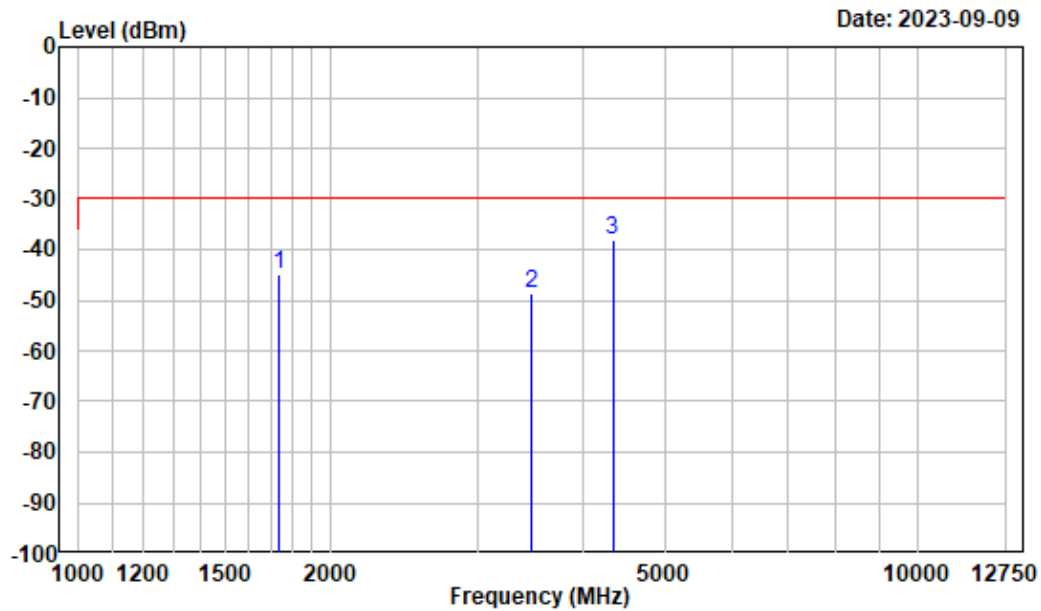
Frequency (MHz)	Receiver Reading (dBμV)	Turn Table Angle Degree	Rx Antenna		Substituted			Absolute Level (dBm)	EN 300 220	
			Height (m)	Polarity (H/V)	SG Level (dBm)	Cable Loss (dB)	Antenna Gain (dBd/dBi)		Limit (dBm)	Margin (dB)
Transmitting Mode										
130kHz 867.1MHz										
950.6	33.69	143	2.3	H	-62.8	1.36	0.0	-64.16	-36	28.16
950.6	32.63	106	1.9	V	-61.4	1.36	0.0	-62.76	-36	26.76
130kHz 867.9MHz										
951.6	33.68	97	2.4	H	-62.8	1.36	0.0	-64.16	-36	28.16
951.6	33.43	191	1.6	V	-60.6	1.36	0.0	-61.96	-36	25.96
130kHz 868.1MHz										
952.2	32.66	100	2.0	H	-63.8	1.36	0.0	-65.16	-36	29.16
952.2	32.84	343	2.0	V	-61.2	1.36	0.0	-62.56	-36	26.56
130kHz 868.5MHz										
953.7	34.05	206	1.2	H	-62.5	1.36	0.0	-63.86	-36	27.86
953.7	32.93	173	2.2	V	-61.1	1.36	0.0	-62.46	-36	26.46
260kHz 868.3MHz										
954.6	32.96	274	2.1	H	-63.5	1.36	0.0	-64.86	-36	28.86
954.6	32.78	276	2.0	V	-61.3	1.36	0.0	-62.66	-36	26.66
Receiving Mode										
130kHz 867.1MHz										
956.8	32.34	287	2.3	H	-64.2	1.36	0.0	-65.56	-57	8.56
956.8	31.70	182	1.8	V	-62.4	1.36	0.0	-63.76	-57	6.76
130kHz 867.9MHz										
954.0	33.17	205	2.3	H	-63.3	1.36	0.0	-64.66	-57	7.66
954.0	32.35	260	2.4	V	-61.7	1.36	0.0	-63.06	-57	6.06
130kHz 868.1MHz										
958.6	33.46	128	1.9	H	-63.0	1.36	0.0	-64.36	-57	7.36
958.6	31.52	218	1.8	V	-62.5	1.36	0.0	-63.86	-57	6.86
130kHz 868.5MHz										
958.2	32.68	110	1.4	H	-63.8	1.36	0.0	-65.16	-57	8.16
958.2	32.10	288	1.3	V	-62.0	1.36	0.0	-63.36	-57	6.36
260kHz 868.3MHz										
956.3	32.83	157	1.4	H	-63.7	1.36	0.0	-65.06	-57	8.06
956.3	31.89	357	1.7	V	-62.2	1.36	0.0	-63.56	-57	6.56
Standby Mode										
956.4	33.06	86	2.5	H	-63.4	1.36	0.0	-64.76	-57	7.76
956.4	31.46	292	1.7	V	-62.6	1.36	0.0	-63.96	-57	6.96

Note:

Absolute Level = SG Level - Cable Loss + Antenna Gain

Margin = Limit - Absolute Level

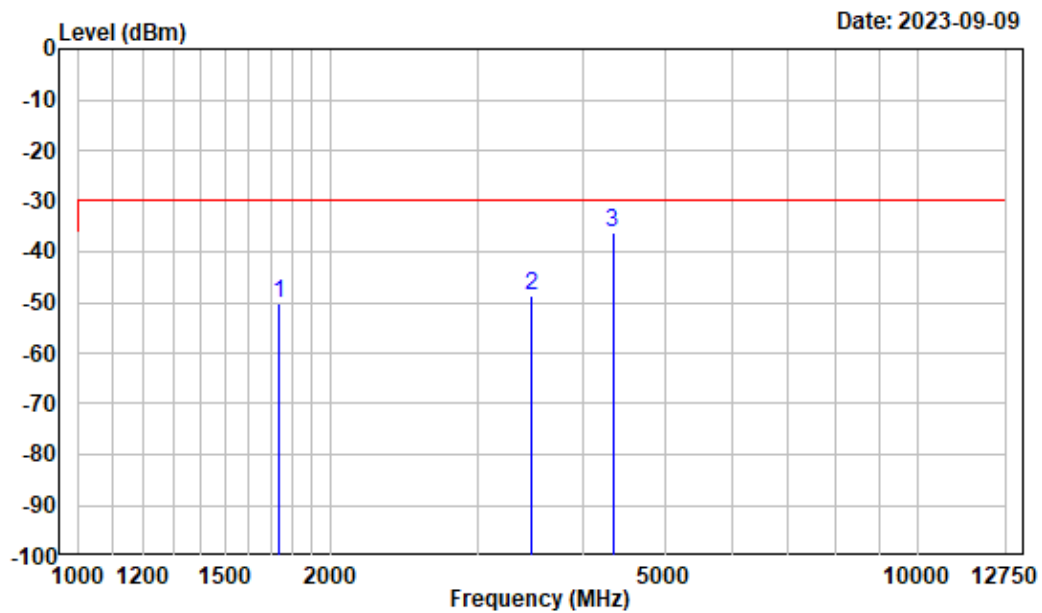
The unit of antenna gain is dBd for below 1GHz frequency range and dBi for above 1GHz frequency range

Above 1GHz:**Test mode: Transmitting****130kHz_867.1MHz****Horizontal**

Site : chamber
Condition : Horizontal
Project Number: SZ1230802-44707E-RF
Test Mode : TX 867.1 power 0

	Freq	Factor	Read Level	Level	Limit	Over Limit	Remark
	MHz	dB	dBm	dBm	dBm	dB	
1	1734.200	6.54	-51.55	-45.01	-30.00	-15.01	Peak
2	3468.400	8.55	-57.27	-48.72	-30.00	-18.72	Peak
3	4335.500	11.17	-49.11	-37.94	-30.00	-7.94	Peak

Vertical

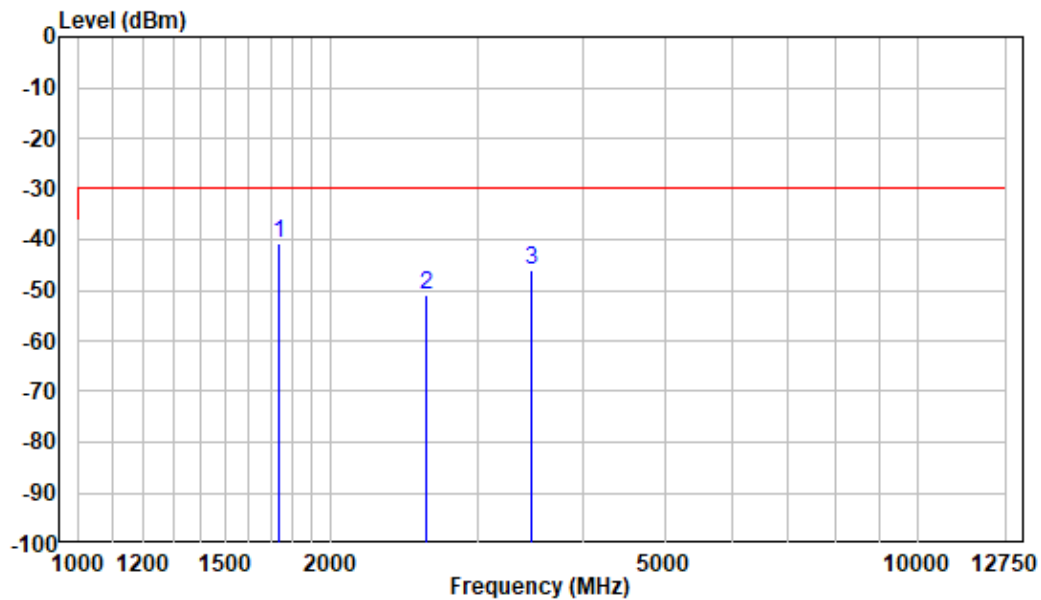


Site : chamber
Condition : Vertical
Project Number: SZ1230802-44707E-RF
Test Mode : TX 867.1 power 0

	Freq	Factor	Read Level	Level	Limit Line	Over Limit	Remark
	MHz	dB	dBm	dBm	dBm	dB	
1	1734.200	5.94	-56.24	-50.30	-30.00	-20.30	Peak
2	3468.400	8.86	-57.56	-48.70	-30.00	-18.70	Peak
3	4335.500	11.37	-47.78	-36.41	-30.00	-6.41	Peak

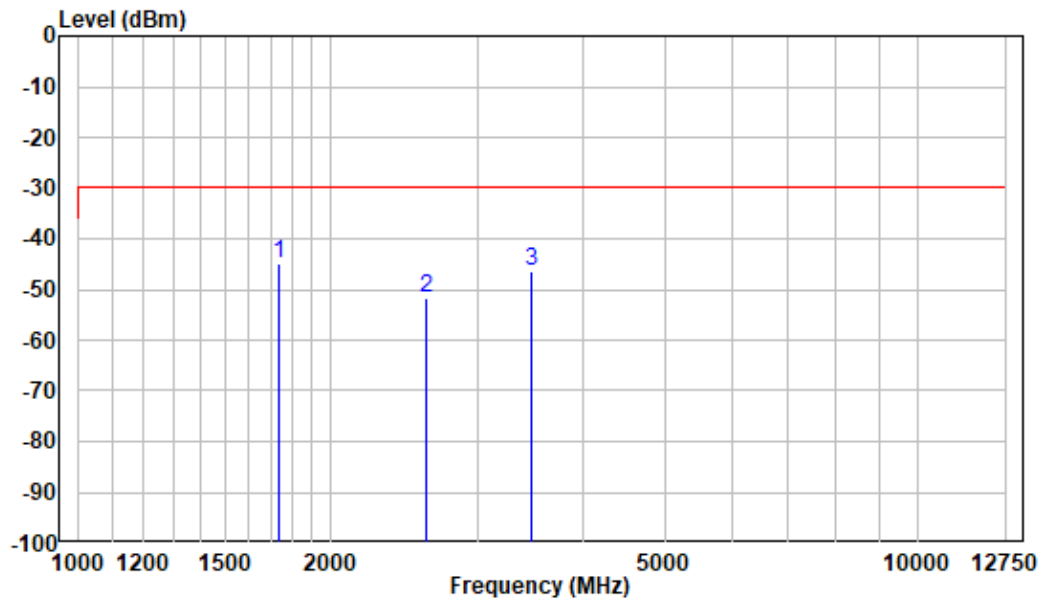
130kHz_867.9MHz

Horizontal



Site : chamber
Condition : Horizontal
Project Number: SZ1230802-44707E-RF
Test Mode : 867.9 TX

	Freq	Factor	Read		Limit	Over	Remark
			Level	Level	Line	Limit	
	MHz	dB	dBm	dBm	dBm	dB	
1	1735.800	6.54	-47.23	-40.69	-30.00	-10.69	Peak
2	2603.700	7.47	-58.34	-50.87	-30.00	-20.87	Peak
3	3471.600	8.55	-54.44	-45.89	-30.00	-15.89	Peak

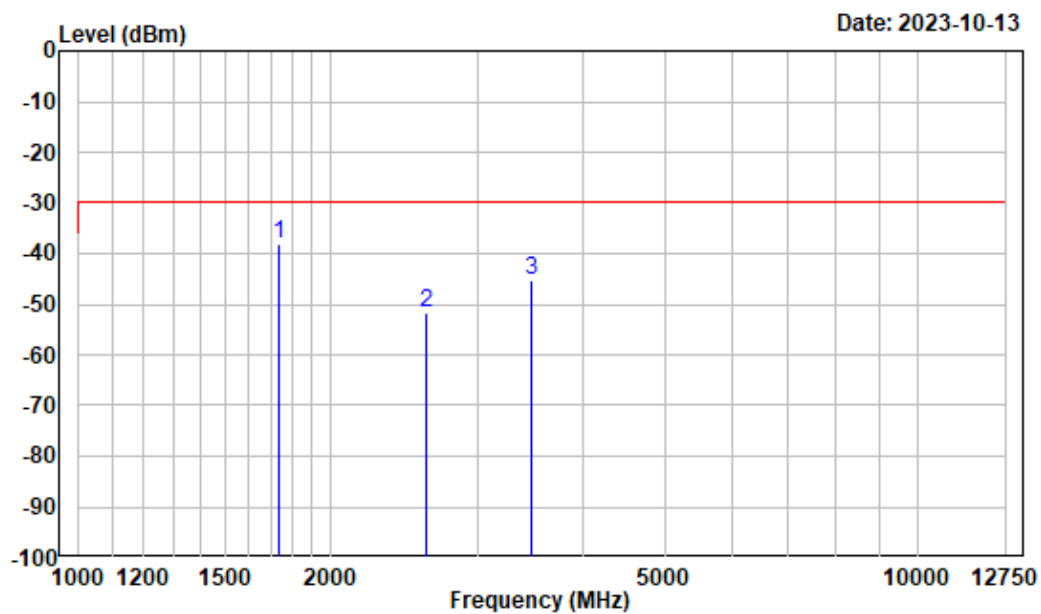
Vertical

Site : chamber
Condition : Vertical
Project Number: SZ1230802-44707E-RF
Test Mode : 867.9 TX

	Freq	Factor	Read Level	Level	Limit Line	Over Limit	Remark
	MHz	dB	dBm	dBm	dBm	dB	
1	1735.800	5.95	-50.99	-45.04	-30.00	-15.04	Peak
2	2603.700	7.72	-59.51	-51.79	-30.00	-21.79	Peak
3	3471.600	8.86	-55.38	-46.52	-30.00	-16.52	Peak

130kHz_868.1MHz

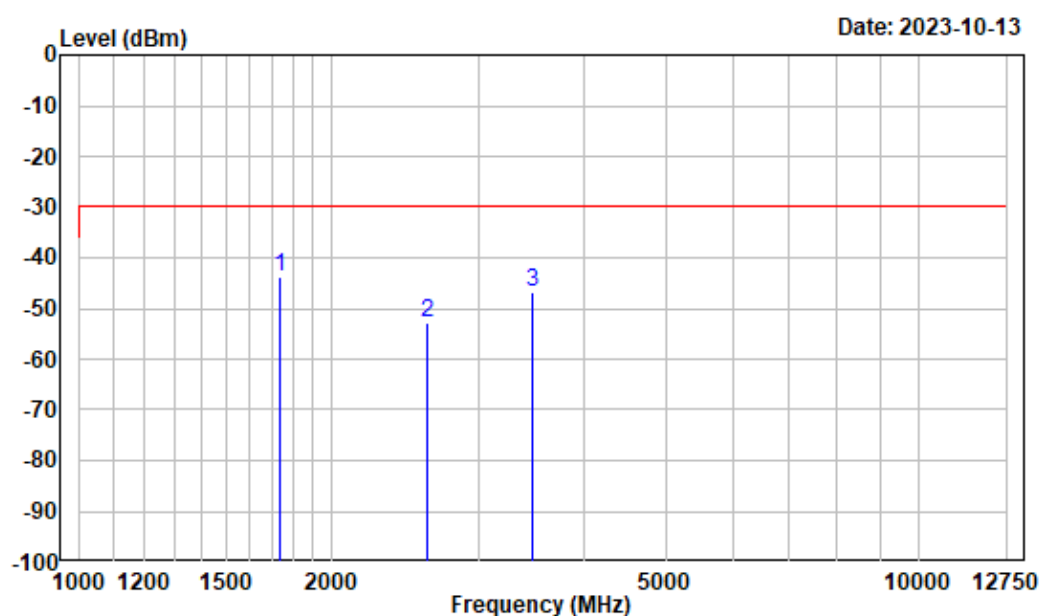
Horizontal



Site : chamber
Condition : Horizontal
Project Number: SZ1230802-44707E-RF
Test Mode : 868.1TX

	Freq	Factor	Read		Limit	Over	Remark
			Level	Level	Line	Limit	
	MHz	dB	dBm	dBm	dBm	dB	
1	1736.200	6.54	-44.51	-37.97	-30.00	-7.97	Peak
2	2601.300	7.47	-59.18	-51.71	-30.00	-21.71	Peak
3	3468.400	8.55	-53.99	-45.44	-30.00	-15.44	Peak

Vertical

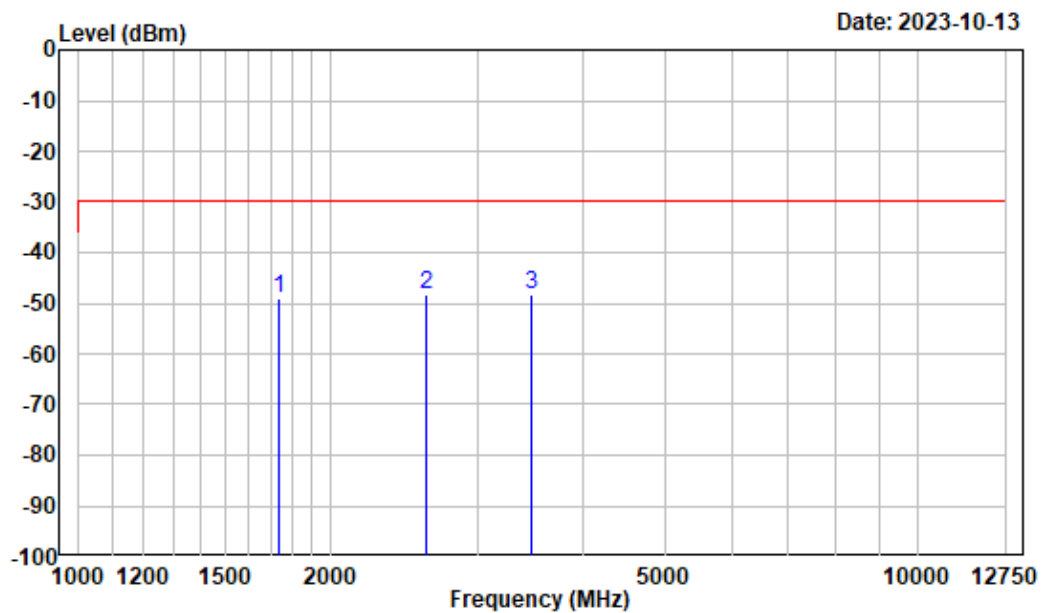


Site : chamber
Condition : Vertical
Project Number: SZ1230802-44707E-RF
Test Mode : 868.1TX

	Freq	Factor	Read Level	Level	Limit Line	Over Limit	Remark
	MHz	dB	dBm	dBm	dBm	dB	
1	1736.200	5.95	-49.80	-43.85	-30.00	-13.85	Peak
2	2601.300	7.72	-60.69	-52.97	-30.00	-22.97	Peak
3	3468.400	8.86	-55.51	-46.65	-30.00	-16.65	Peak

130kHz_868.5MHz

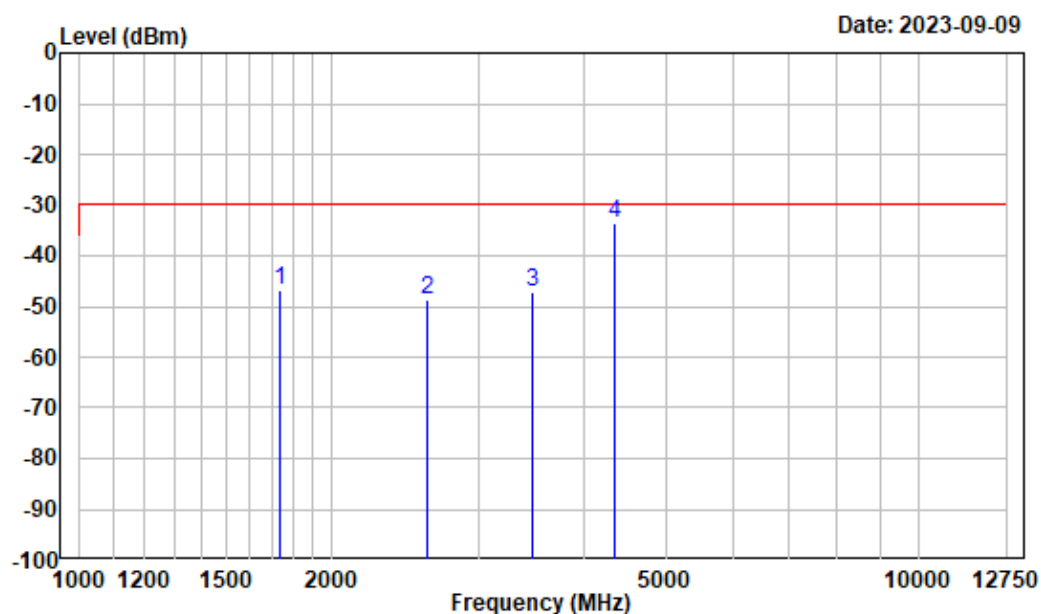
Horizontal



Site : chamber
Condition : Horizontal
Project Number: SZ1230802-44707E-RF
Test Mode : TX 868.5 power 0

	Freq	Factor	Read Level	Level	Limit Line	Over Limit	Remark
	MHz	dB	dBm	dBm	dBm	dB	
1	1736.000	6.54	-55.77	-49.23	-30.00	-19.23	Peak
2	2605.000	7.48	-55.69	-48.21	-30.00	-18.21	Peak
3	3473.500	8.55	-56.97	-48.42	-30.00	-18.42	Peak

Vertical

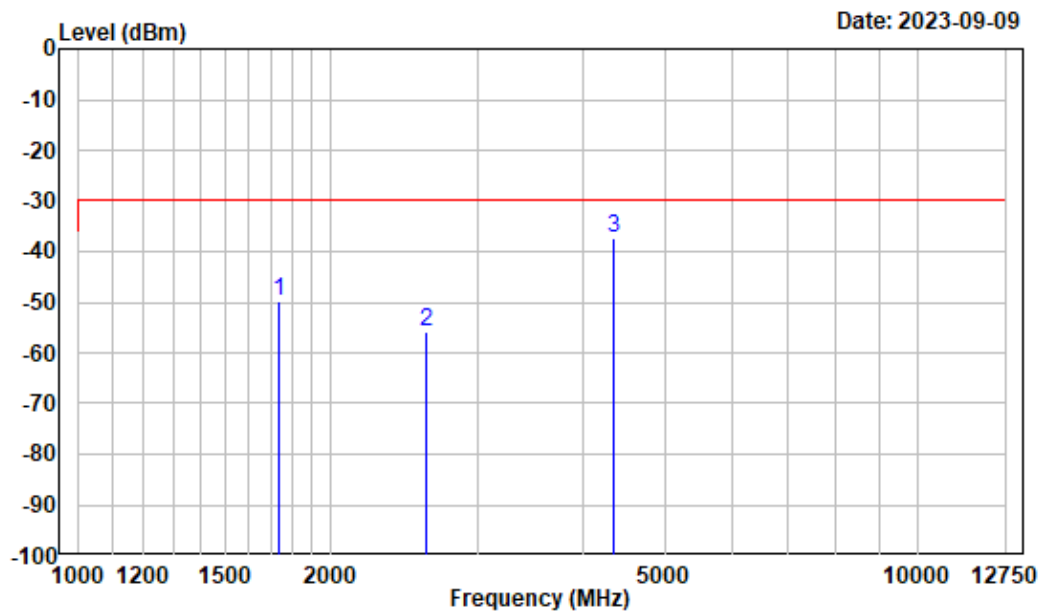


Site : chamber
Condition : Vertical
Project Number: SZ1230802-44707E-RF
Test Mode : TX 868.5 power 0

	Freq	Factor	Read Level	Level	Limit Line	Over Limit	Remark
	MHz	dB	dBm	dBm	dBm	dB	
1	1736.000	5.95	-52.80	-46.85	-30.00	-16.85	Peak
2	2605.000	7.85	-56.64	-48.79	-30.00	-18.79	Peak
3	3473.500	8.87	-56.18	-47.31	-30.00	-17.31	Peak
4	4342.500	11.40	-44.82	-33.42	-30.00	-3.42	Peak

260kHz_868.3MHz

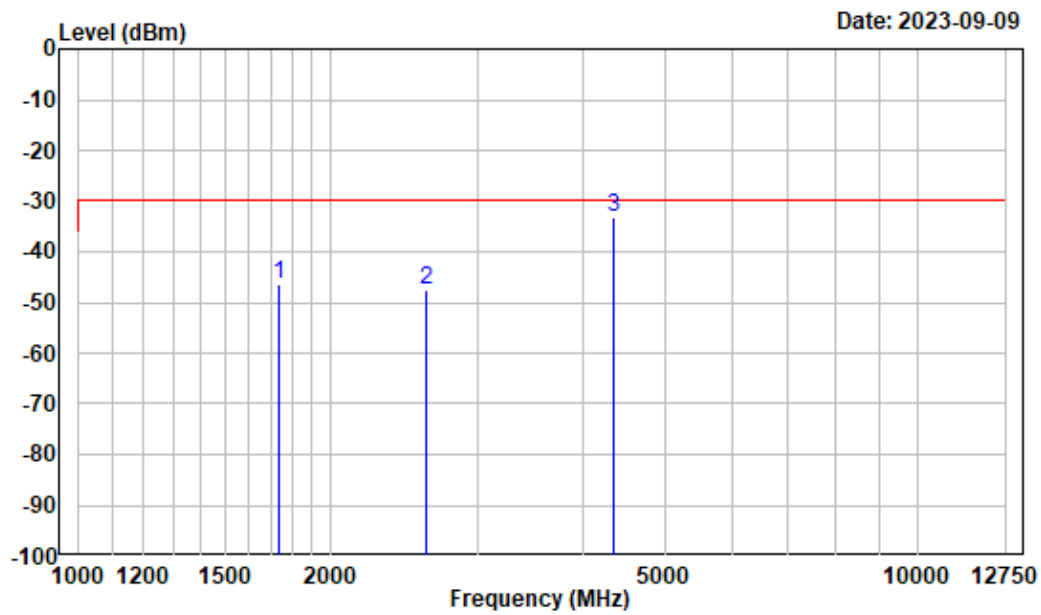
Horizontal



Site : chamber
Condition : Horizontal
Project Number: SZ1230802-44707E-RF
Test Mode : TX 868.3 power 0

	Freq	Factor	Read Level	Level	Limit Line	Over Limit	Remark
	MHz	dB	dBm	dBm	dBm	dB	
1	1736.600	6.54	-56.28	-49.74	-30.00	-19.74	Peak
2	2605.000	7.48	-63.26	-55.78	-30.00	-25.78	Peak
3	4341.500	11.20	-48.56	-37.36	-30.00	-7.36	Peak

Vertical



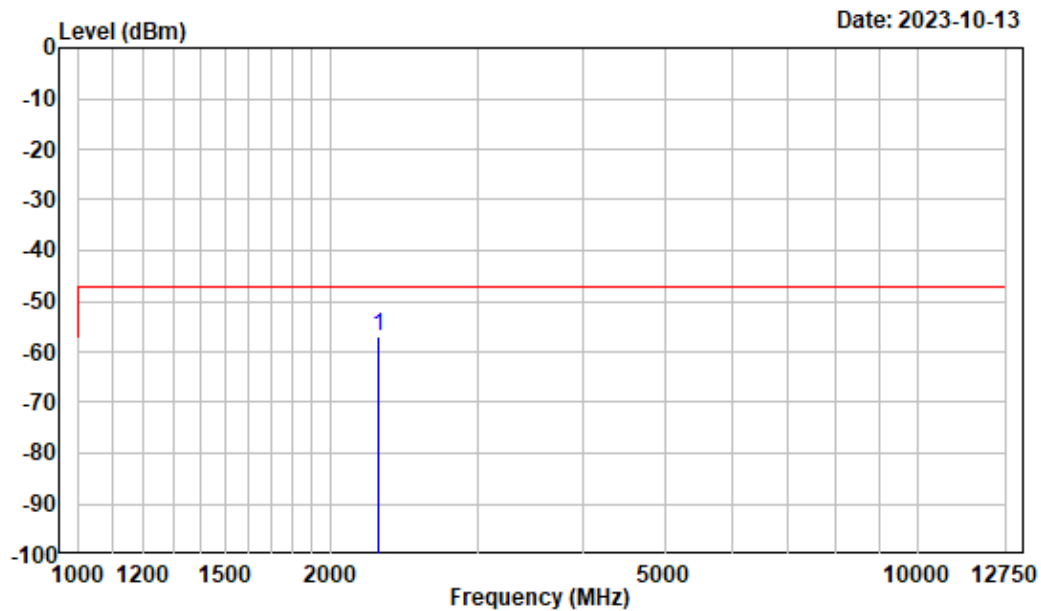
Site : chamber
Condition : Vertical
Project Number: SZ1230802-44707E-RF
Test Mode : TX 868.3 power 0

	Freq	Factor	Read Level	Level	Limit Line	Over Limit	Remark
	MHz	dB	dBm	dBm	dBm	dB	
1	1736.600	5.95	-52.44	-46.49	-30.00	-16.49	Peak
2	2604.900	7.85	-55.40	-47.55	-30.00	-17.55	Peak
3	4341.500	11.40	-44.76	-33.36	-30.00	-3.36	Peak

Test mode: Receiving

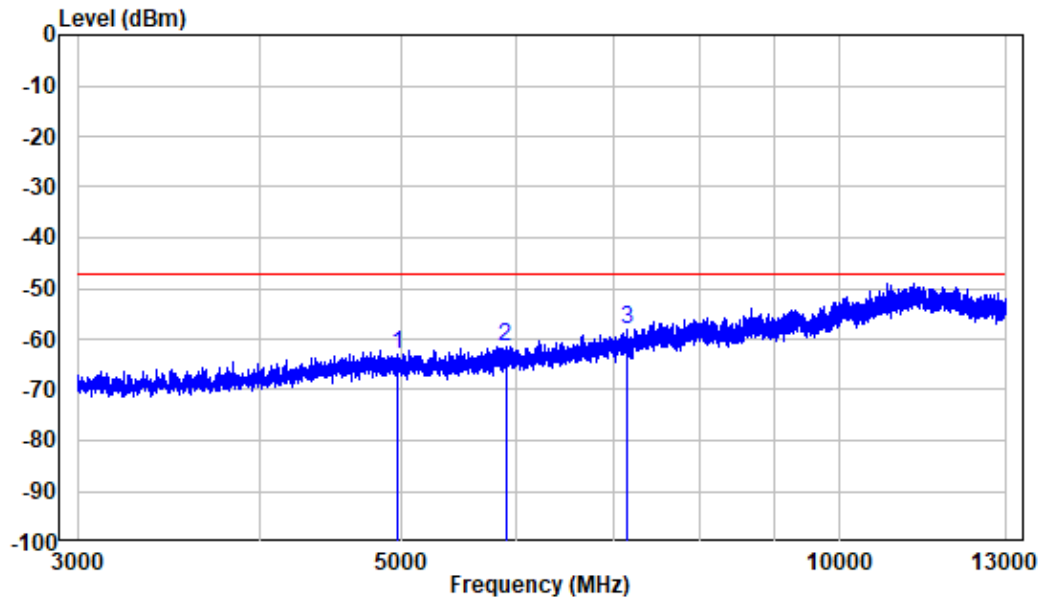
130kHz_867.1MHz

Horizontal



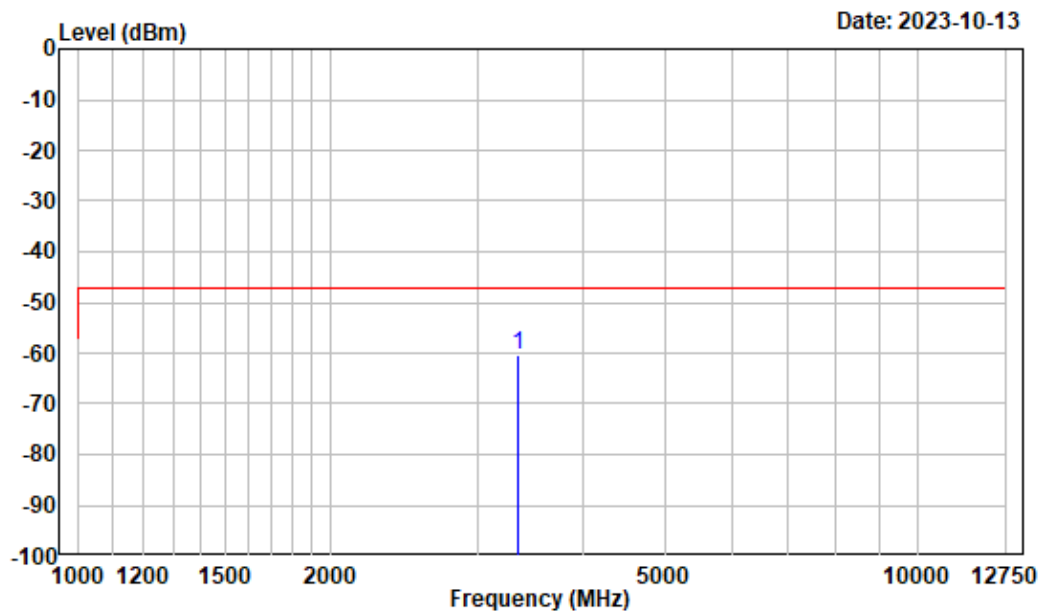
Site : chamber
Condition : Horizontal
Project Number: SZ1230802-44707E-RF
Test Mode : 867.1-RX

	Freq	Factor	Read Level	Level	Limit Line	Over Limit	Remark
	MHz	dB	dBm	dBm	dBm	dB	
1	2286.625	7.06	-63.97	-56.91	-47.00	-9.91	Peak

Vertical

Site : chamber
Condition : Vertical
Project Number: SZ1230802-44707E-RF
Test Mode : RX

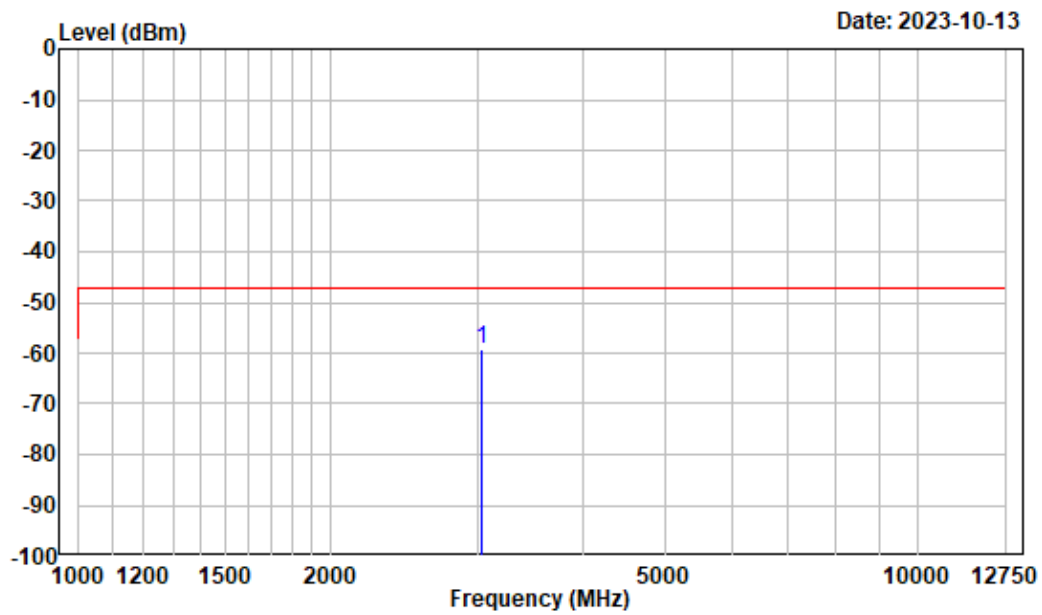
	Freq	Factor	Read Level	Level	Limit Line	Over Limit	Remark
	MHz	dB	dBm	dBm	dBm	dB	
1	4973.750	12.60	-75.73	-63.13	-47.00	-16.13	Peak
2	5897.500	13.79	-75.23	-61.44	-47.00	-14.44	Peak
3	7140.000	18.25	-76.29	-58.04	-47.00	-11.04	Peak

130kHz_867.9MHz**Horizontal**

Site : chamber
Condition : Horizontal
Project Number: SZ1230802-44707E-RF
Test Mode : 867.9-RX

Freq Factor		Read Level	Level	Limit Line	Over Limit	Remark
MHz	dB	dBm	dBm	dBm	dB	
1	3341.188	8.50	-68.90	-60.40	-47.00	-13.40 Peak

Vertical

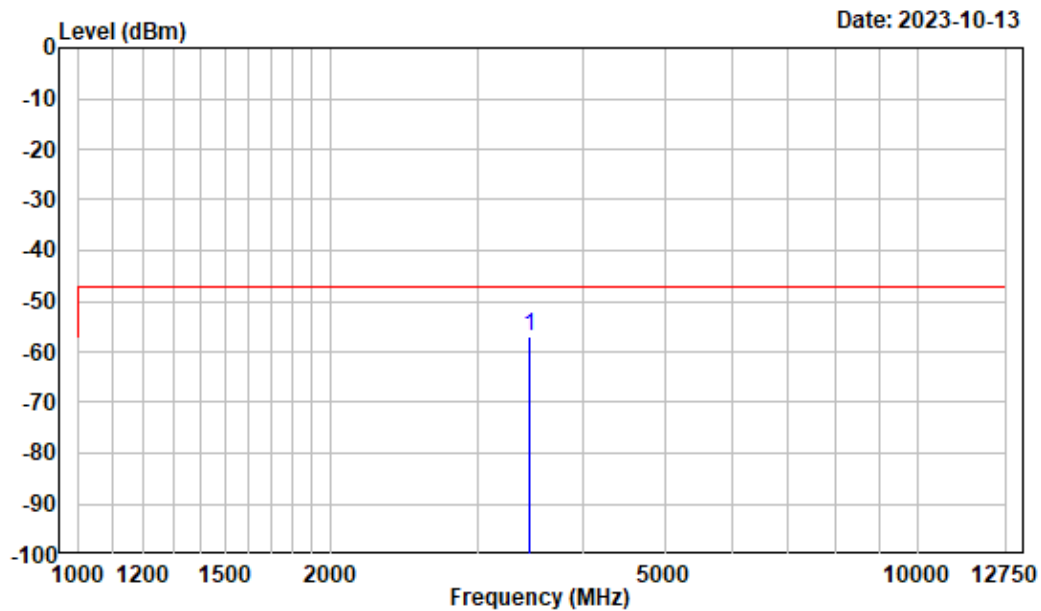


Site : chamber
Condition : Vertical
Project Number: SZ1230802-44707E-RF
Test Mode : 867.9-RX

	Freq	Factor	Read Level	Level	Limit Line	Over Limit	Remark
	MHz	dB	dBm	dBm	dBm	dB	
1	3032.750	8.63	-67.99	-59.36	-47.00	-12.36	Peak

130kHz_868.1MHz

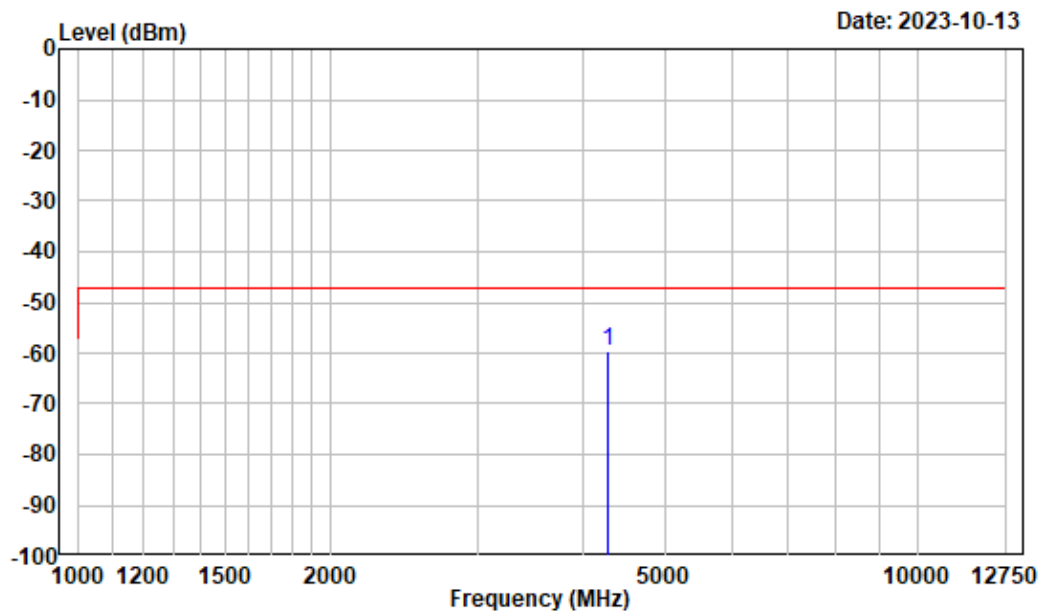
Horizontal



Site : chamber
Condition : Horizontal
Project Number: SZ1230802-44707E-RF
Test Mode : 868.1

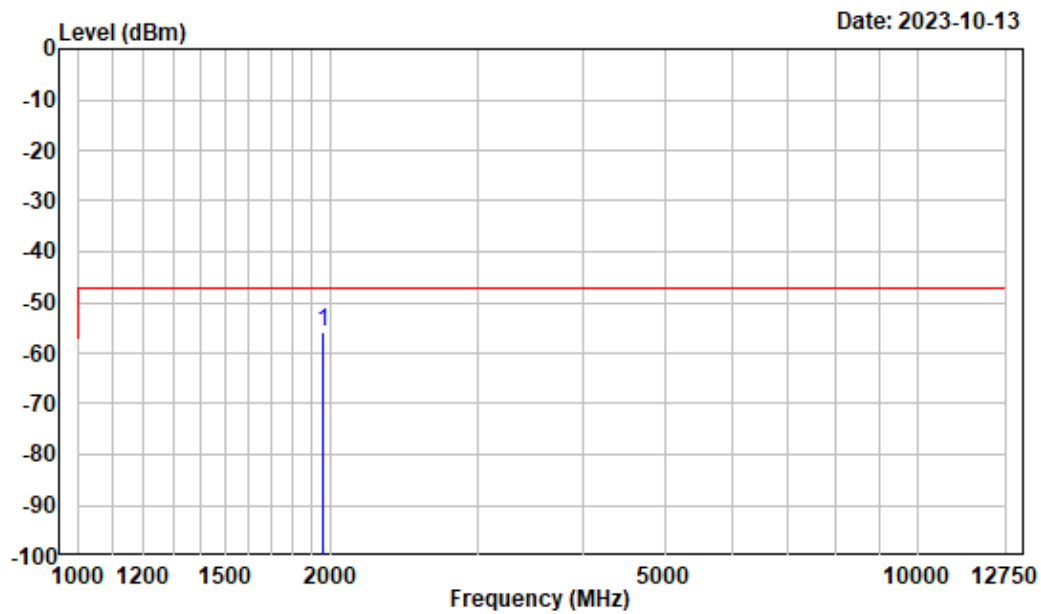
	Freq	Factor	Read Level	Level	Limit Line	Over Limit	Remark
	MHz	dB	dBm	dBm	dBm	dB	
1	3456.880	6.54	-63.47	-56.93	-47.00	-9.93	

Vertical



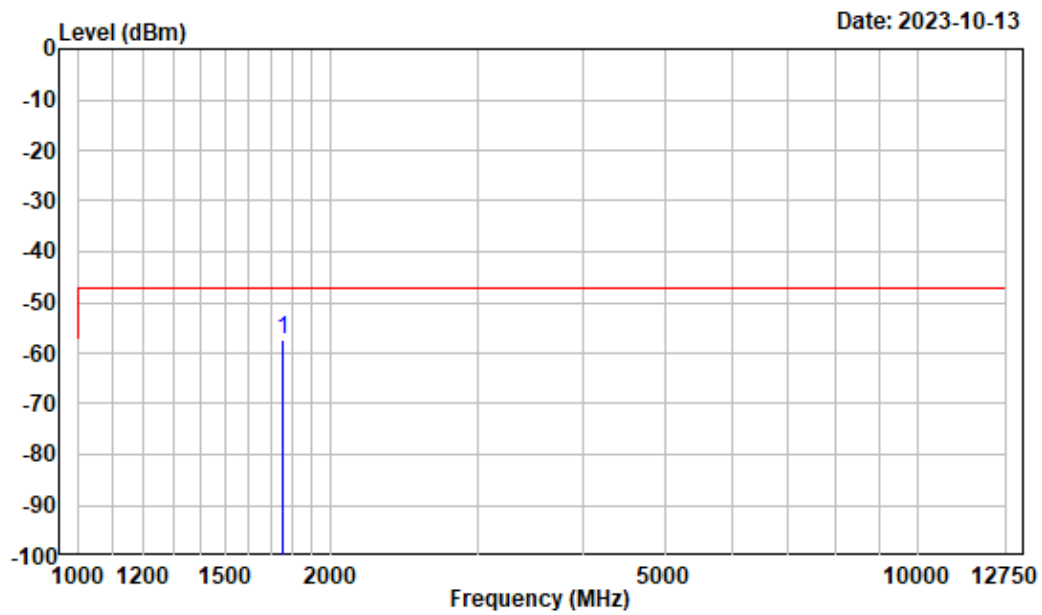
Site : chamber
Condition : Vertical
Project Number: SZ1230802-44707E-RF
Test Mode : 868.1

	Freq	Factor	Read Level	Level	Limit Line	Over Limit	Remark
	MHz	dB	dBm	dBm	dBm	dB	
1	4289.660	5.95	-65.72	-59.77	-47.00	-12.77	

130kHz_868.5MHz**Horizontal**

Site : chamber
Condition : Horizontal
Project Number: SZ1230802-44707E-RF
Test Mode : 868.5-RX

Freq Factor		Read Level	Level	Limit Line	Over Limit	Remark
MHz	dB	dBm	dBm	dBm	dB	
1	1957.625	6.78	-62.80	-56.02	-47.00	-9.02 Peak

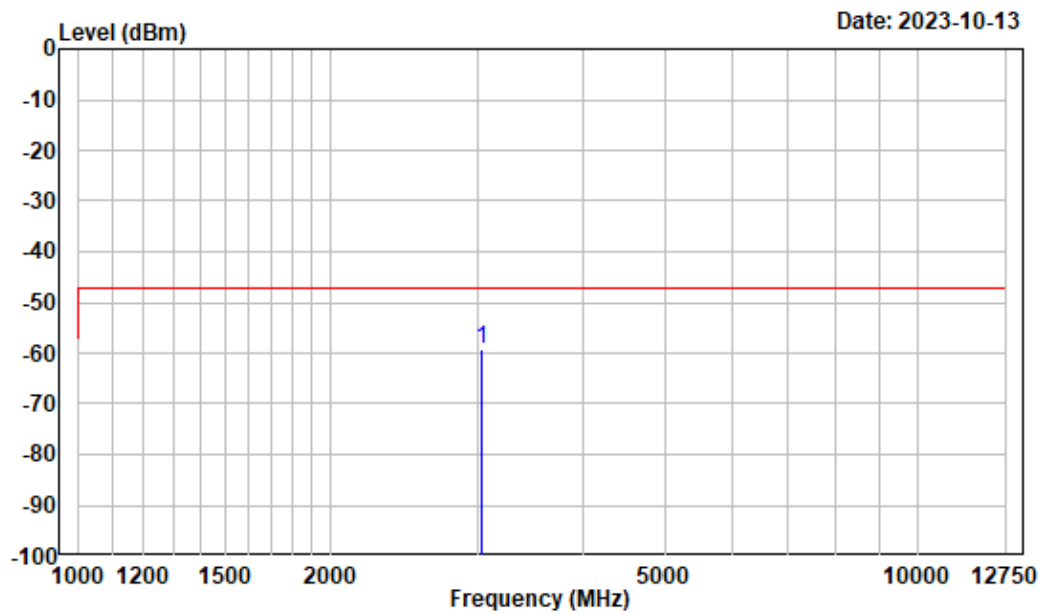
Vertical

Site : chamber
Condition : Vertical
Project Number: SZ1230802-44707E-RF
Test Mode : 868.5-RX

	Freq	Factor	Read Level	Level	Limit Line	Over Limit	Remark
	MHz	dB	dBm	dBm	dBm	dB	
1	1754.938	5.96	-63.39	-57.43	-47.00	-10.43	Peak

260kHz_868.3MHz

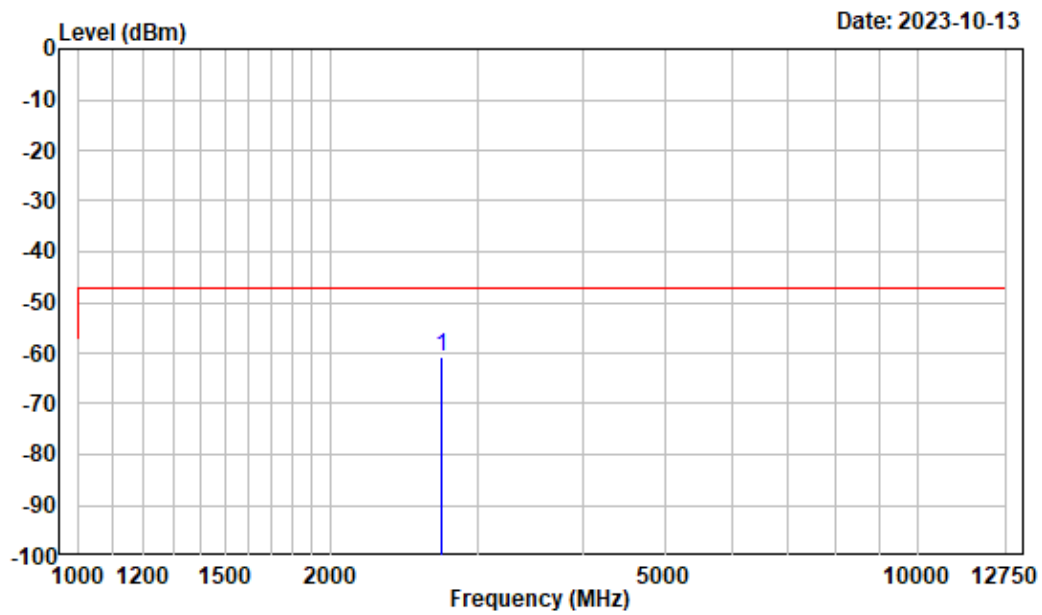
Horizontal



Site : chamber
Condition : Horizontal
Project Number: SZ1230802-44707E-RF
Test Mode : 868.3-RX

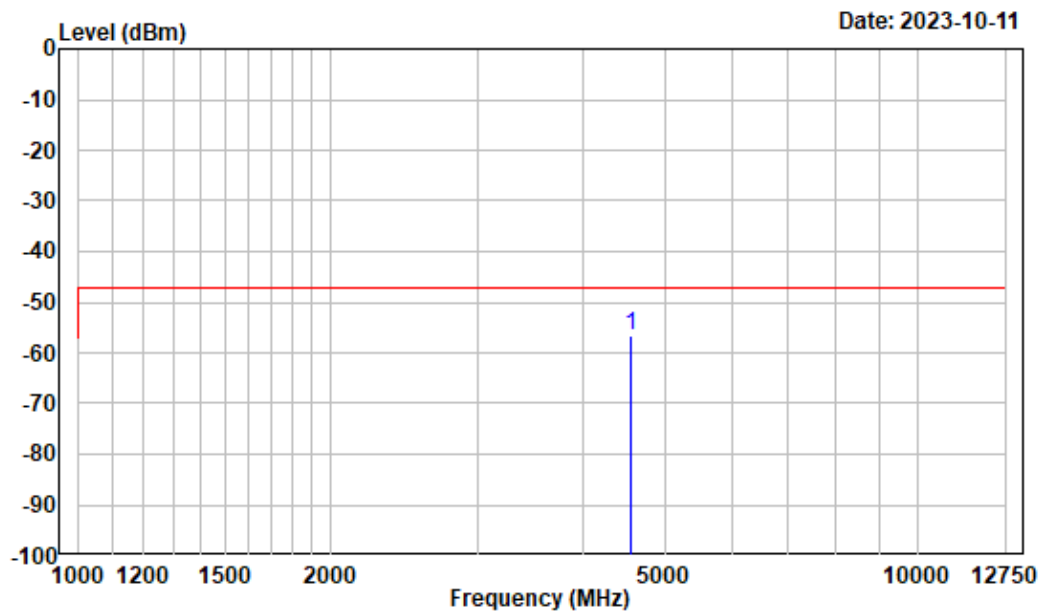
	Freq	Factor	Read Level	Level	Limit Line	Over Limit	Remark
	MHz	dB	dBm	dBm	dBm	dB	
1	3026.875	8.37	-67.58	-59.21	-47.00	-12.21	Peak

Vertical



Site : chamber
Condition : Vertical
Project Number: SZ1230802-44707E-RF
Test Mode : 868.3-RX

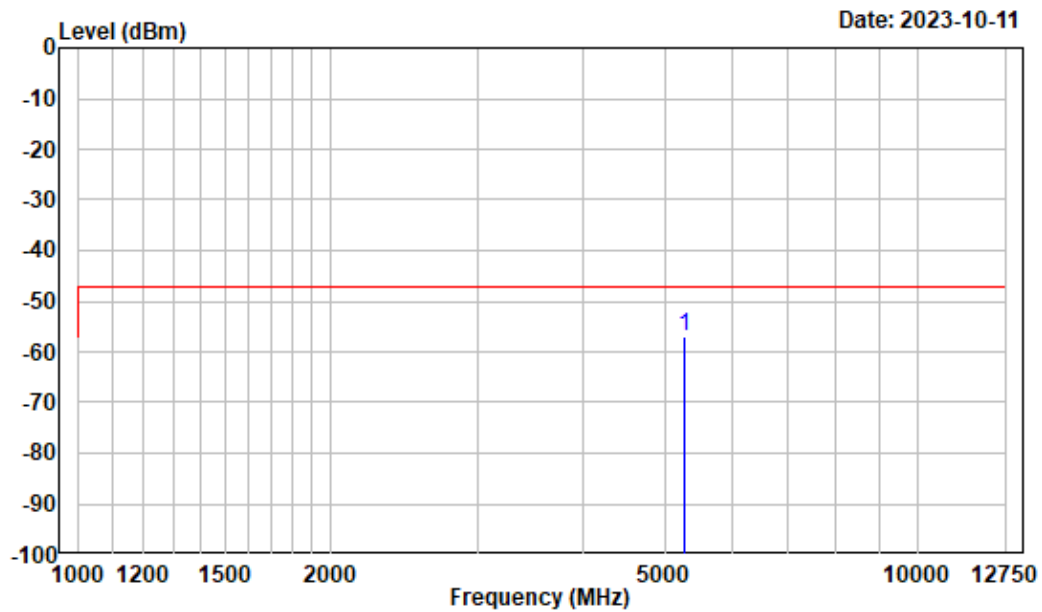
	Freq	Factor	Read Level	Level	Limit Line	Over Limit	Remark
	MHz	dB	dBm	dBm	dBm	dB	
1	2712.563	7.97	-68.64	-60.67	-47.00	-13.67	Peak

Test mode: Standby**Horizontal**

Site : chamber
Condition : Horizontal
Project Num: SZ1230802-44707E-RF
Test Mode : standby

	Freq	Factor	Read Level	Level	Limit Line	Over Limit	Remark
	MHz	dB	dBm	dBm	dBm	dB	
1	4566.820	11.81	-68.55	-56.74	-47.00	-9.74	

Vertical



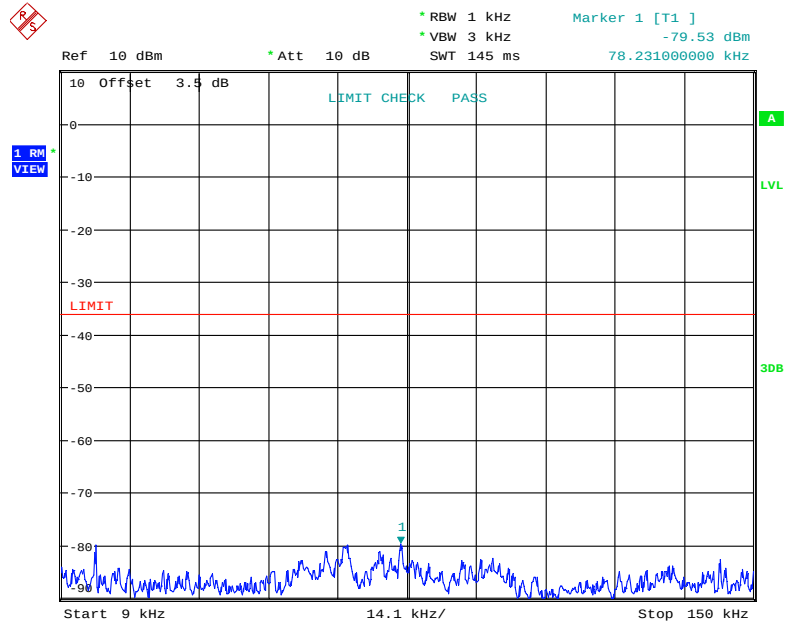
Site : chamber
Condition : Vertical
Project Num: SZ1230802-44707E-RF
Test Mode : standby

	Freq	Factor	Read Level	Level	Limit Line	Over Limit	Remark
	MHz	dB	dBm	dBm	dBm	dB	
1	5268.320	12.74	-69.77	-57.03	-47.00	-10.03	

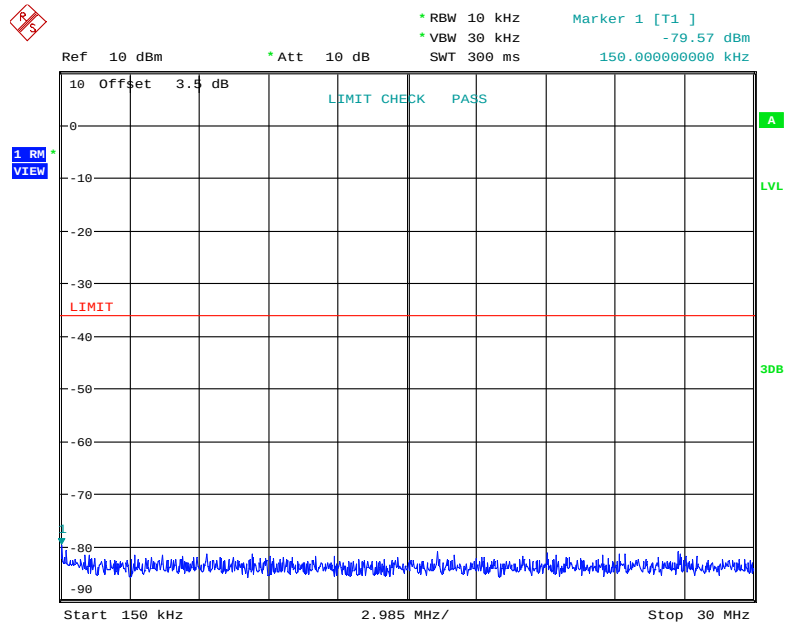
Conducted spurious emissions:

Test mode: Transmitting

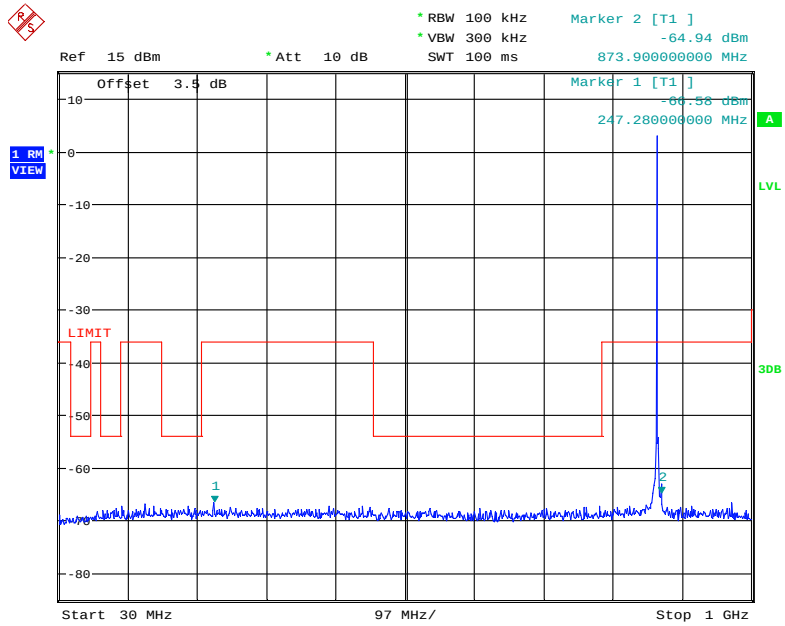
130kHz_867.1MHz



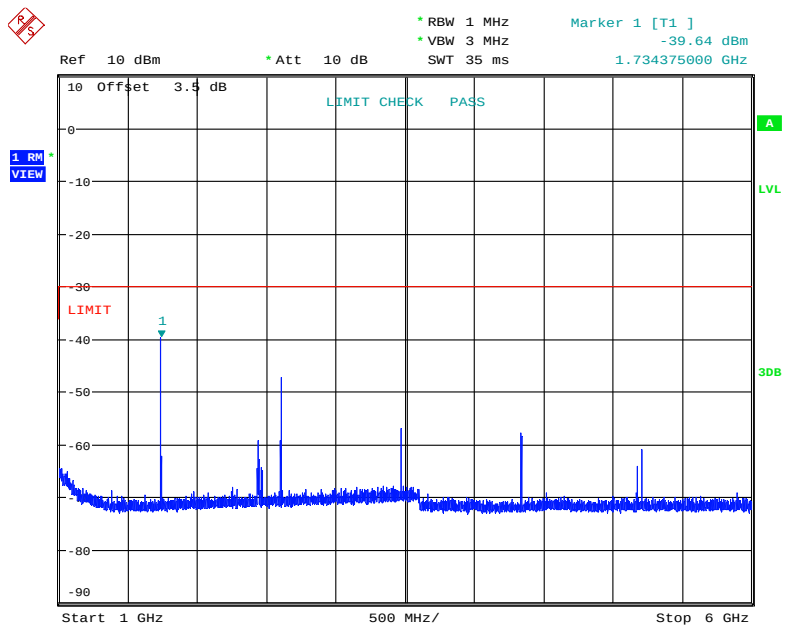
Date: 11.OCT.2023 14:37:54



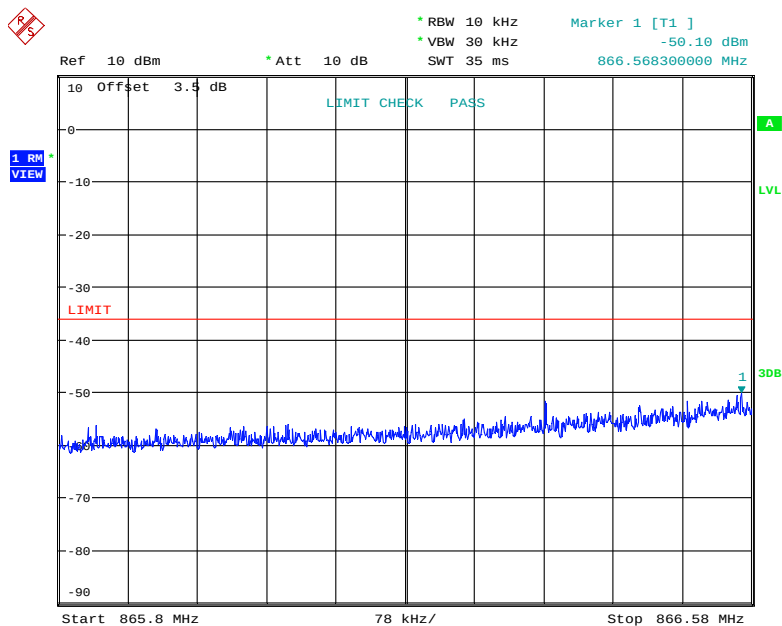
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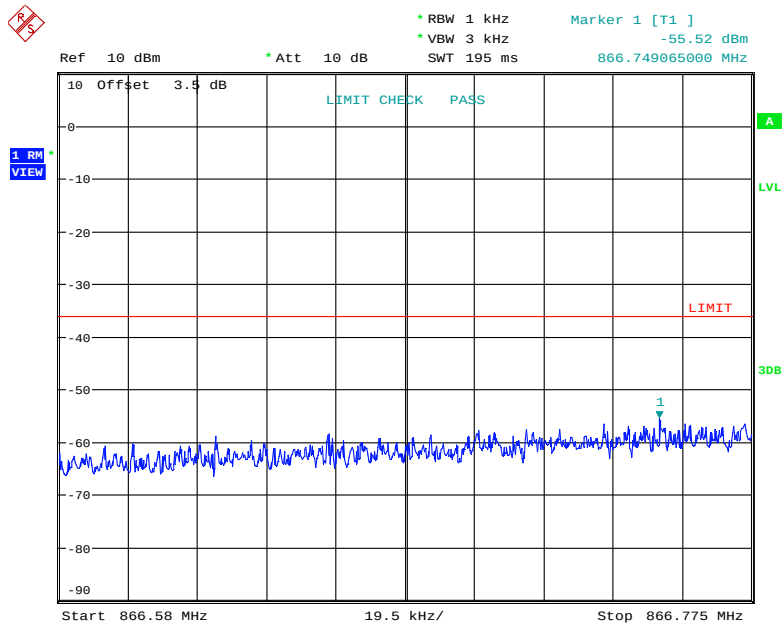
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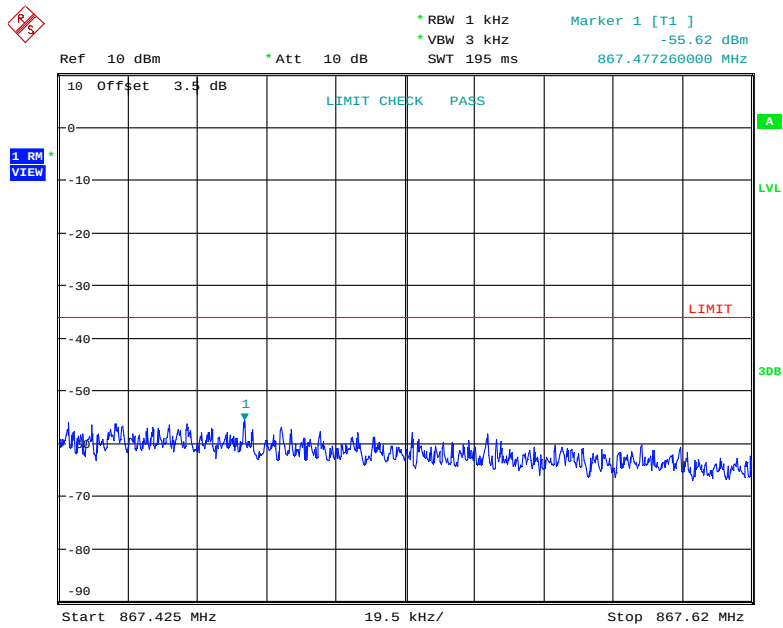
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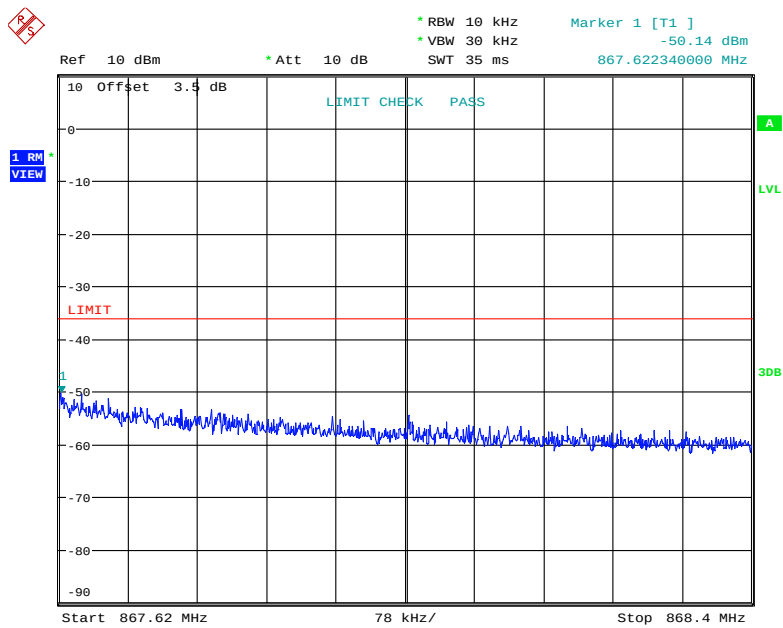
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Date: 11.OCT.2023 14:39:37

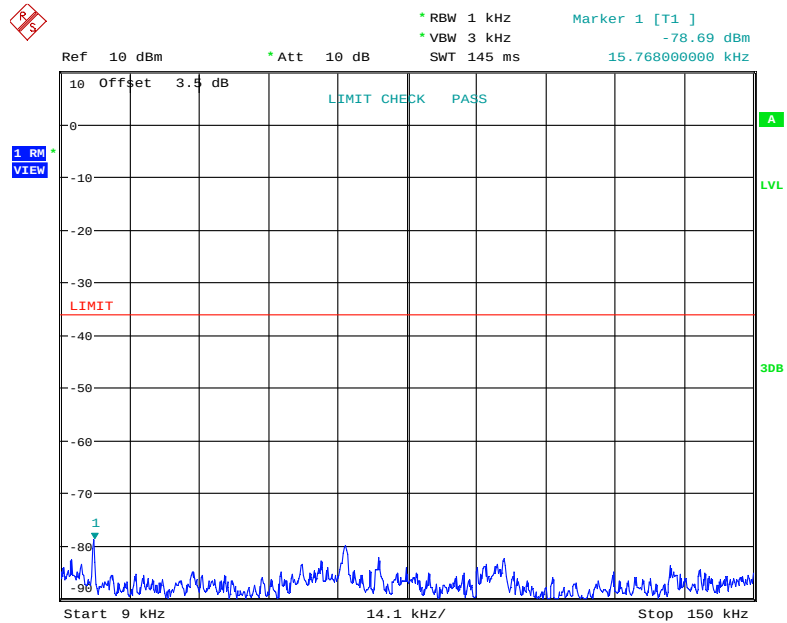


Date: 11.OCT.2023 14:39:56

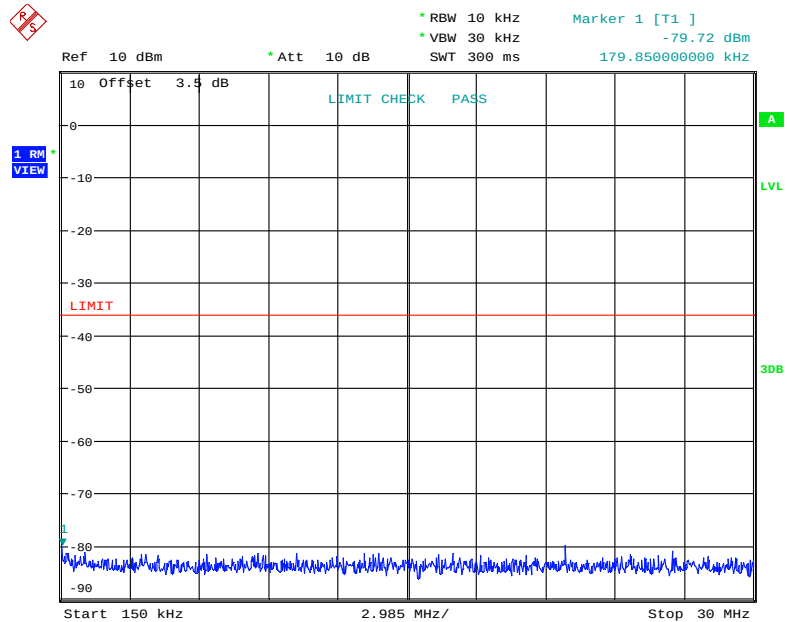


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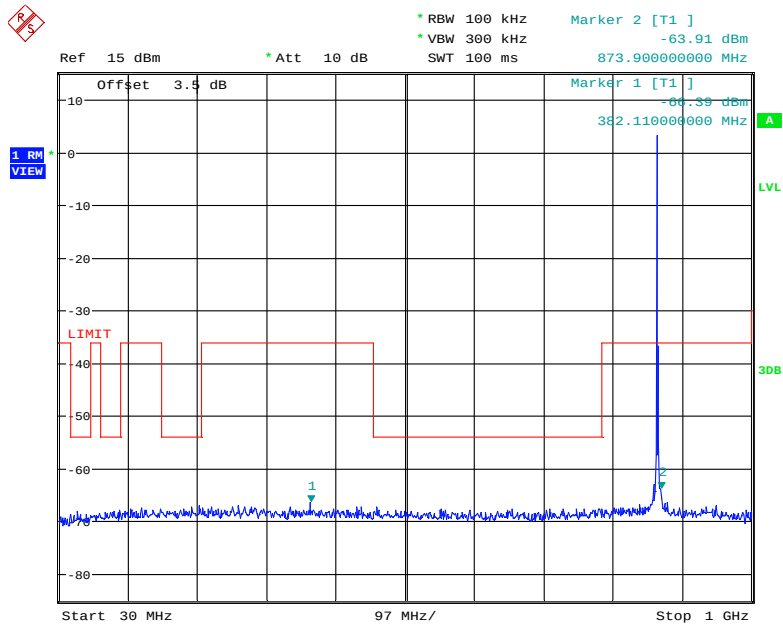
130kHz_867.9MHz



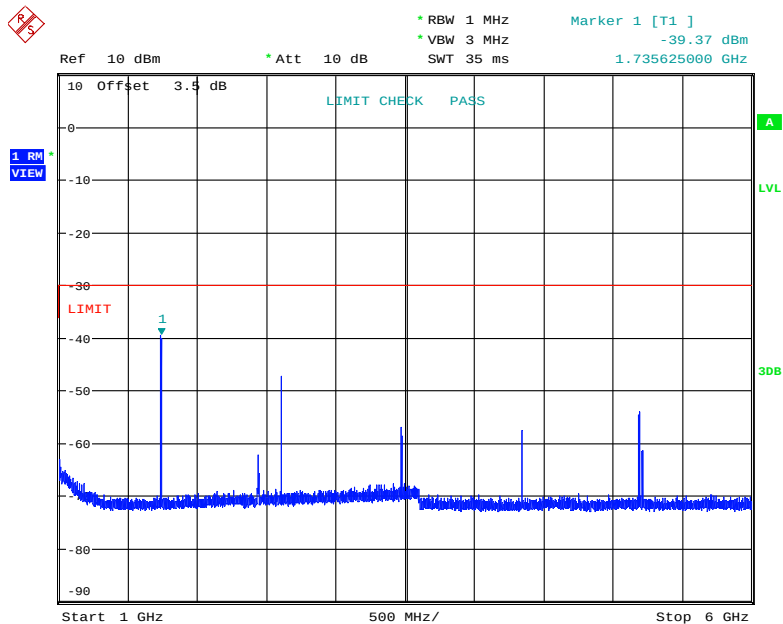
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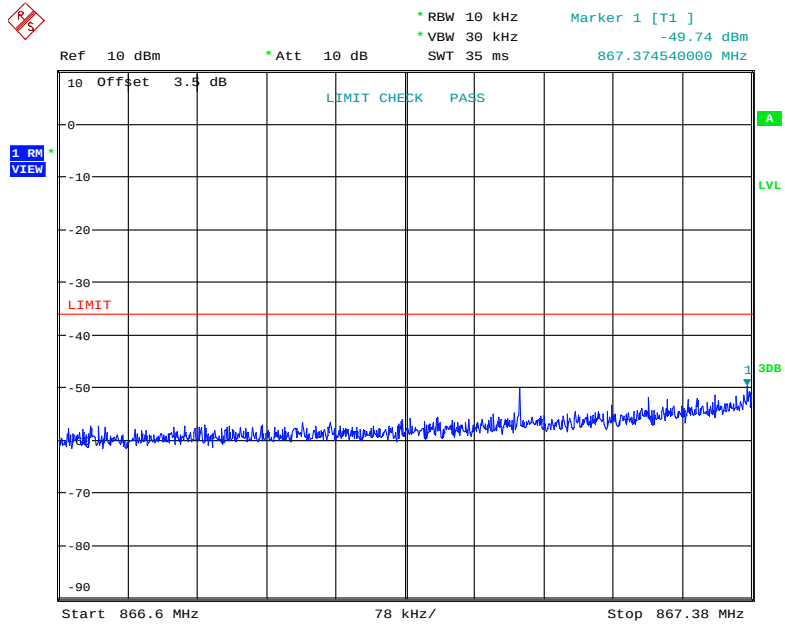
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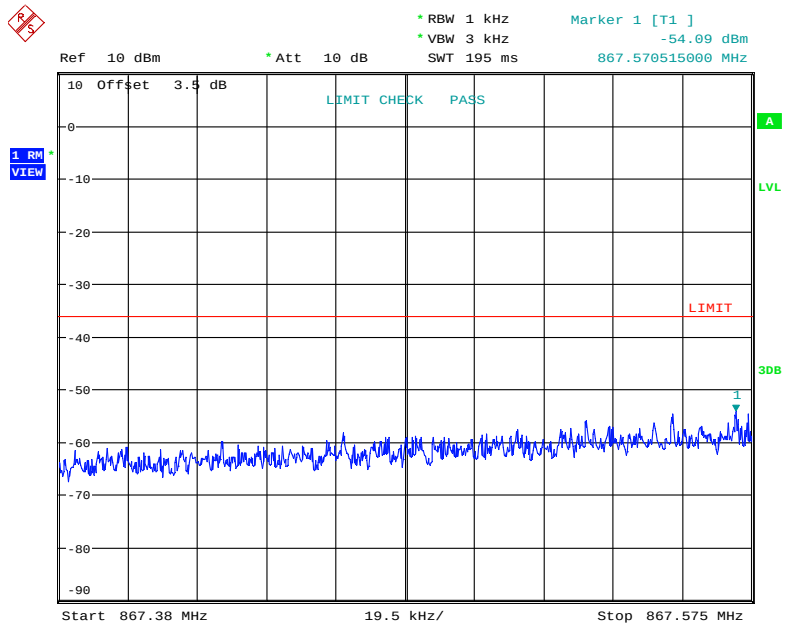
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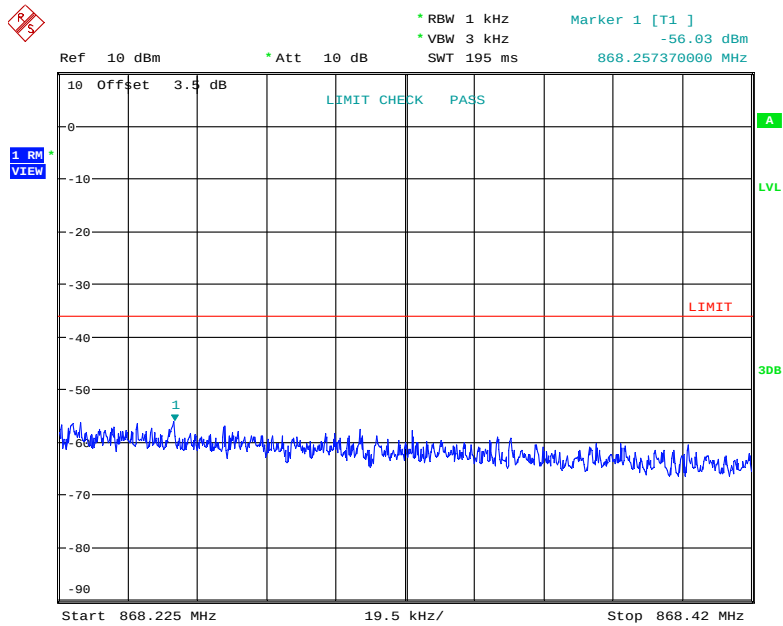
Date: 11.OCT.2023 14:47:02



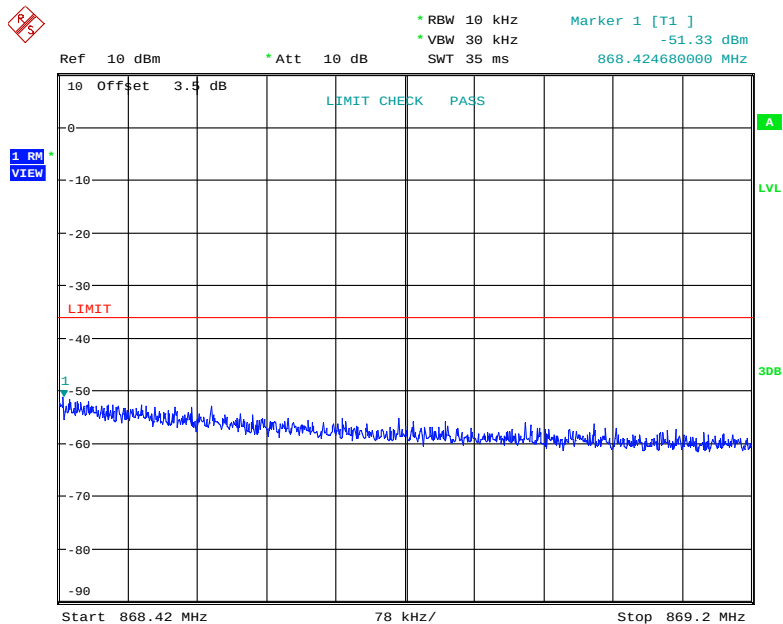
Date: 11.OCT.2023 14:47:21



Date: 11.OCT.2023 14:47:39

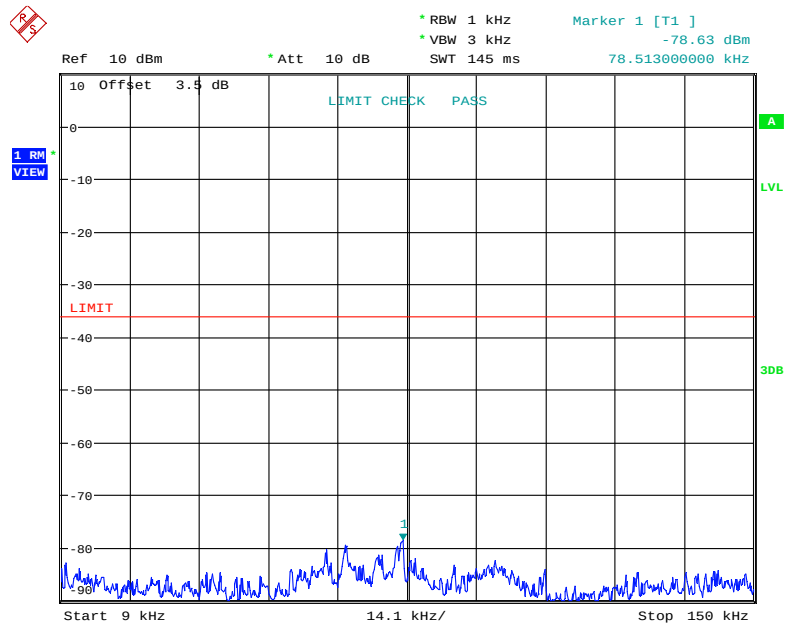


Date: 11.OCT.2023 14:47:58

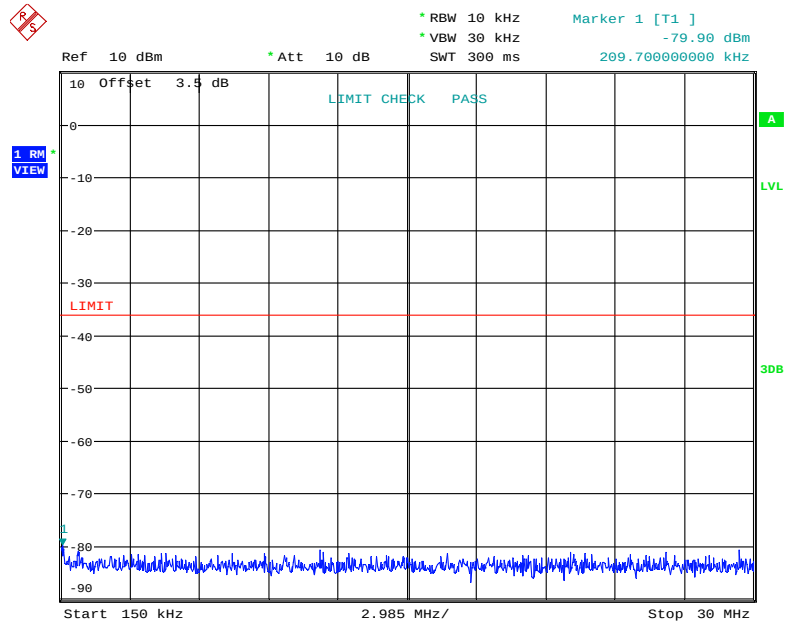


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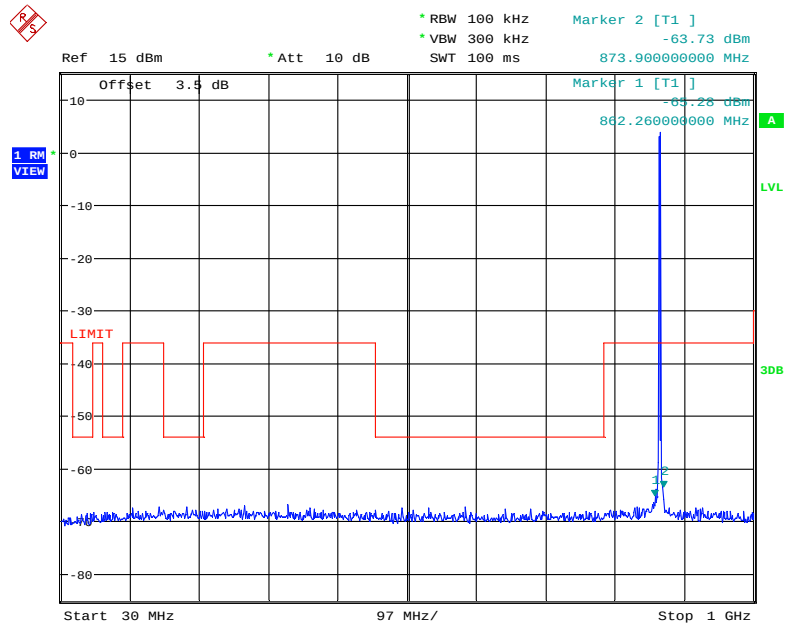
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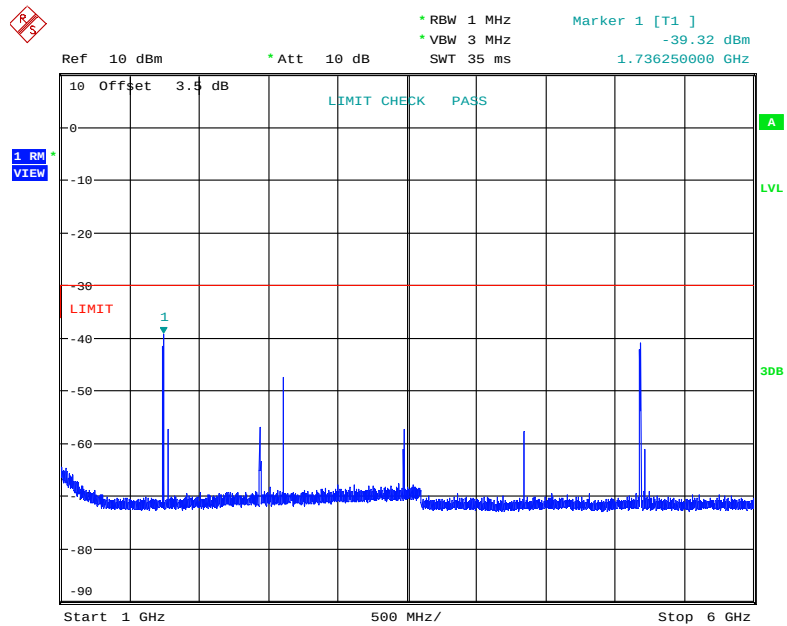
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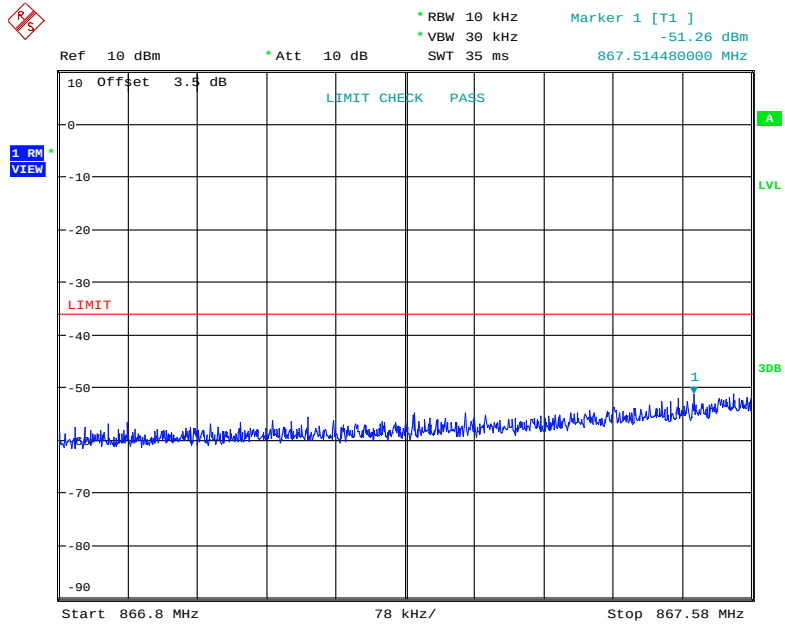
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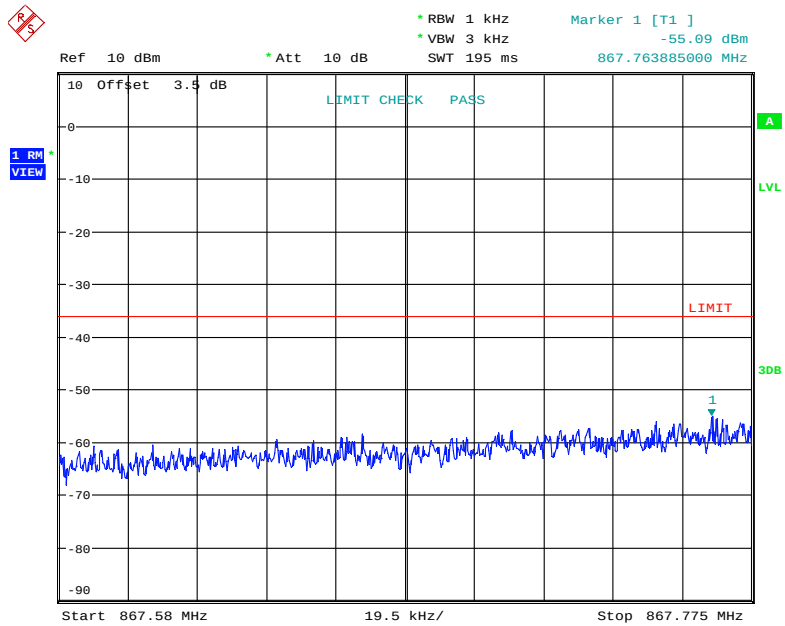
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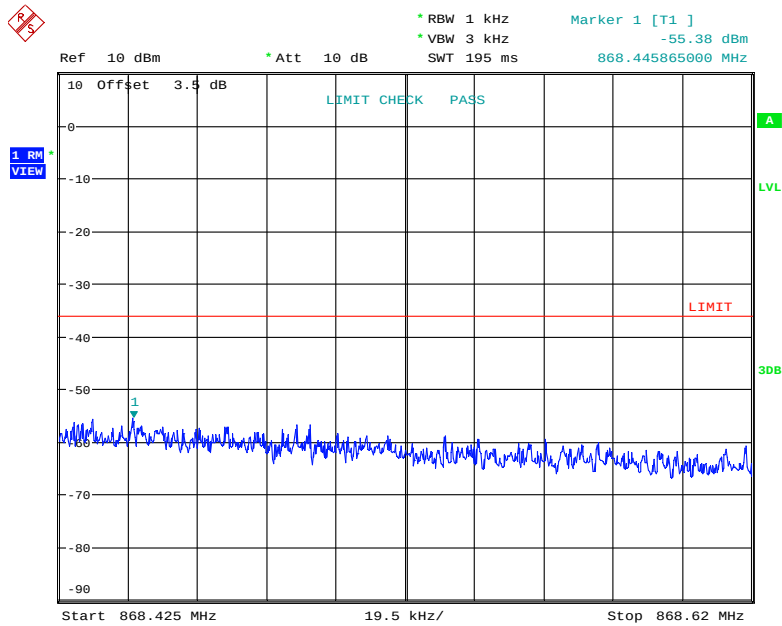
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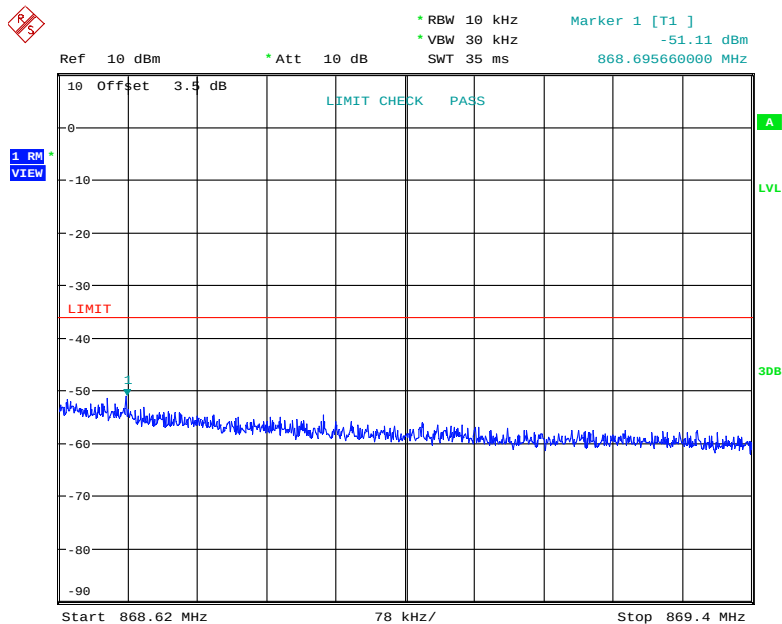
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Date: 11.OCT.2023 15:25:15

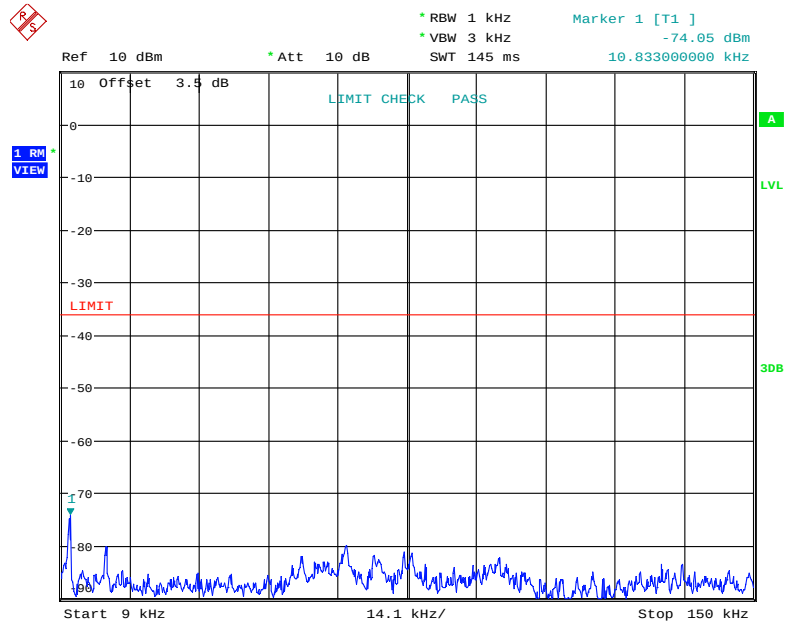


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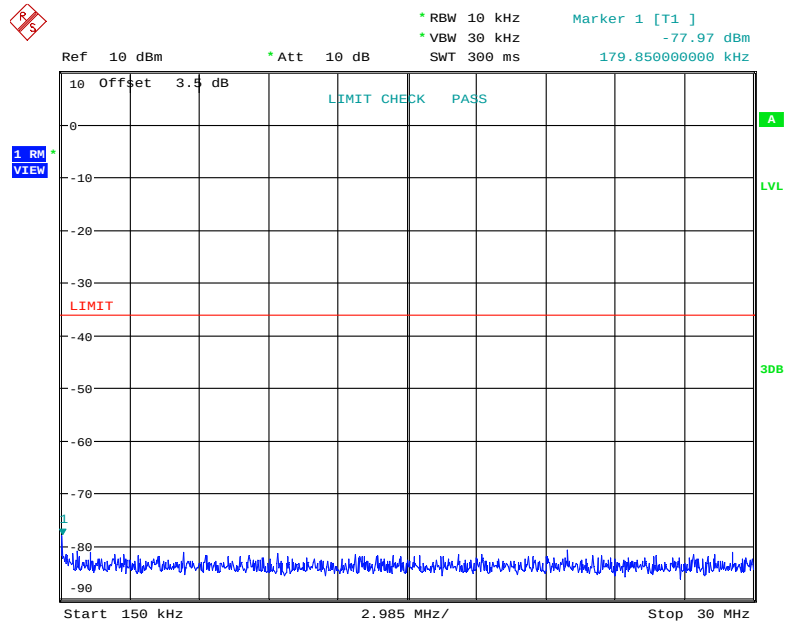


Date: 11.OCT.2023 15:25:52

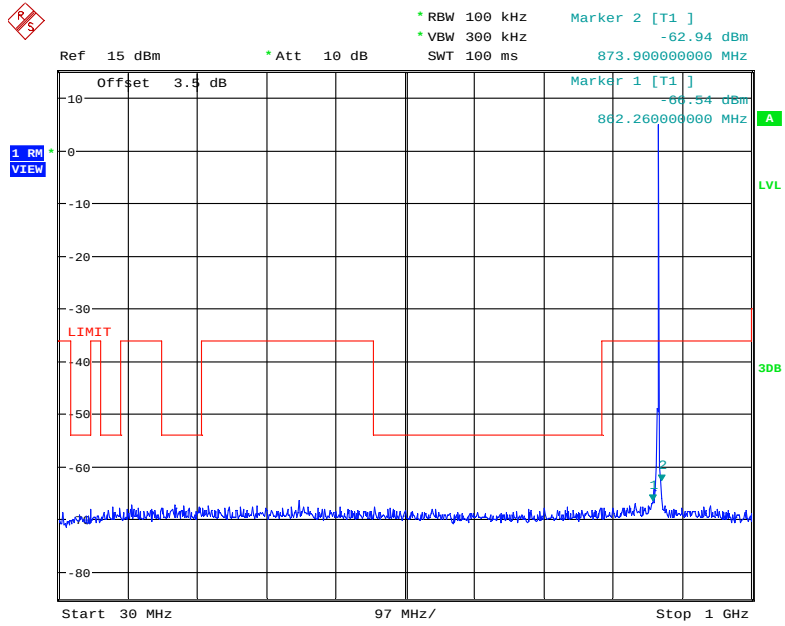
130kHz_868.5MHz



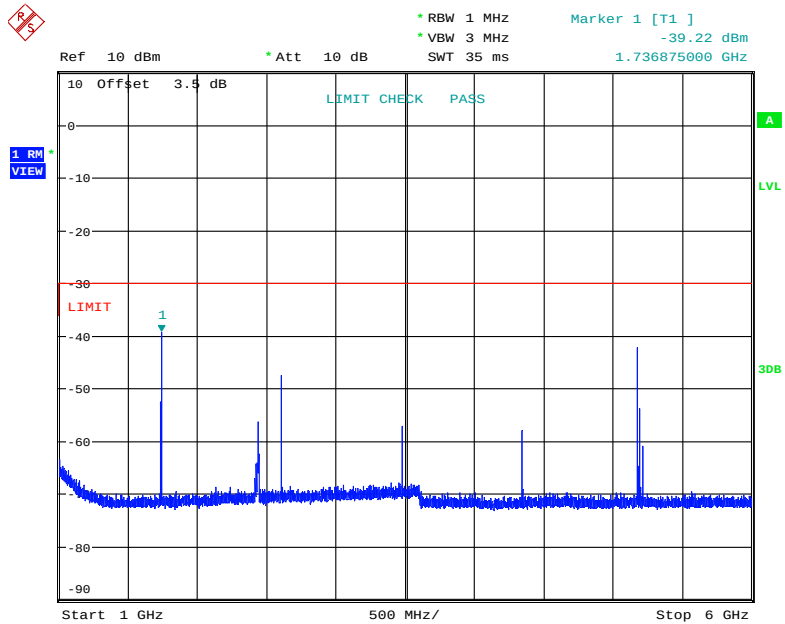
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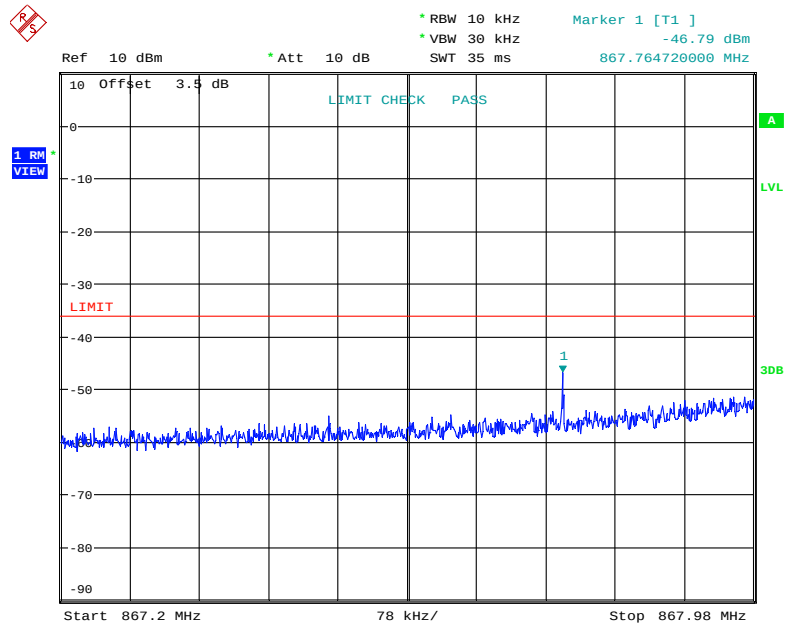
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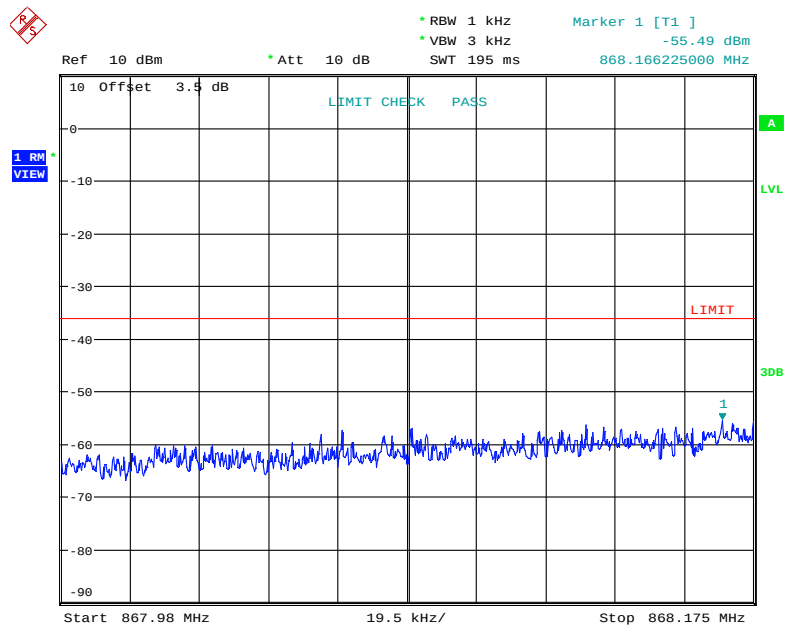
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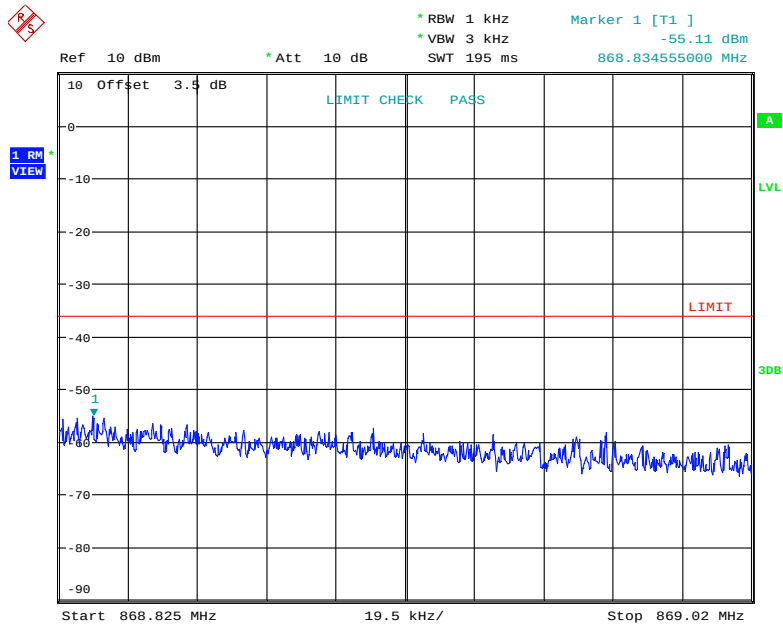
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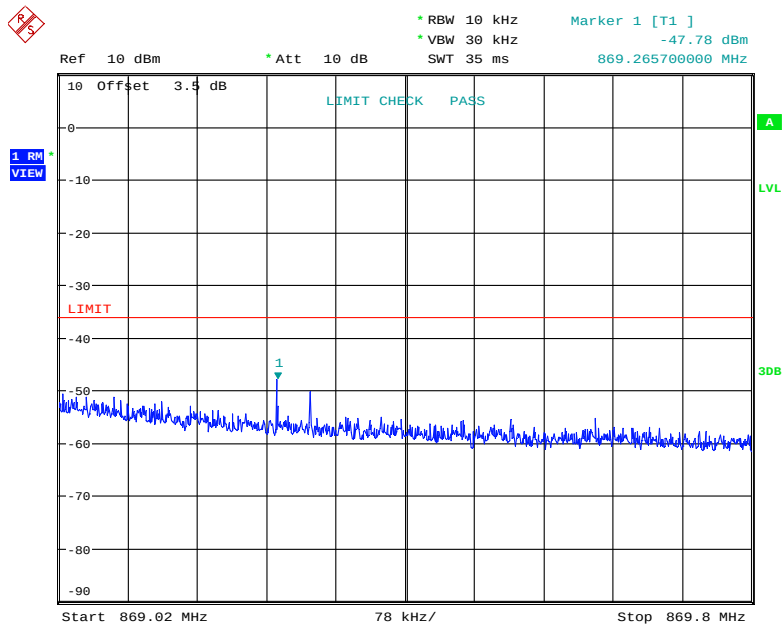
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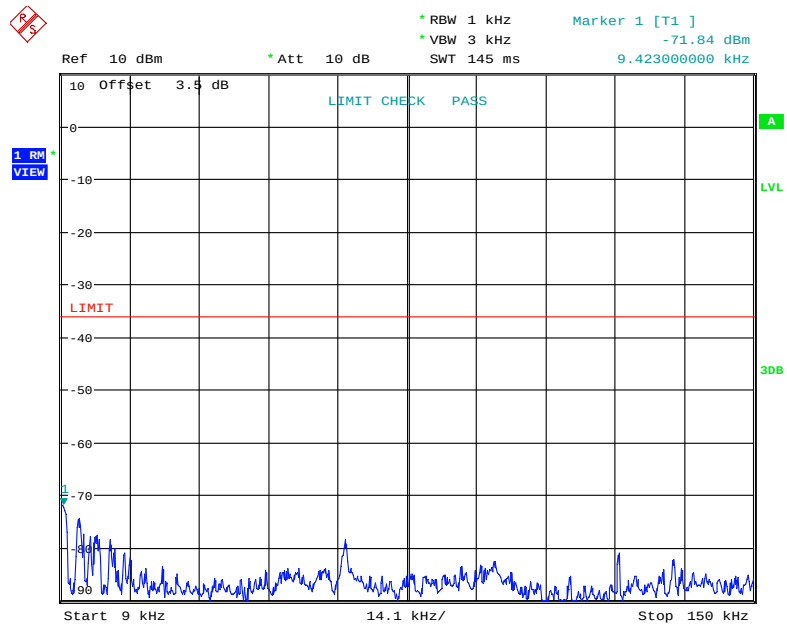


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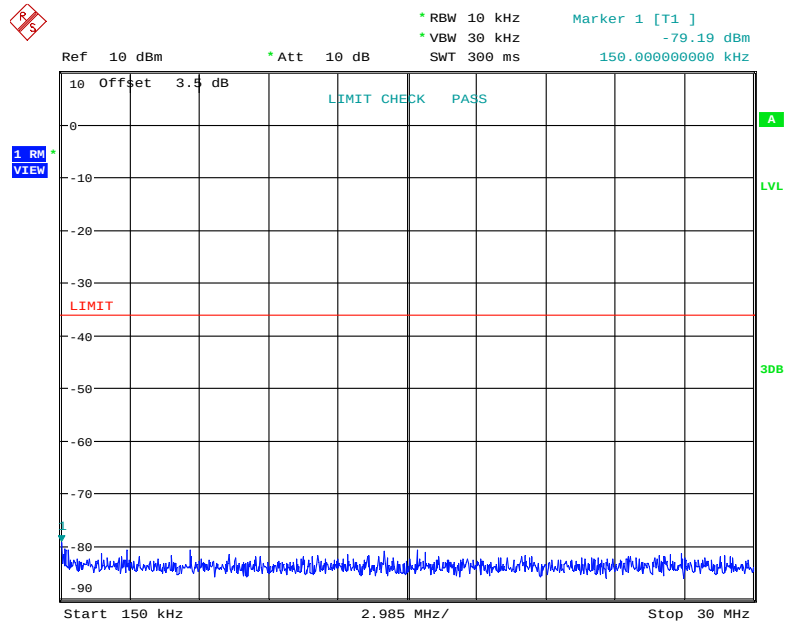


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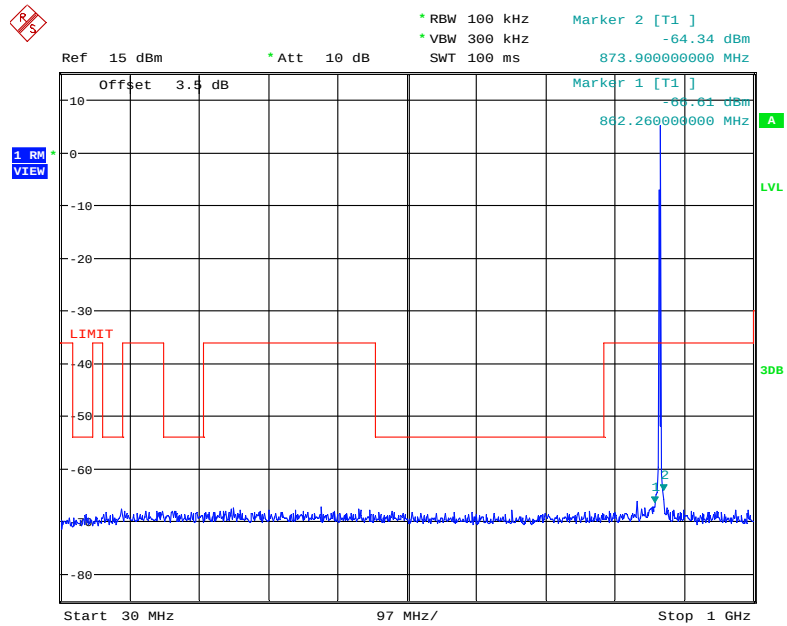
260kHz_868.3MHz



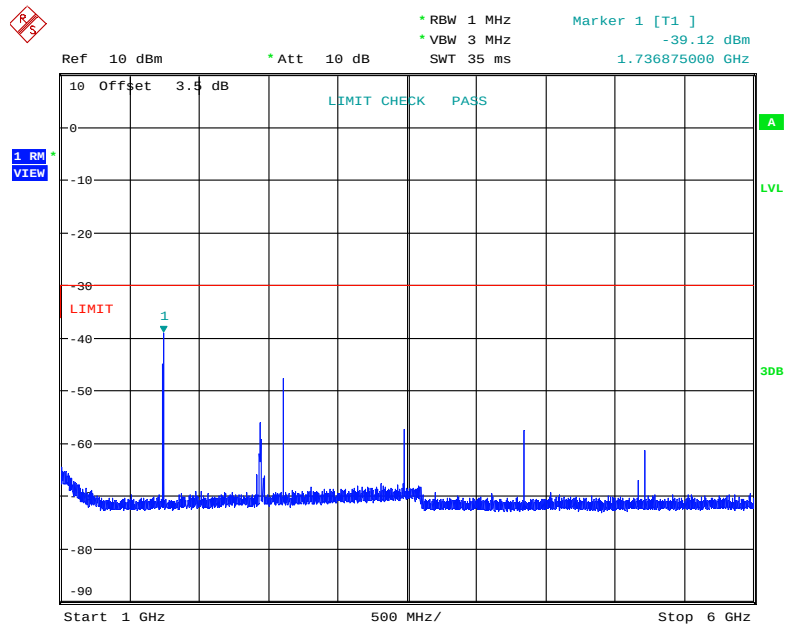
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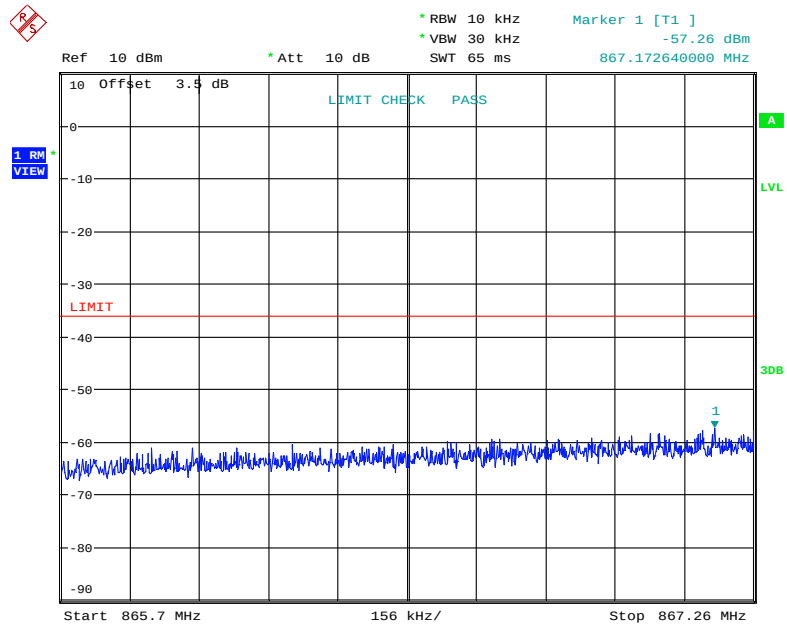
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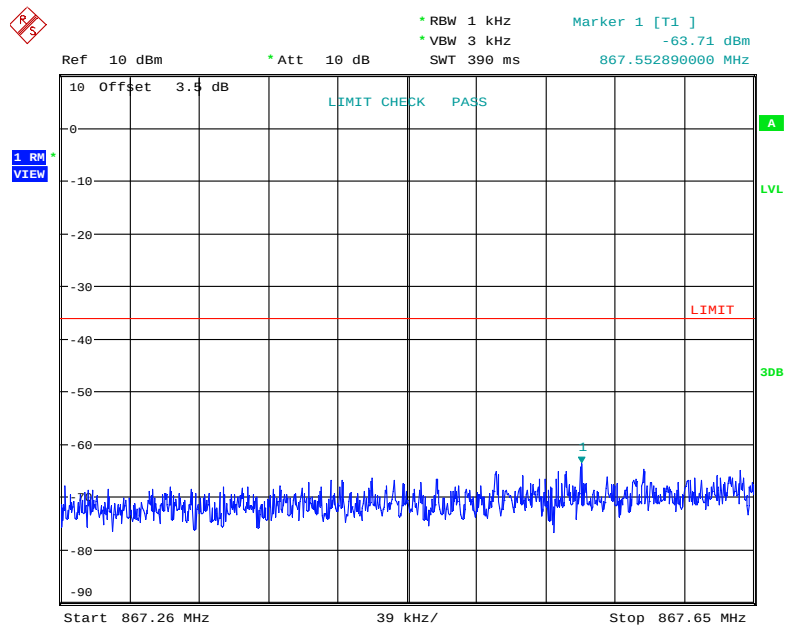
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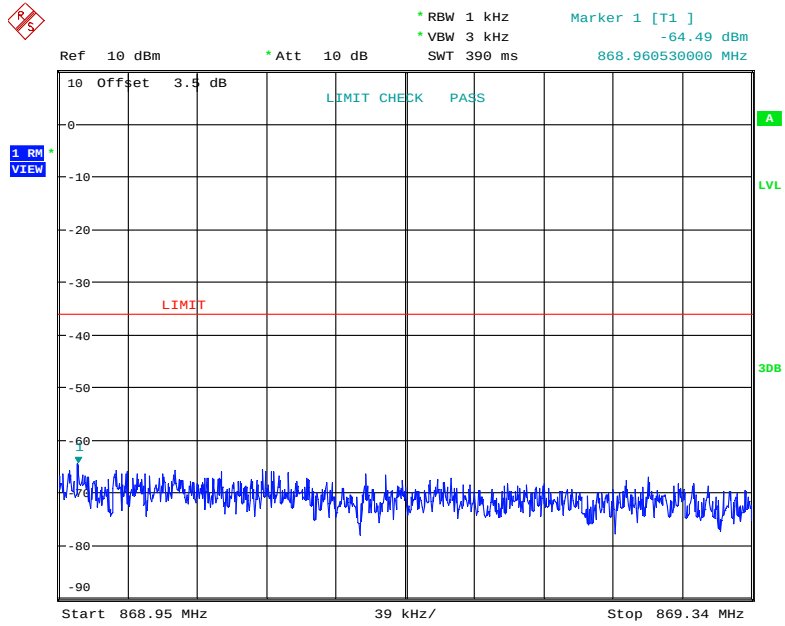
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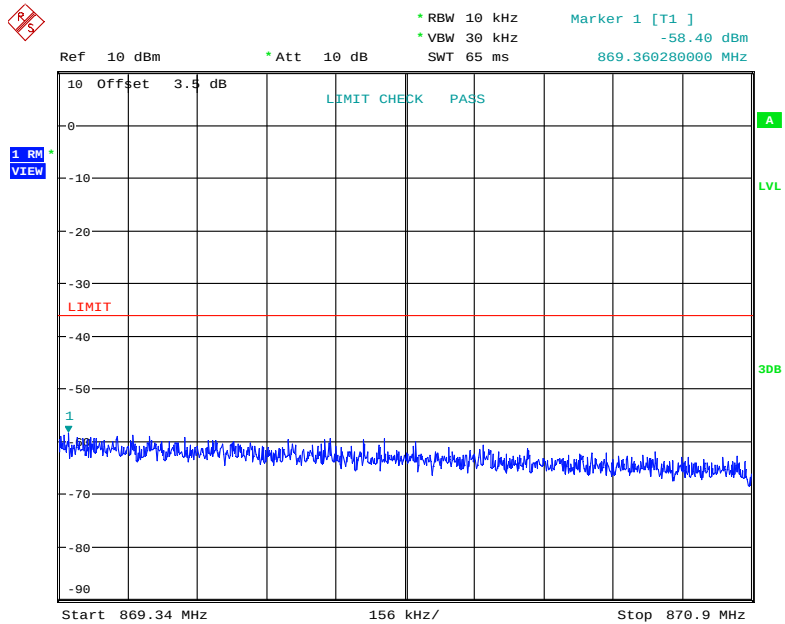
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Date: 11.OCT.2023 15:30:58



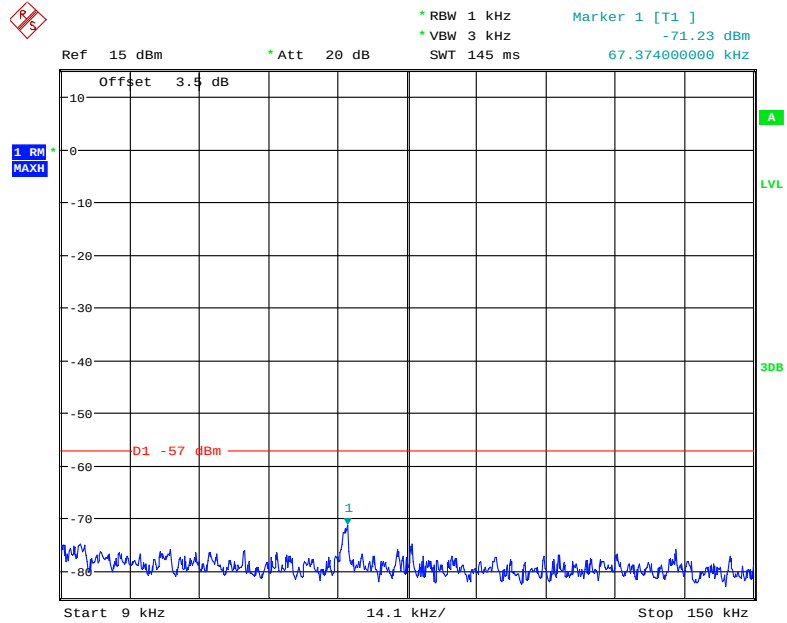
Date: 11.OCT.2023 15:31:16



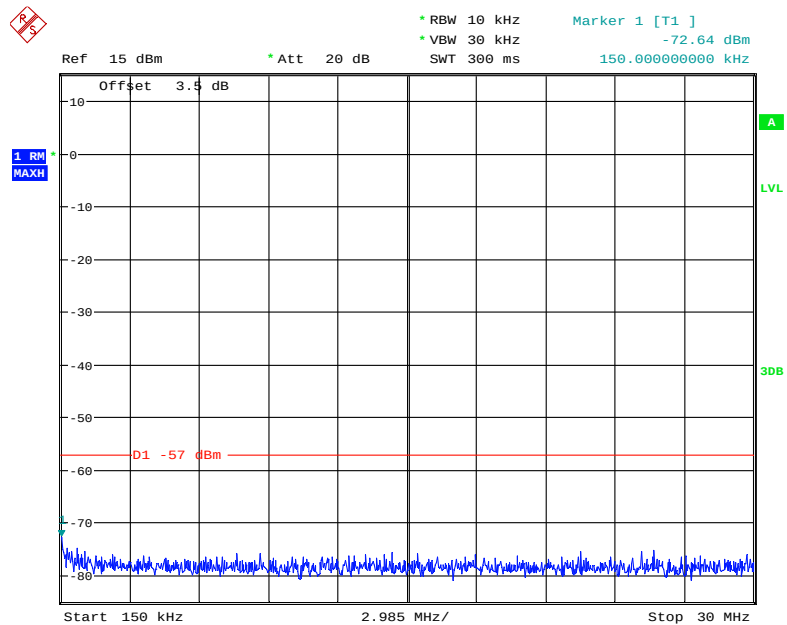
Date: 11.OCT.2023 15:31:35

Test mode: Receiving

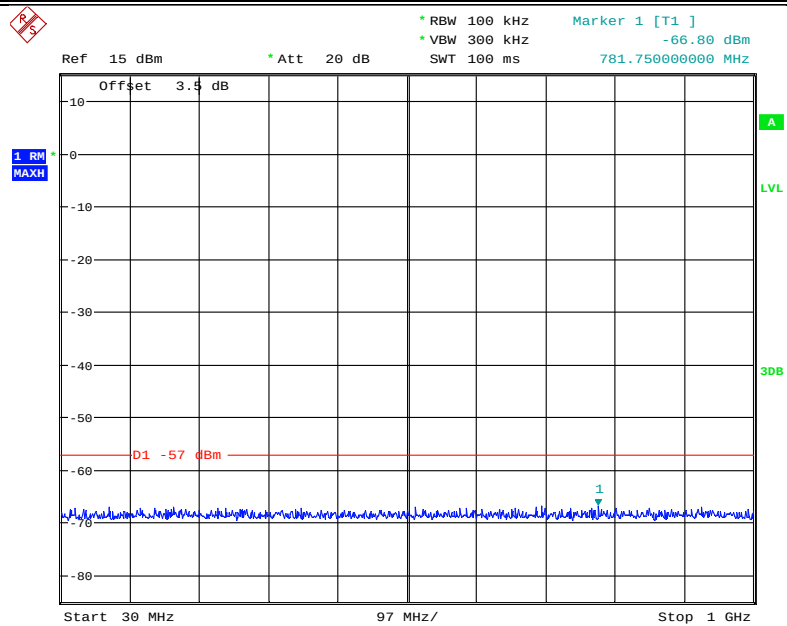
130kHz_867.1MHz



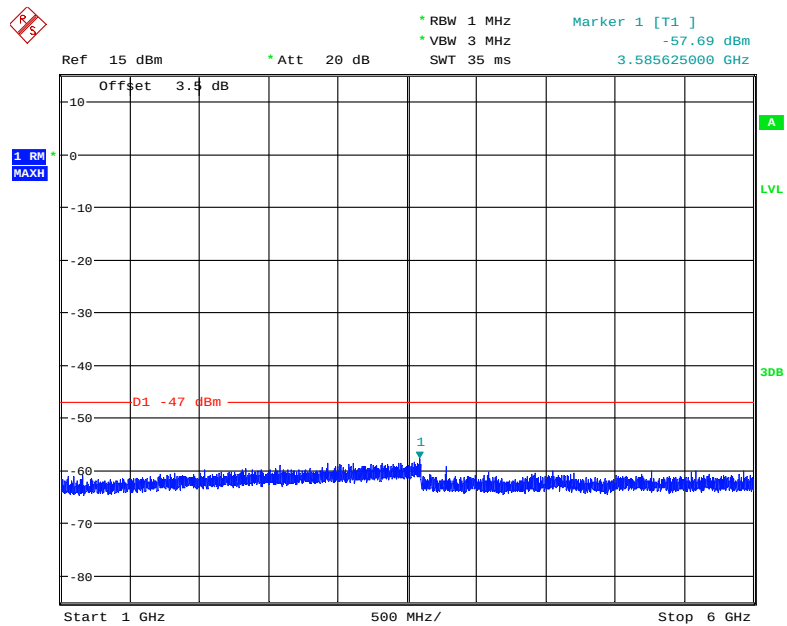
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Date: 7.SEP.2023 15:11:50

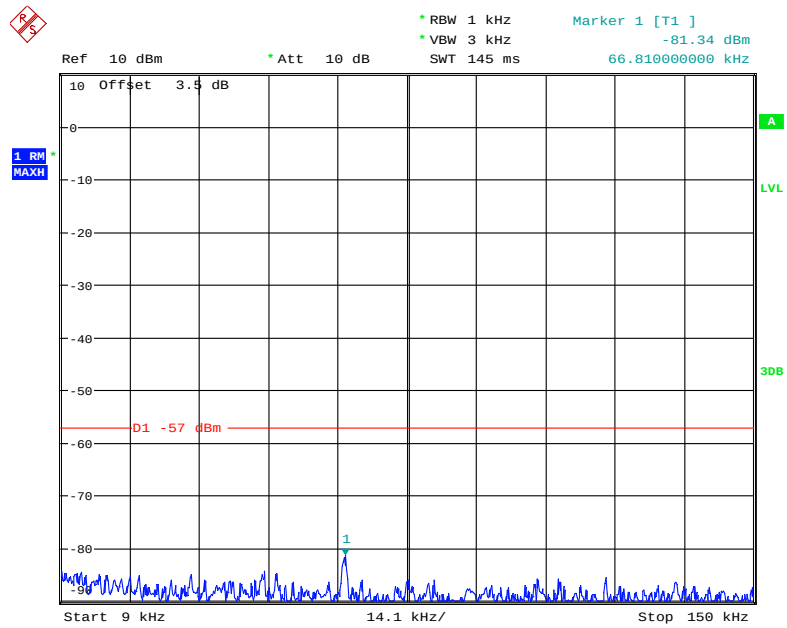


Date: 7.SEP.2023 15:12:08

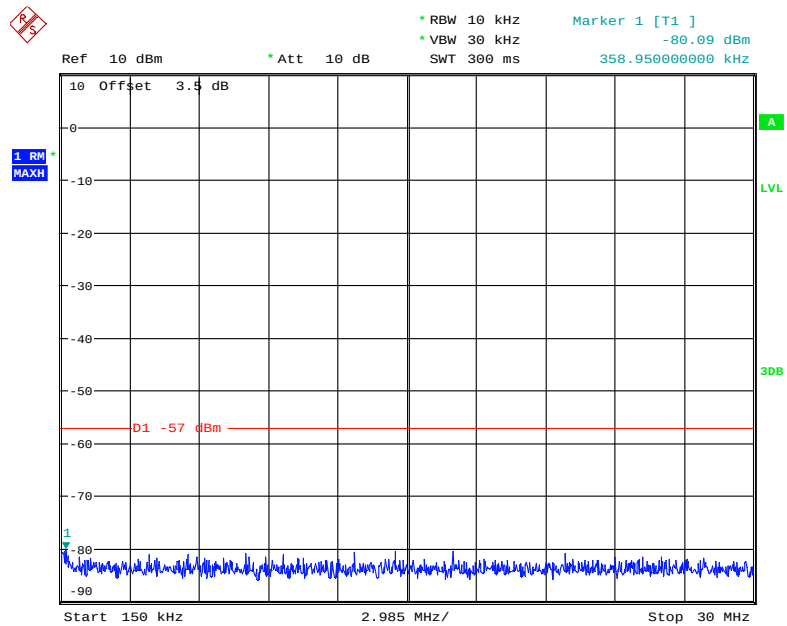


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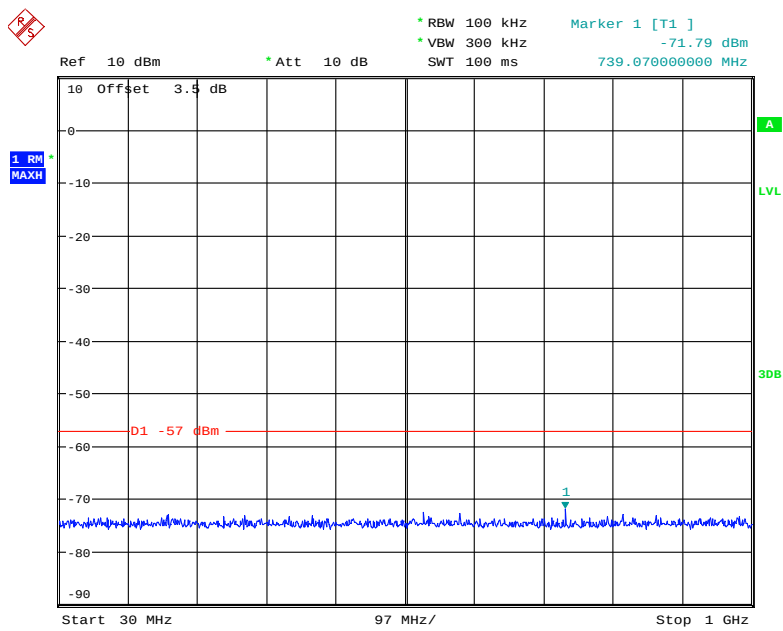
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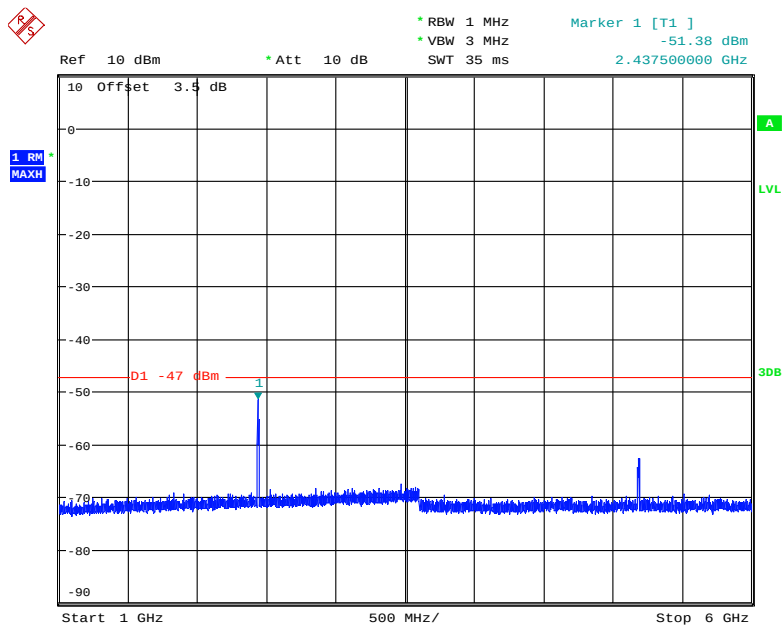
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Date: 13.SEP.2023 10:51:03

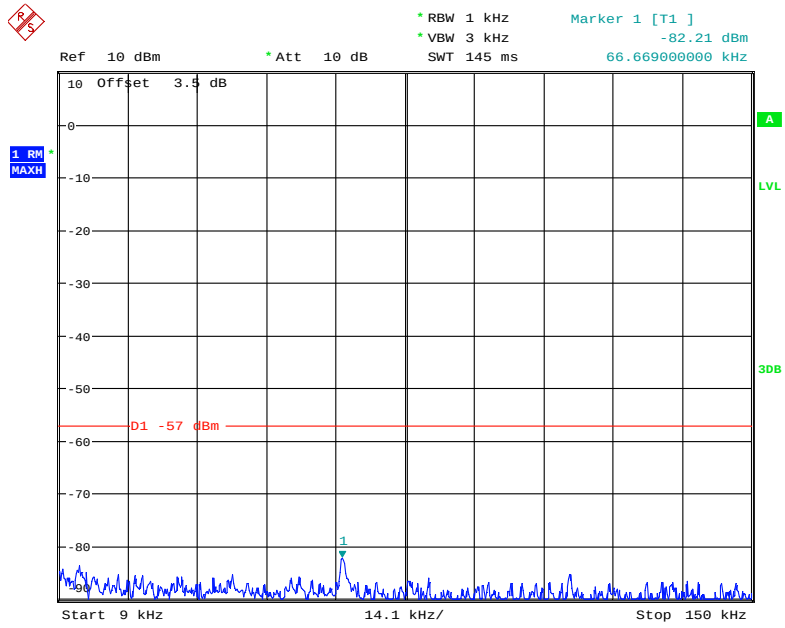


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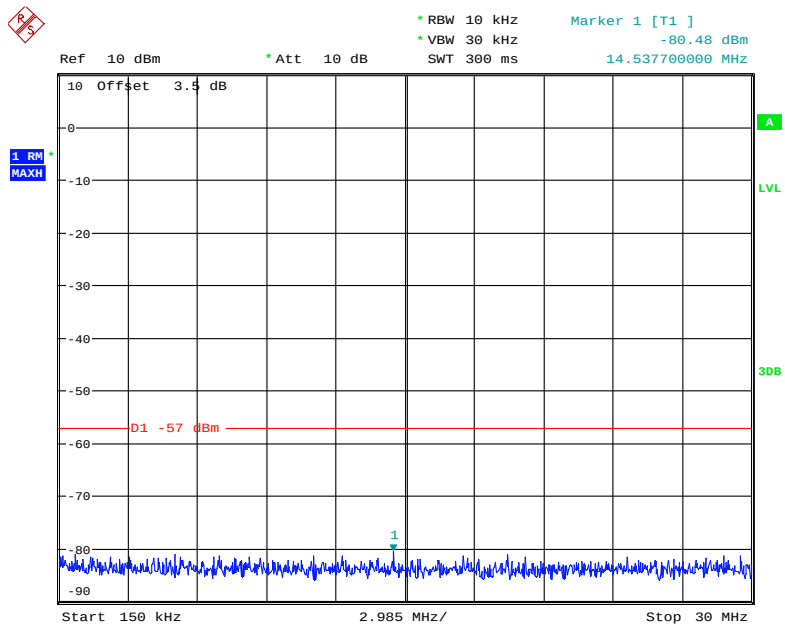


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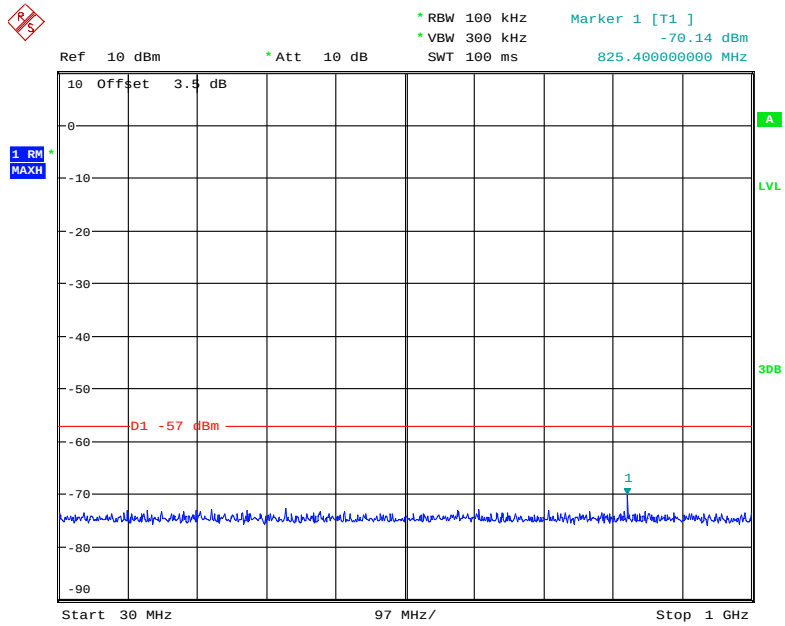
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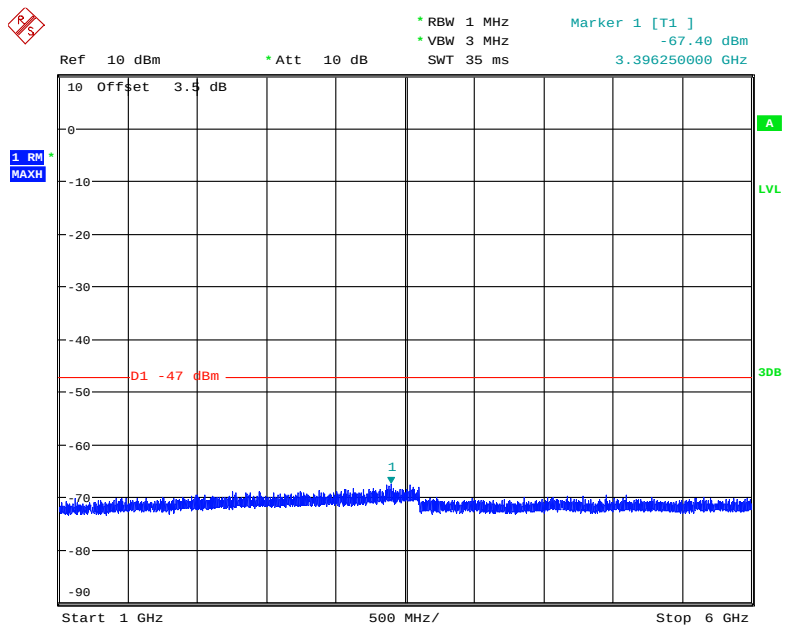
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Date: 13.SEP.2023 11:14:29

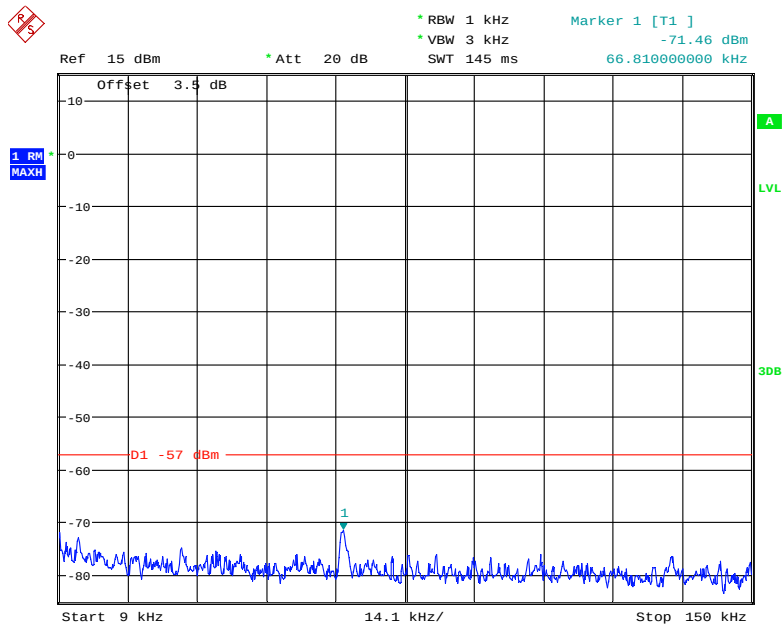


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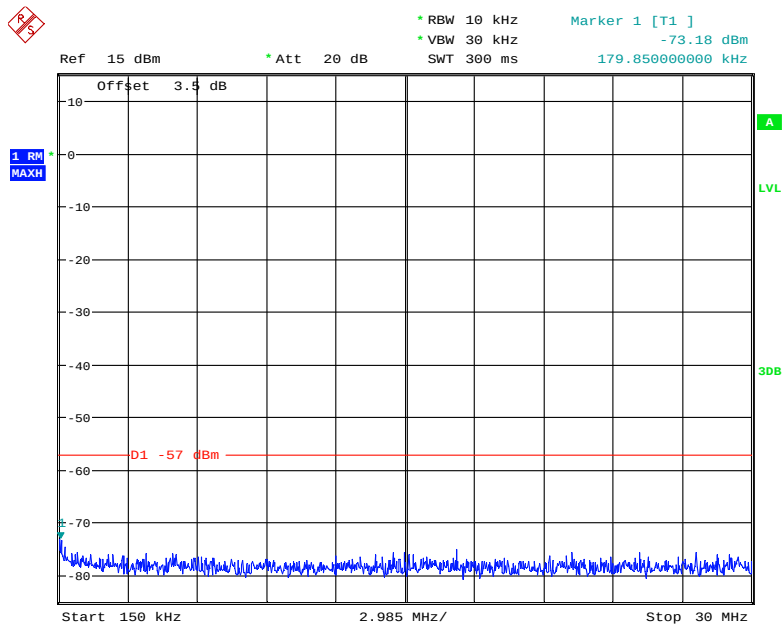


Date: 13.SEP.2023 11:15:04

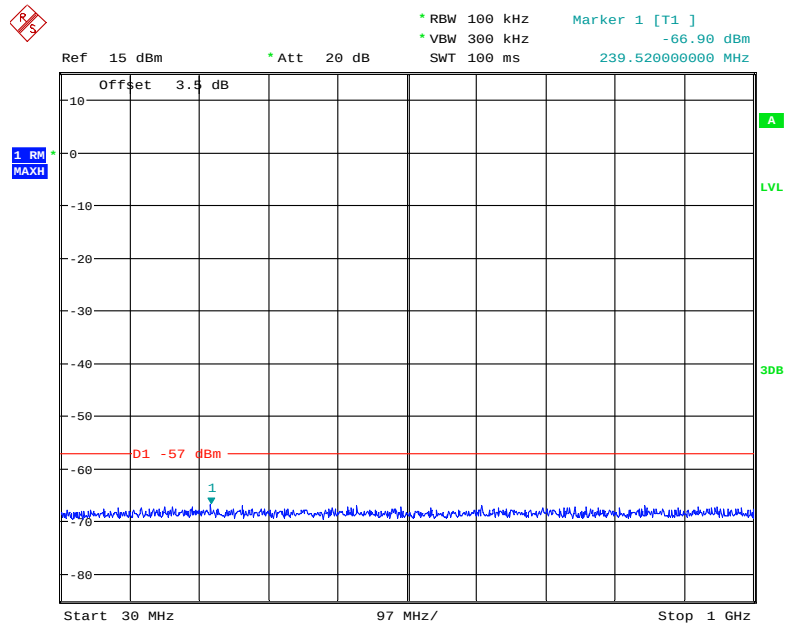
130kHz_868.5MHz



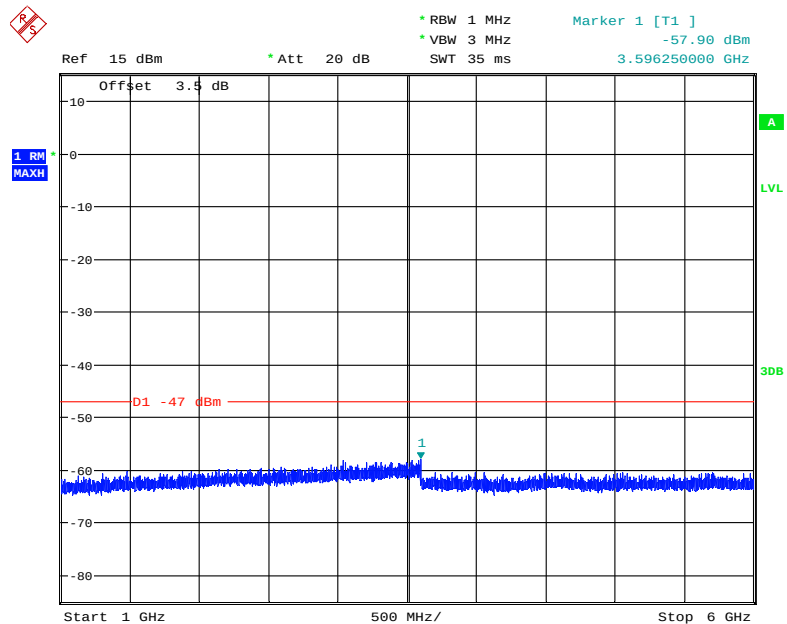
Date: 7.SEP.2023 15:15:20



Date: 7.SEP.2023 15:15:38

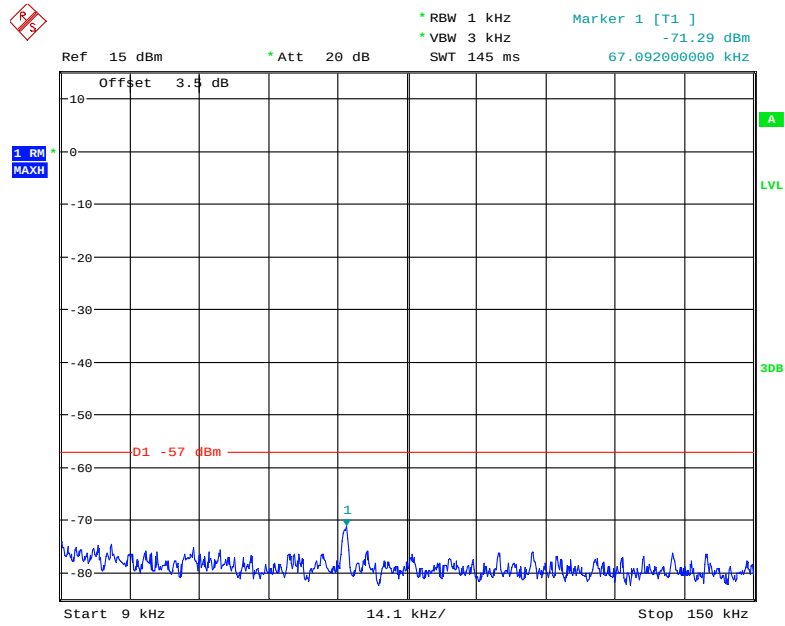


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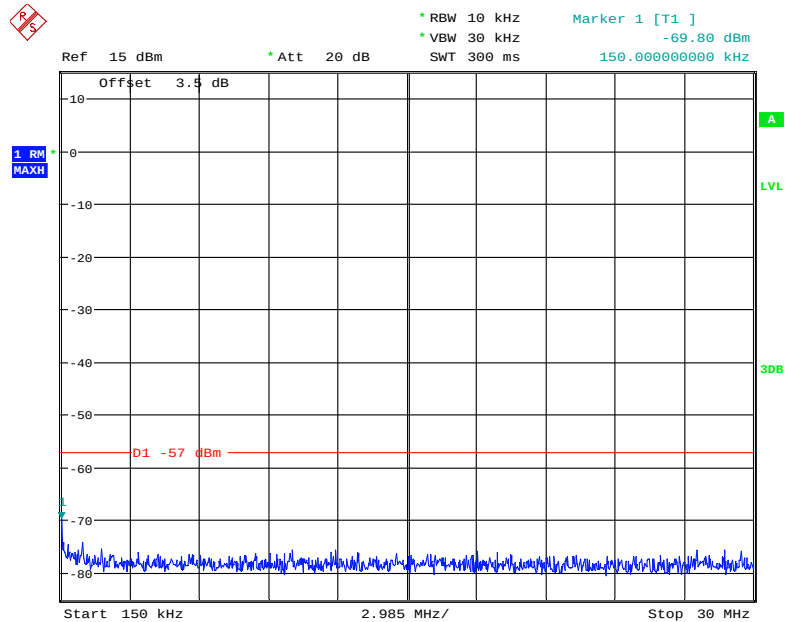


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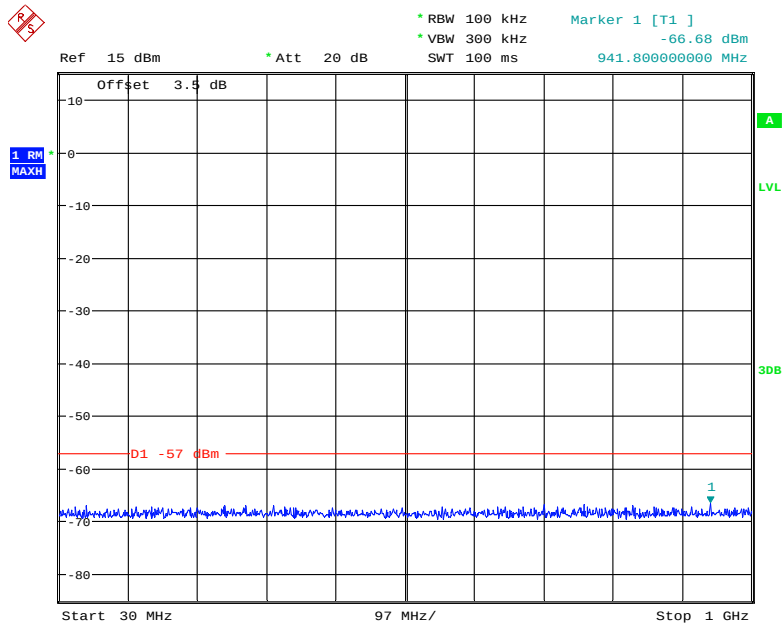
260kHz_868.3MHz



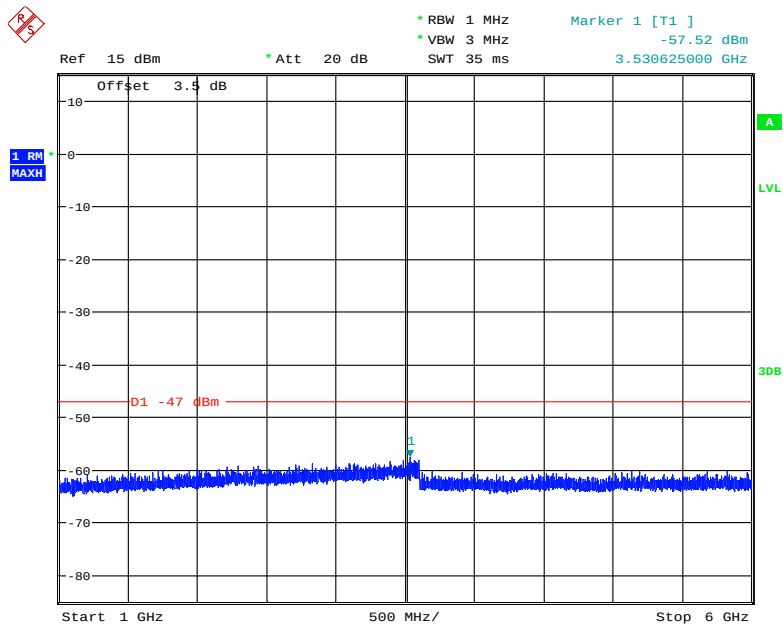
Date: 7.SEP.2023 15:17:11



Date: 7.SEP.2023 15:17:29

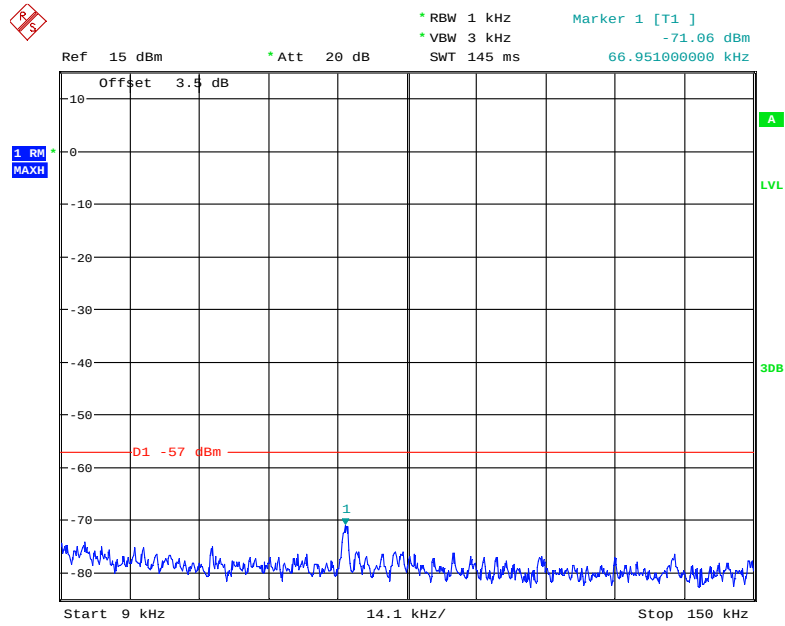


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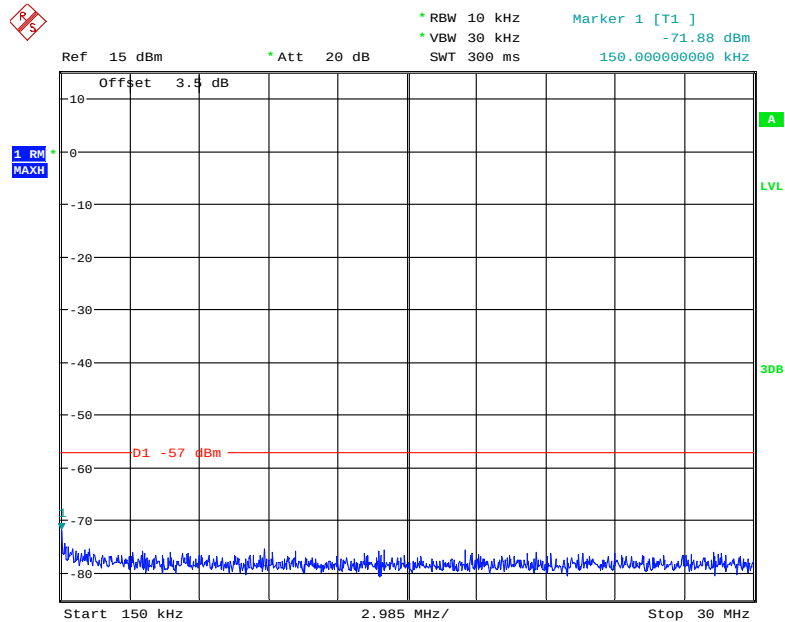


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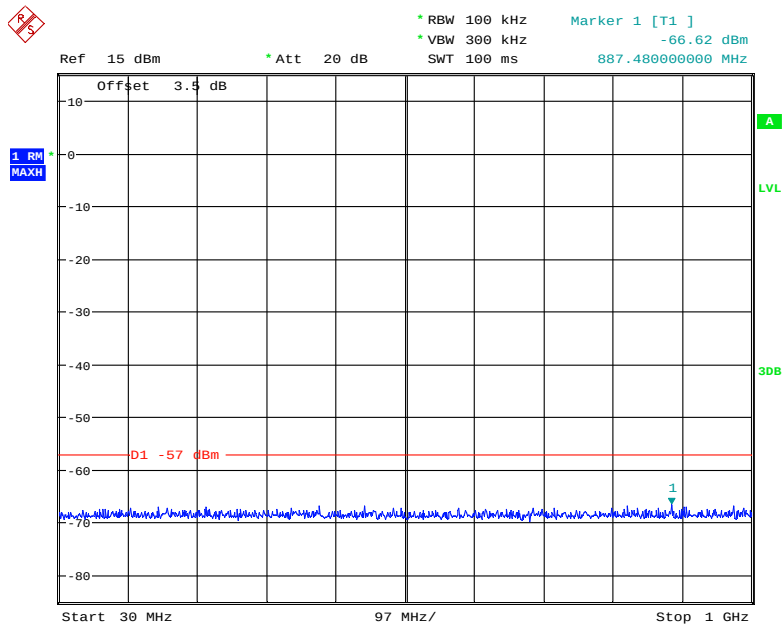
Test mode: Standby



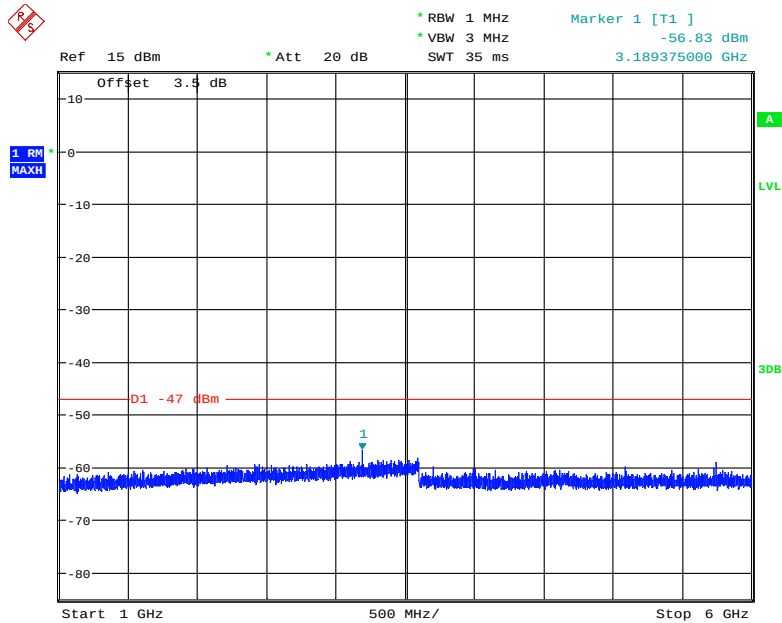
Date: 7.SEP.2023 15:21:33



Date: 7.SEP.2023 15:21:51



Date: 7.SEP.2023 15:22:08



Date: 7.SEP.2023 15:22:26

ETSI EN 300 220-2 V3.2.1 (2018-06) §4.3.1 - EFFECTIVE RADIATED POWER

Applicable Standard

According to ETSI EN 300 220-1 V3.1.1 (2017-02) clause 5.2.1:

The effective radiated power (e.r.p) is the power radiated in the direction of the maximum radiated power under specified conditions of measurements for any condition of modulation. For equipment with a permanent or temporary antenna connection it may be taken as the power delivered from that connector taking into account the antenna gain.

According to ETSI EN 300 220-2 V3.2.1 (2018-06) clause 4.3.1.2:

Limit: The effective radiated power shall not be greater than the value allowed in annexes B or C for the chosen operational frequency band(s).

Method of Measurement

According to ETSI EN 300 220-1 V3.1.1 (2017-02) clause 5.2.2.1:

Effective Radiated Power (conducted measurement):

This method applies only to EUT with a permanent external antenna connector.

The transmitter shall be connected to a dummy load as described in clause 4.3.7 and the conducted power delivered shall be measured with a measurement receiver according to clause 4.3.10.

In the case of non-constant envelope modulation, a peak detector shall be used.

The maximum gain of the antenna to be used together with the equipment shall be declared by the manufacturer and this shall be recorded in the test report.

Perp, the radiated power (e.r.p.) limit applies to the maximum measured conducted power ($P_{\text{conducted}}$) value adjusted by the antenna gain (relative to a dipole) ($P_{\text{erp}} = P_{\text{conducted}} + \text{antenna gain}$).

The information shown in Table 7 shall be recorded in the test report.

**Table 7: Information Recorded in the Test Report
for conducted Effective Radiated Power**

Value	Notes
Test environment	Normal operation or unmodulated carrier
Centre frequency	Nominal Operating Frequency
Measured Effective Radiated Power	maximum measured conducted power value adjusted by the antenna gain (relative to a dipole)
NOTE: In case of a dedicated antenna the antenna gain (in dB, i.e. relative to a dipole) is declared by the manufacturer.	

According to ETSI EN 300 220-1 V3.1.1 (2017-02) clause 5.2.2.2:

Effective radiated power (radiated measurement):

This measurement method applies to EUT other than those measured using clause 5.2.2.1.

A suitable test site shall be selected from those described in clause C.1 and the radiated power established using the procedures described in clause C.5.1 (or clause C.5.2) depending on the test site, followed by clause C.5.3.

In the case of non-constant envelope modulation, a peak detector shall be used.

The information shown in Table 8 shall be recorded in the test report.

Table 8: Information Recorded in the Test Report for Effective Radiated Power

Value	Notes
Test environment	Normal operation or unmodulated carrier
Centre frequency	Nominal Operating Frequency
Measure of Effective Radiated Power	Larger value from horizontal and vertical measurement equivalent radiated power, plus equipment antenna gain
NOTE: In case of a removable antenna the antenna gain (in dB, i.e. relative to a dipole) is declared by the manufacturer.	

Test Data

Environmental Conditions

Temperature:	25.9~26.5℃
Relative Humidity:	46~48%
ATM Pressure:	101.0~102.1 kPa

The testing was performed by Hanic Pan from 2023-08-30 to 2023-09-13.

Test Mode: Transmitting

Test Frequency (MHz)	Test Condition					Result
867.1	Normal	L.V. L.T.	L.V. H.T.	H.V.L.T	H.V. H.T	Compliance
867.9	Normal	L.V. L.T.	L.V. H.T.	H.V.L.T	H.V. H.T	Compliance
868.1	Normal	L.V. L.T.	L.V. H.T.	H.V.L.T	H.V. H.T	Compliance
868.5	Normal	L.V. L.T.	L.V. H.T.	H.V.L.T	H.V. H.T	Compliance
868.3	Normal	L.V. L.T.	L.V. H.T.	H.V.L.T	H.V. H.T	Compliance

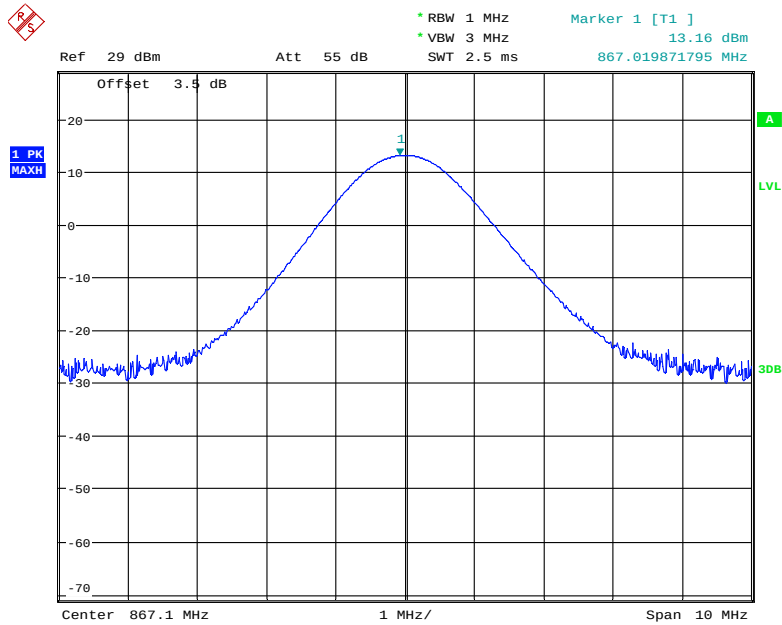
Test Condition			Conducted Power (dBm)	Antenna Gain (dBd)	ERP (dBm)	Limit (dBm)	Result
Frequency (MHz)	Temperature (°C)	Voltage (V _{DC})					
130kHz							
867.1	L.T	L.V.	13.15	-0.15	13.00	14	Pass
		H.V.	13.12	-0.15	12.97	14	Pass
	N.T.	N.V.	13.16	-0.15	13.01	14	Pass
	H.T.	L.V.	13.14	-0.15	12.99	14	Pass
		H.V.	13.13	-0.15	12.98	14	Pass
867.9	L.T	L.V.	13.32	-0.15	13.17	14	Pass
		H.V.	13.29	-0.15	13.14	14	Pass
	N.T.	N.V.	13.28	-0.15	13.13	14	Pass
	H.T.	L.V.	13.25	-0.15	13.10	14	Pass
		H.V.	13.23	-0.15	13.08	14	Pass
868.1	L.T	L.V.	13.31	-0.15	13.16	14	Pass
		H.V.	13.26	-0.15	13.11	14	Pass
	N.T.	N.V.	13.28	-0.15	13.13	14	Pass
	H.T.	L.V.	13.25	-0.15	13.10	14	Pass
		H.V.	13.20	-0.15	13.05	14	Pass
868.5	L.T	L.V.	13.02	-0.15	12.87	14	Pass
		H.V.	13.05	-0.15	12.90	14	Pass
	N.T.	N.V.	13.10	-0.15	12.95	14	Pass
	H.T.	L.V.	13.11	-0.15	12.96	14	Pass
		H.V.	13.12	-0.15	12.97	14	Pass
260kHz							
868.3		L.V.	13.20	-0.15	13.05	14	Pass
		H.V.	13.15	-0.15	13.00	14	Pass
	N.T.	N.V.	13.18	-0.15	13.03	14	Pass
	H.T.	L.V.	13.14	-0.15	12.99	14	Pass
		H.V.	13.15	-0.15	13.00	14	Pass

Note:

Maximum Antenna Gain=2.dBi (-0.15dBd), which was declared by manufacturer.
0dBd=2.15dBi

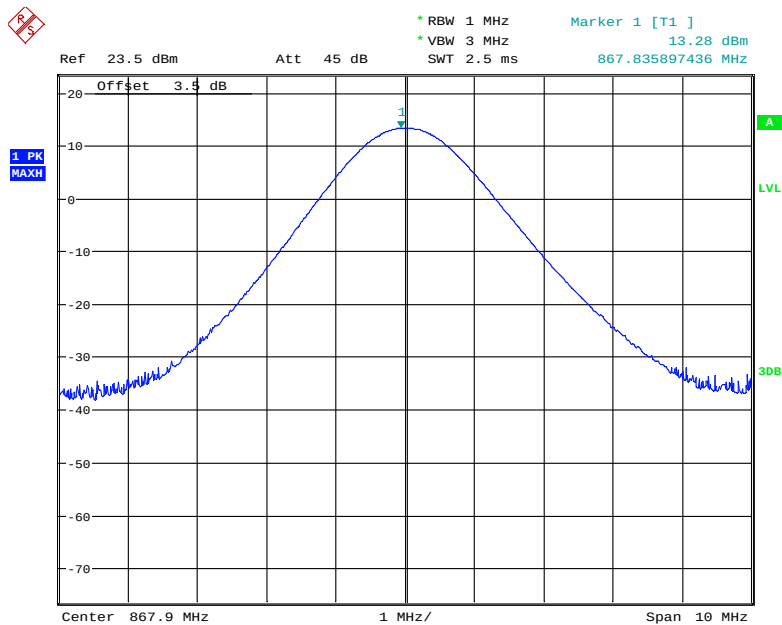
Please refer below plots for normal condition:

130kHz_867.1 MHz



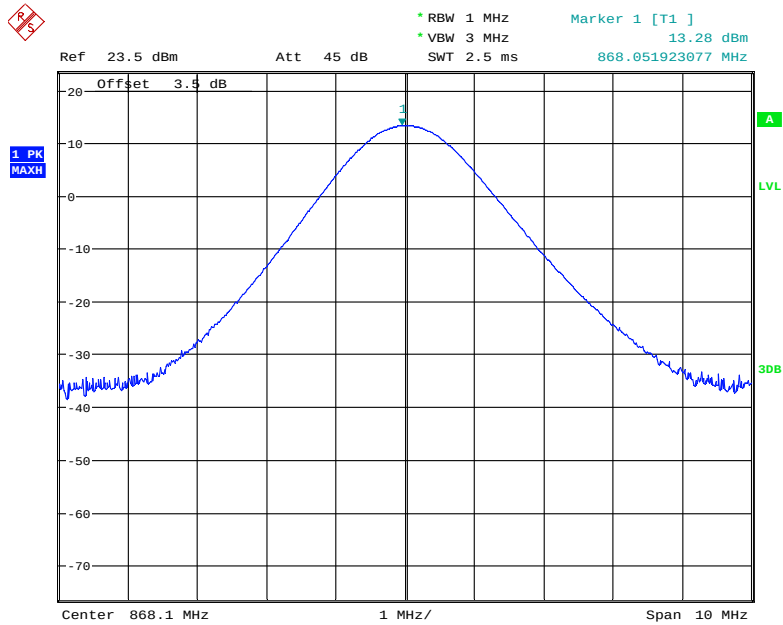
Date: 30.AUG.2023 09:24:59

130kHz_867.9 MHz



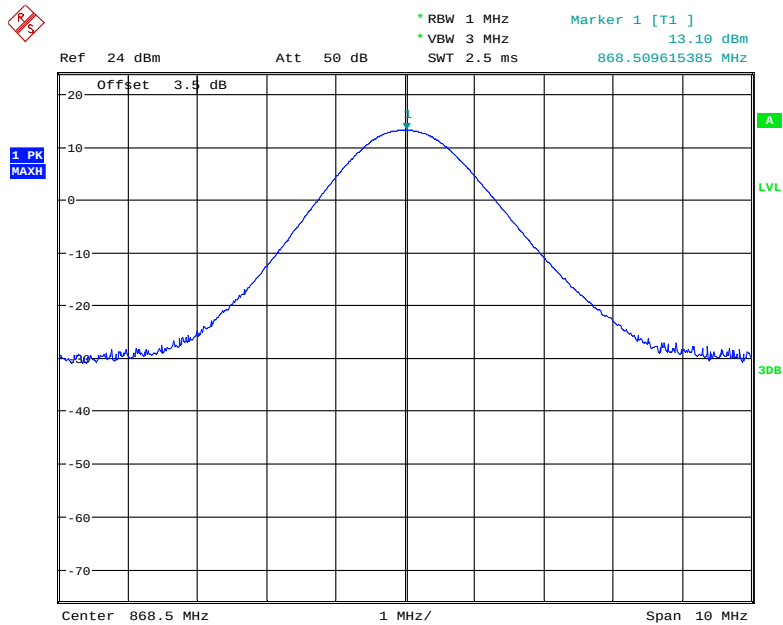
Date: 13.SEP.2023 16:06:43

130kHz_868.1 MHz

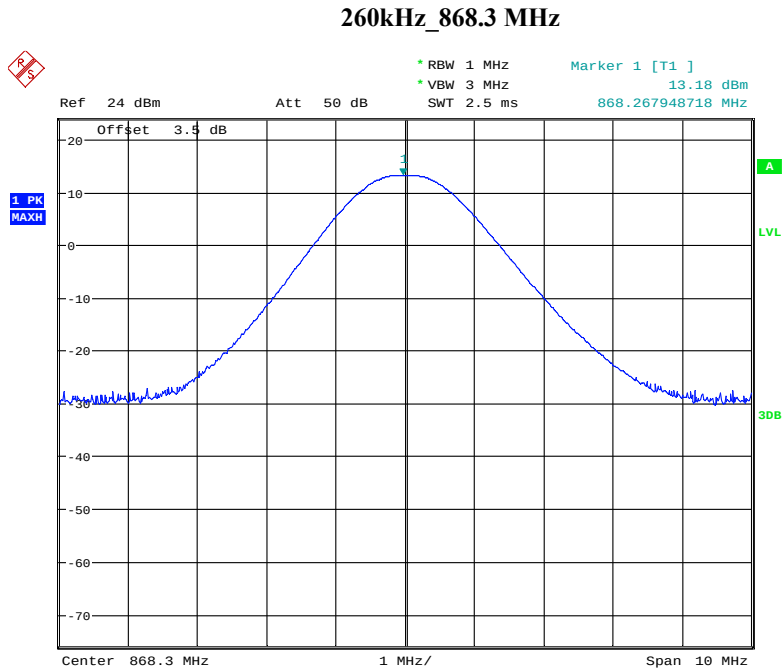


Date: 13.SEP.2023 16:08:38

130kHz_868.5 MHz



Date: 30.AUG.2023 09:12:48



Date: 30.AUG.2023 09:38:55

ETSI EN 300 220-2 V3.2.1 (2018-06) §4.3.4 - OCCUPIED BANDWIDTH**Applicable Standard**

According to ETSI EN 300 220-1 V3.1.1 (2017-02) clause 5.6:

The occupied bandwidth (OBW) is the Frequency Range in which 99 % of the total mean power of a given emission falls. The residual part of the total power being denoted as β , which, in cases of symmetrical spectra, splits up into $\beta/2$ on each side of the spectrum. Unless otherwise specified, $\beta/2$ is taken as 0,5 % as described in Figure 3.

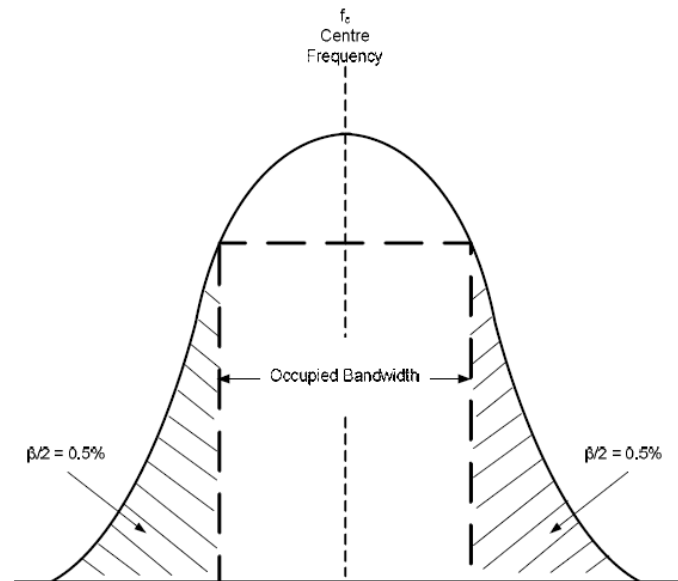


Figure 3: Signal occupied bandwidth

The maximum occupied bandwidth includes all associated side bands above the appropriate emissions level and the frequency error or drift under extreme test conditions.

Limit:

The Operating Channel shall be declared and shall reside entirely within the Operational Frequency Band.

The Maximum Occupied Bandwidth at 99 % shall reside entirely within the Operating Channel defined by F_{low} and F_{high} .

Method of measurement

According to ETSI EN 300 220-1 V3.1.1 (2017-02) clause 5.6.3:

The spectrum analyser shall be configured as appropriate for the parameters shown in Table 12.

Table 12: Test Parameters for Max Occupied Bandwidth Measurement

Setting	Value	Notes
Centre frequency	The nominal Operating Frequency	The highest or lowest Operating Frequency as declared by the manufacturer
RBW	1 % to 3 % of OCW without being below 100 Hz	
VBW	3 x RBW	Nearest available analyser setting to 3 x RBW
Span	At least 2 x Operating Channel width	Span should be large enough to include all major components of the signal and its side bands
Detector Mode	RMS	
Trace	Max hold	

If the equipment is capable of producing an unmodulated carrier and the test in clause 5.7 is performed, then the OBW measurements need only be performed under normal test conditions. Any required results for Maximum OBW under extreme conditions are obtained by addition and subtraction of the upper and lower frequency error results to each bandwidth measurement obtained in this test.

Step 1: Operation of the EUT shall be started, on the highest operating frequency as declared by the manufacturer, with the appropriate test signal.

The signal attenuation shall be adjusted to ensure that the signal power envelope is sufficiently above the noise floor of the analyser to avoid the noise signals on either side of the power envelope being included in the measurement.

Step 2: When the trace is completed the peak value of the trace shall be located and the analyser marker placed on this peak.

Step 3: The 99 % occupied bandwidth function of the spectrum analyser shall be used to measure the occupied bandwidth of the signal.

Test Data

Environmental Conditions

Temperature:	25.9~26.5 °C
Relative Humidity:	46~48%
ATM Pressure:	101.0~102.1 kPa

The testing was performed by Hanic Pan from 2023-08-30 to 2023-09-13.

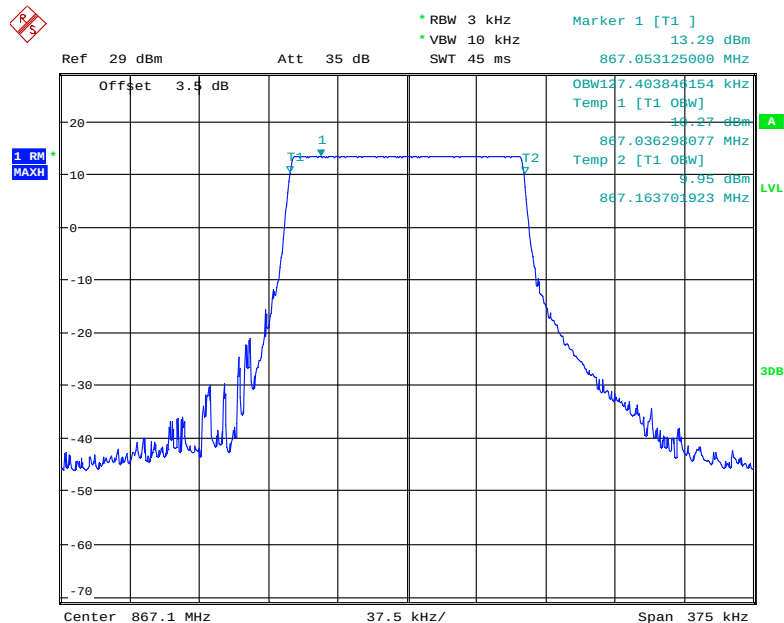
Test Mode: Transmitting

Test Condition					Result
Normal	L.V. L.T.	L.V. H.T.	H.V.H.T	H.V.L.T	Pass

Normal Condition Test plots as below:

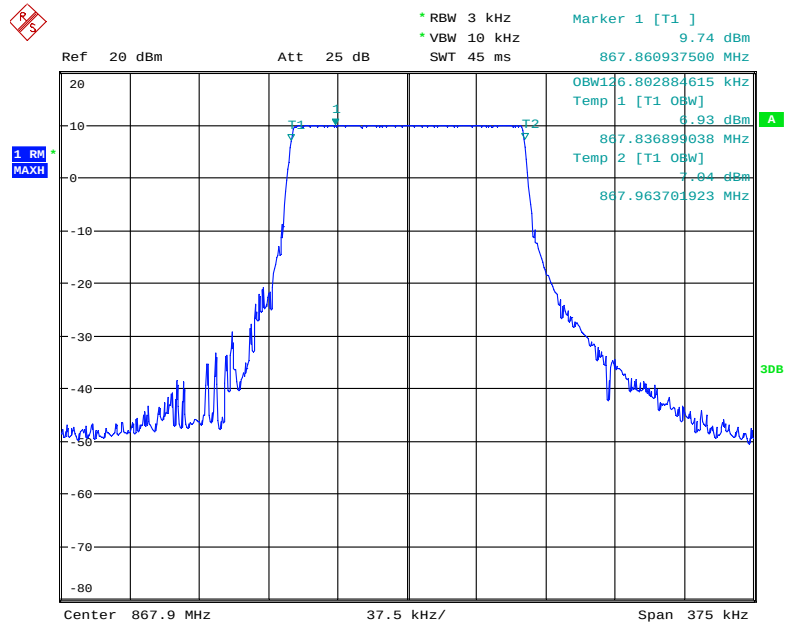
Voltage Supply (V _{DC})	Temperature (°C)	Bandwidth (kHz)	Frequency (MHz)	Occupied Bandwidth (kHz)	Frequency of Lower point (MHz)	Frequency of Upper point (MHz)	Limit (MHz)
N.V.	N.T.	130	867.1	127.404	867.036	867.163	Within 865-868
			867.9	126.803	867.837	867.964	
			868.1	128.005	868.036	868.164	Within 868-868.6
			868.5	127.404	868.436	868.564	
N.V.	N.T.	260	868.3	258.413	868.169	868.427	Within 868-868.6

130kHz_867.1 MHz



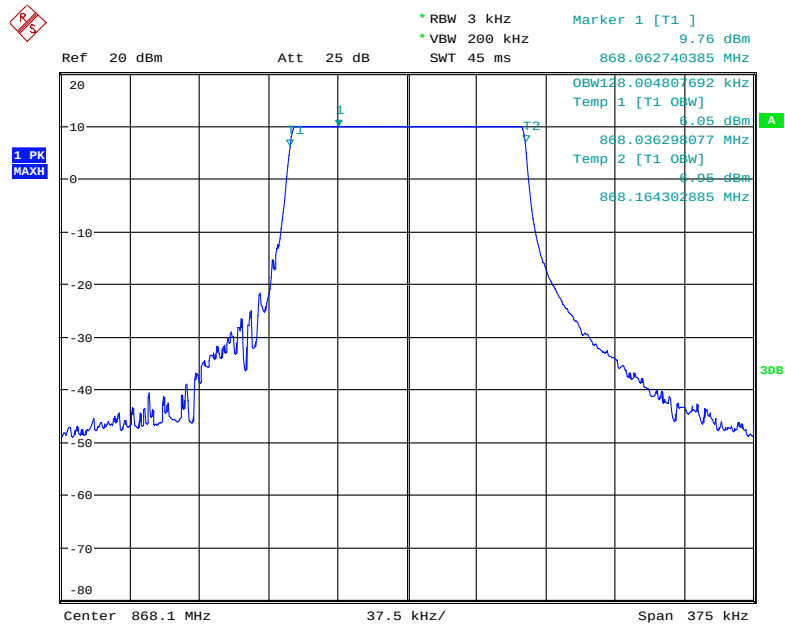
Date: 30.AUG.2023 09:23:58

130kHz_867.9 MHz



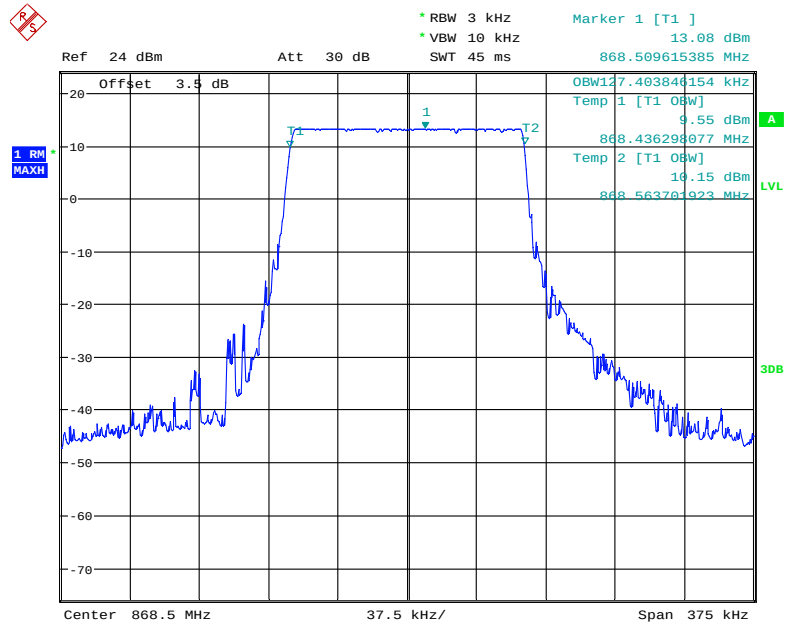
Date: 13.SEP.2023 10:56:45

130kHz_868.1 MHz



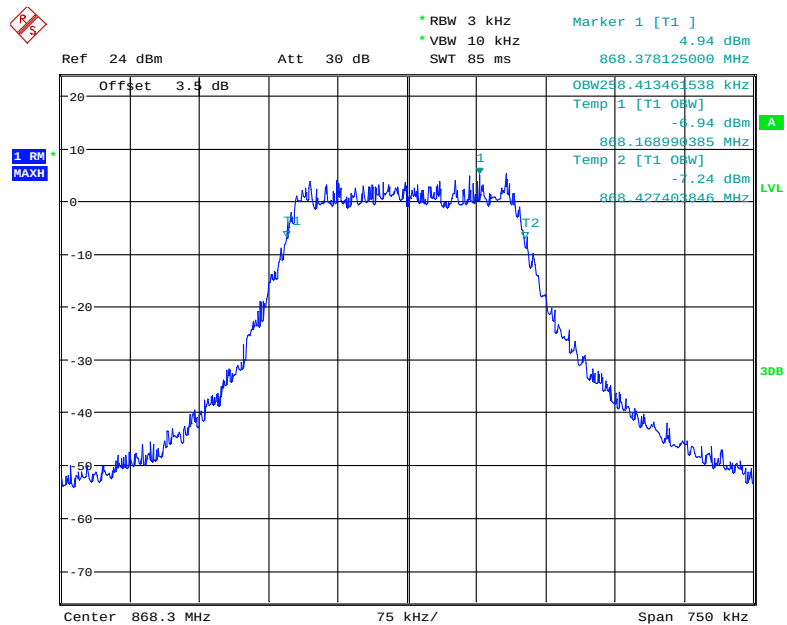
Date: 13.SEP.2023 11:37:18

130kHz_868.5 MHz



Date: 30.AUG.2023 09:10:50

260kHz_868.3 MHz



Date: 30.AUG.2023 09:37:02

ETSI EN 300 220-2 V3.2.1 (2018-06) §4.3.5–TX OUT OF BAND EMISSIONS

Applicable Standard

According to ETSI EN 300 220-1 V3.1.1 (2017-02) clause 5.8:

Two OOB domains are defined, one for OC (see Figure 5) and one for Operational Frequency band (see Figure 6). The spectrum masks for these two OOB domains may overlap.

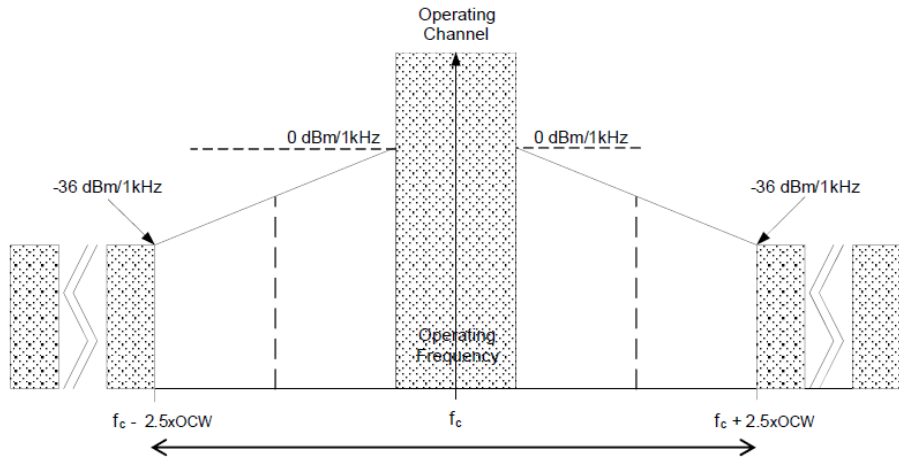


Figure 5: Out Of Band Domain for Operating Channel with reference BW

Unwanted emissions in the Out Of Band domain are those falling in the frequency range immediately below the lower, and above the upper, frequency of the Operating Channel. The OOB domain includes both frequencies outside the Operating Channel within the Operational Frequency Band and frequencies outside the Operational Frequency Band.

The relevant Out Of Band domain is shown in Figure 5 and applies within the Operational Frequency Band.

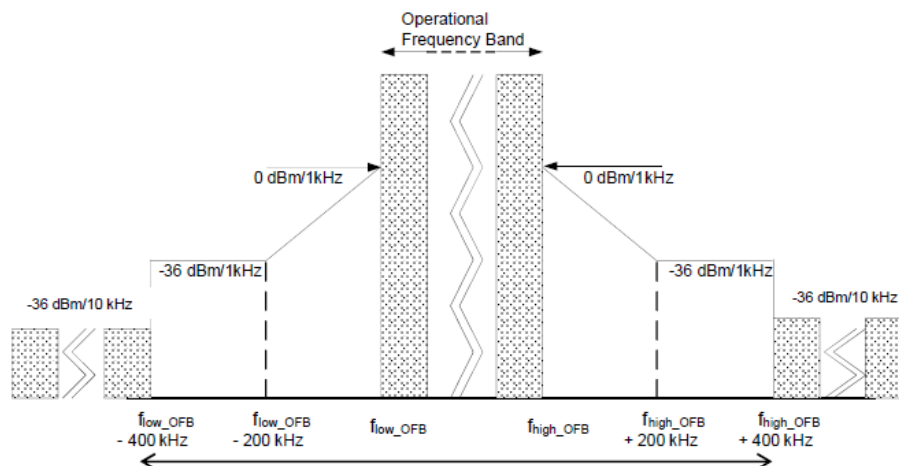


Figure 6: Out Of Band Domain for Operational Frequency Band with reference BW

Specific limits apply at frequencies immediately above and below the Operational Frequency Band as shown in Figure 6.

NOTE: $f_{\text{low_OFB}}$ is the lower edge of the Operational Frequency Band.

$f_{\text{high_OFB}}$ is the upper edge of the Operational Frequency Band.

Limit: The EUT emissions level in OOB domains for the Operating Channel and the Operational Frequency Band shall be less or equal to Table 15 spectrum mask.

Table 15: Emission limits in the Out Of Band domains

Domain	Frequency Range	RBW _{REF}	Max power limit
OOB limits applicable to Operational Frequency Band (See Figure 6)	$f \leq f_{\text{low_OFB}} - 400 \text{ kHz}$	10 kHz	-36 dBm
	$F_{\text{low_OFB}} - 400 \text{ kHz} \leq f \leq f_{\text{low_OFB}} - 200 \text{ kHz}$	1 kHz	-36 dBm
	$f_{\text{low}} - 200 \text{ kHz} \leq f < f_{\text{low_OFB}}$	1 kHz	See Figure 6
	$f = f_{\text{low_OFB}}$	1 kHz	0 dBm
	$f = f_{\text{high_OFB}}$	1 kHz	0 dBm
	$F_{\text{high_OFB}} < f \leq f_{\text{high_OFB}} + 200 \text{ kHz}$	1 kHz	See Figure 6
	$F_{\text{high_OFB}} + 200 \text{ kHz} \leq f \leq f_{\text{high_OFB}} + 400 \text{ kHz}$	1 kHz	-36 dBm
	$F_{\text{high_OFB}} + 400 \text{ kHz} \leq f$	10 kHz	-36 dBm
OOB limits applicable to Operating Channel (See Figure 5)	$f = f_c - 2.5 \times \text{OCW}$	1 kHz	-36 dBm
	$f_c - 2.5 \times \text{OCW} \leq f \leq f_c - 0.5 \times \text{OCW}$	1 kHz	See Figure 5
	$f = f_c - 0.5 \times \text{OCW}$	1 kHz	0 dBm
	$f = f_c + 0.5 \times \text{OCW}$	1 kHz	0 dBm
	$f_c + 0.5 \times \text{OCW} \leq f \leq f_c + 2.5 \times \text{OCW}$	1 kHz	See Figure 5
	$f = f_c + 2.5 \times \text{OCW}$	1 kHz	-36 dBm
NOTE: f is the measurement frequency. f _c is the Operating Frequency. F _{low_OFB} is the lower edge of the Operational Frequency Band. F _{high_OFB} is the upper edge of the Operational Frequency Band. OCW is the operating channel bandwidth.			

Method of measurement

According to ETSI EN 300 220-1 V3.1.1 (2017-02) clause 5.8.3.

Test Data

Environmental Conditions

Temperature:	26.2°C
Relative Humidity:	47%
ATM Pressure:	101.0~102.1 kPa

The testing was performed by Hanic Pan on 2023-09-25.

Test Mode: Transmitting

• Test Condition					Result
Normal	L.V. L.T.	L.V. H.T.	N.V.L.T	N.V. H.T	Pass

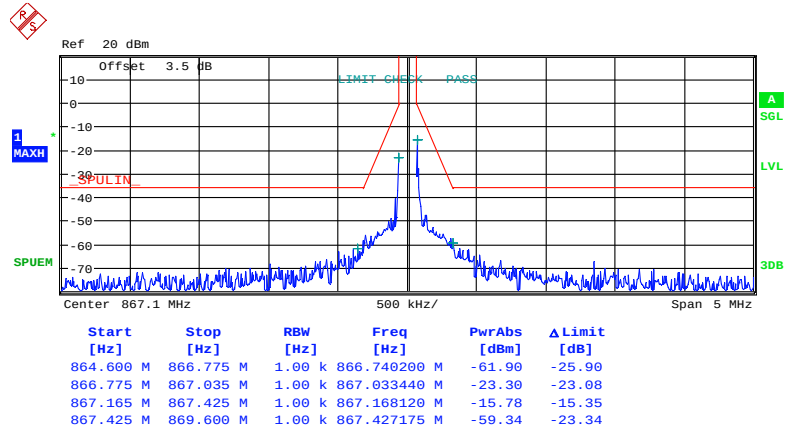
Normal Condition Test Data as below:

Test with conducted measurement.

Please refer to the below plots:

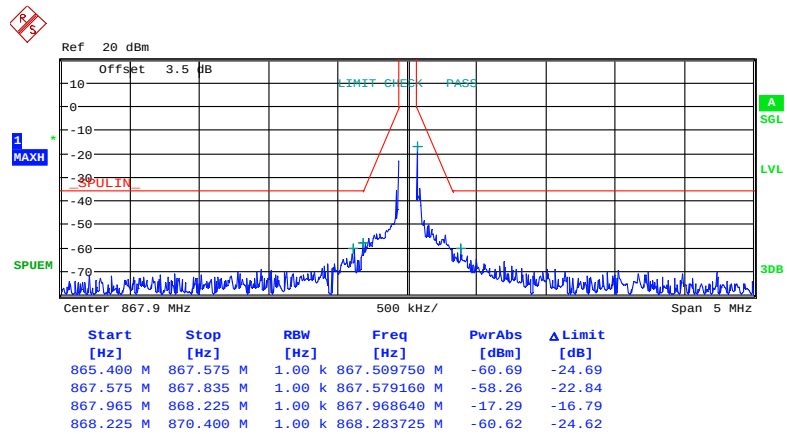
OOB for OC:

130kHz_867.1MHz



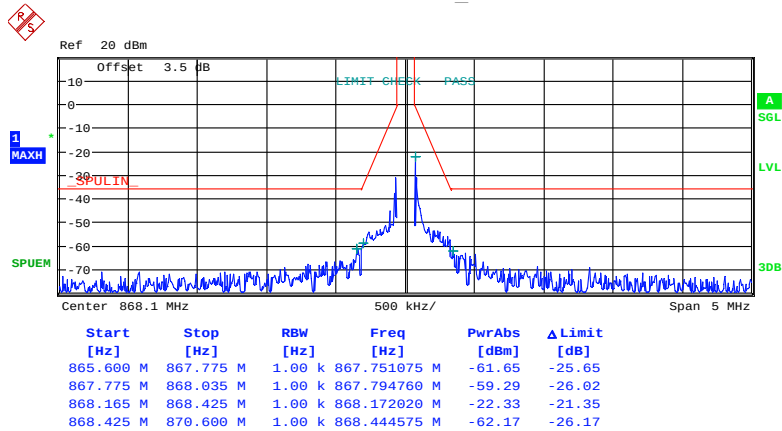
Date: 25.SEP.2023 16:05:22

130kHz_867.9MHz



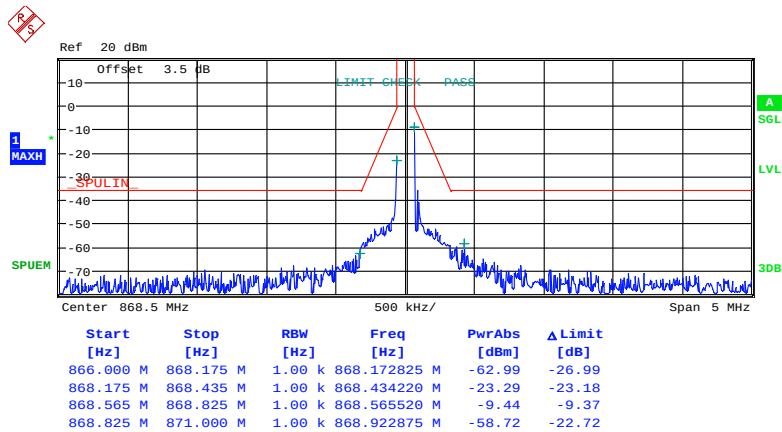
Date: 25.SEP.2023 16:18:38

130kHz_868.1MHz



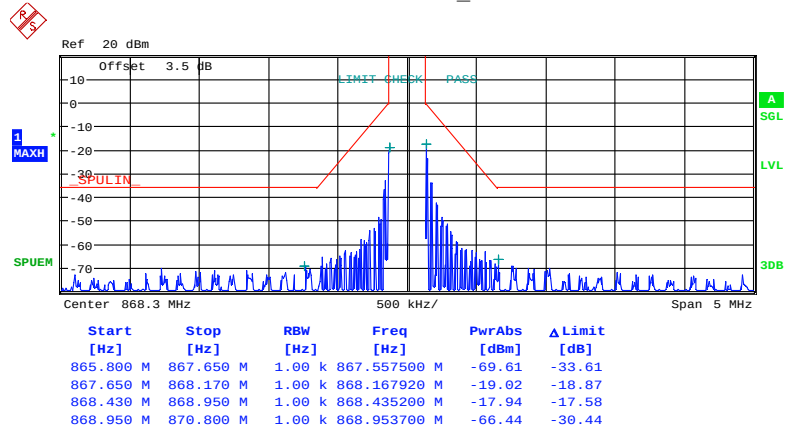
Date: 25.SEP.2023 16:25:29

130kHz_868.5MHz



Date: 25.SEP.2023 16:38:01

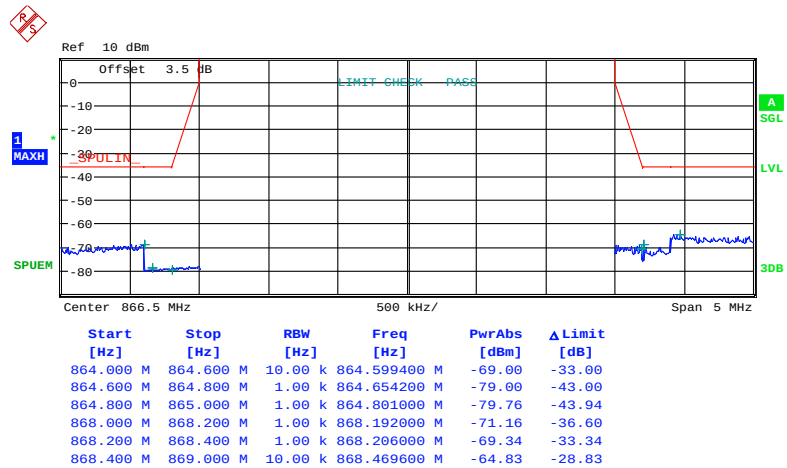
260kHz_868.3MHz



Date: 25.SEP.2023 16:47:22

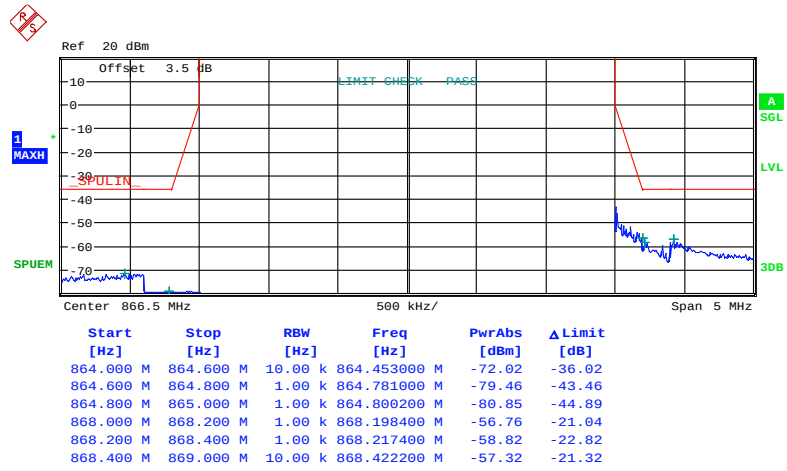
OOB for OFB

130kHz_867.1MHz



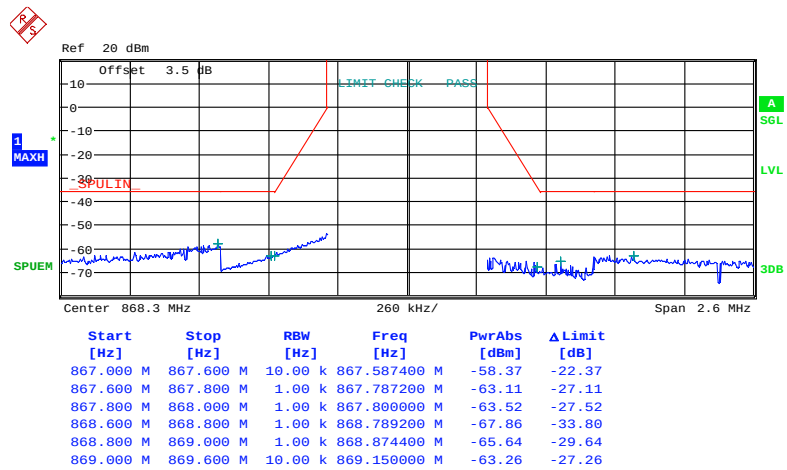
Date: 25.SEP.2023 16:03:35

130kHz_867.9MHz



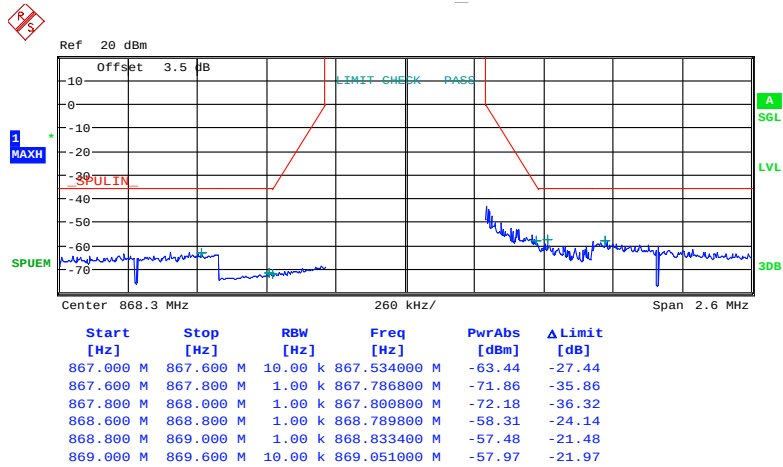
Date: 25.SEP.2023 16:17:45

130kHz_868.1MHz



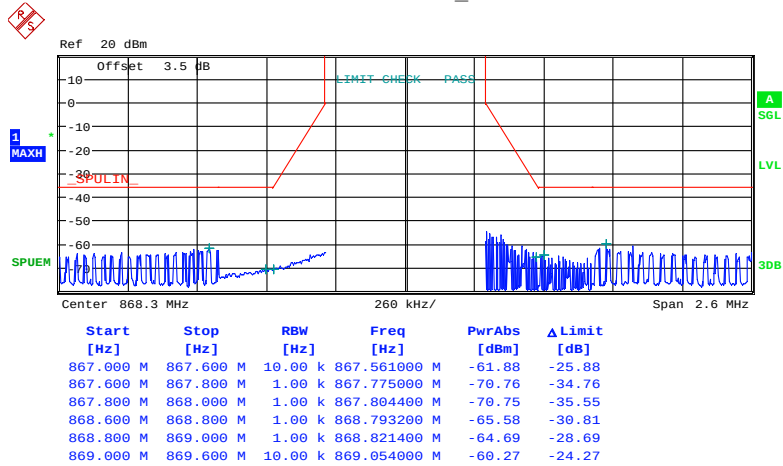
Date: 25.SEP.2023 16:24:47

130kHz_868.5MHz



Date: 25.SEP.2023 16:37:33

260kHz_868.3MHz



Date: 25.SEP.2023 16:46:49

ETSI EN 300 220-2 V3.2.1 (2018-06) §4.3.6 - TRANSIENT POWER

Applicable Standard

According to ETSI EN 300 220-1 V3.1.1 (2017-02) clause 5.10:

Transmitter transient power is power falling into frequencies other than the operating channel as a result of the transmitter being switched on and off.

Limit: The transient power shall not exceed the values given in Table 23.

Table 23: Transmitter Transient Power limits

Absolute offset from centre frequency	RBW _{REF}	Peak power limit applicable at measurement points
≤ 400 kHz	1 kHz	0 dBm
> 400 kHz	1 kHz	-27 dBm

Method of measurement

The output of the EUT shall be connected to a spectrum analyser or equivalent measuring equipment.

The measurement shall be undertaken in zero span mode. The analyser's centre frequency shall be set to an offset from the operating centre frequency. These offset values and their corresponding RBW configurations are listed in Table 24.

Table 24: RBW for Transient Measurement

Measurement points: offset from centre frequency	Analyser RBW	RBW _{REF}
-0,5 x OCW - 3 kHz 0,5 x OCW + 3 kHz Not applicable for OCW < 25 kHz	1 kHz	1kHz
±12,5 kHz or ±OCW whichever is the greater	Max (RBW pattern 1, 3, 10 kHz) ≤ Offset frequency/6 (see note)	1 kHz
-0,5 x OCW - 400 kHz 0,5 x OCW + 400 kHz	100 kHz	1 kHz
-0,5 x OCW -1 200 kHz 0,5 x OCW + 1 200 kHz	300 kHz	1 kHz
NOTE: Max (RBW pattern 1, 3, 10 kHz) means the maximum bandwidth that falls into the commonly implemented 1, 3, 10 kHz RBW filter bandwidth incremental pattern of spectrum analysers. EXAMPLE: If OCW is 25 kHz then the RBW value corresponding to one OCW offset frequency is 3 kHz. The rest of the analyser settings are listed in Table 25, and if OCW is 250 kHz then the RBW value corresponding to one OCW offset frequency is 30 kHz.		

The used modulation shall be D-M3. The analyser shall be set to the settings of Table 25 and a measurement shall be started for each offset frequency. The EUT shall transmit at least five D-M3 test signal. The peak value shall be recorded and the measurement shall be repeated at each offset frequency mentioned in Table 24.

The recorded power values shall be converted to power values measured in RBW_{REF} by the formula in clause 4.3.10.1.

Table 25: Parameters for Transient Measurement

Spectrum Analyser Setting	Value	Notes
VBW/RBW	10	At higher RBW values VBW may be clipped to its maximum value
Sweep time	500 ms	
RBW filter	Gaussian	
Trace Detector Function	RMS	
Trace Mode	Max hold	
Sweep points	501	
Measurement mode	Continuous sweep	
NOTE: The ratio between the number of sweep points and the sweep time shall be the same ratio as above if different number of sweep points is used.		

The used modulation shall be D-M3. The analyser shall be set to the settings of Table 25 and a measurement shall be started for each offset frequency. The EUT shall transmit at least five D-M3 test signal. The peak value shall be recorded and the measurement shall be repeated at each offset frequency mentioned in Table 24.

The recorded power values shall be converted to power values measured in RBWREF by the formula in clause 4.3.10.1.

When $RBW_{measured} > RBW_{REF}$ the result for broadband emissions should be normalized to the bandwidth Ratio according to the formula (2):

$$B = A + 10 \log \frac{RBW_{ref}}{RBW_{MEASURED}} \quad (2)$$

Where:

- A is the measured value at the wider measurement bandwidth $RBW_{measured}$;
- B is the corresponding value at RBW_{REF} .

Test Data

Environmental Conditions

Temperature:	25.9~26.5℃
Relative Humidity:	46~48%
ATM Pressure:	101.0~102.1 kPa

The testing was performed by Hanic Pan from 2023-09-13 to 2023-10-11.

Test Mode: Transmitting

Please refer to the below tables and plots

130kHz 867.1MHz

Item	Test Frequency Offset From Centre Frequency	Measure Transient Power (dBm)	RBW _{ref} / RBW _{meas} (kHz)	Correct Factor (dB)	Transient Power (dBm)	Limit (dBm)	Result
Offset *1	-0.5*OCW-3kHz	-14.64	1/1	0	-14.64	0	Pass
Offset *2	-OCW	-37.63	1/10	-10	-47.63	0	Pass
Offset *3	-0.5*OCW-400kHz	-46.91	1/100	-20	-66.91	-27	Pass
Offset *4	-0.5*OCW-1200kHz	-40.04	1/300	-24.8	-64.84	-27	Pass
Offset *5	+0.5*OCW+3kHz	-15.24	1/1	0	-15.24	0	Pass
Offset *6	+OCW	-37.35	1/10	-10	-47.35	0	Pass
Offset *7	+0.5*OCW+400kHz	-46.96	1/100	-20	-66.96	-27	Pass
Offset *8	+0.5*OCW+1200kHz	-39.75	1/300	-24.8	-64.55	-27	Pass

130kHz 867.9MHz

Item	Test Frequency Offset From Centre Frequency	Measure Transient Power (dBm)	RBW _{ref} / RBW _{meas} (kHz)	Correct Factor (dB)	Transient Power (dBm)	Limit (dBm)	Result
Offset *1	-0.5*OCW-3kHz	-13.67	1/1	0	-13.67	0	Pass
Offset *2	-OCW	-38.64	1/10	-10	-48.64	0	Pass
Offset *3	-0.5*OCW-400kHz	-46.74	1/100	-20	-66.74	-27	Pass
Offset *4	-0.5*OCW-1200kHz	-40.03	1/300	-24.8	-64.83	-27	Pass
Offset *5	+0.5*OCW+3kHz	-13.95	1/1	0	-13.95	0	Pass
Offset *6	+OCW	-37.36	1/10	-10	-47.36	0	Pass
Offset *7	+0.5*OCW+400kHz	-46.91	1/100	-20	-66.91	-27	Pass
Offset *8	+0.5*OCW+1200kHz	-39.76	1/300	-24.8	-64.56	-27	Pass

130kHz 868.1MHz

Item	Test Frequency Offset From Centre Frequency	Measure Transient Power (dBm)	RBW _{ref} / RBW _{meas} (kHz)	Correct Factor (dB)	Transient Power (dBm)	Limit (dBm)	Result
Offset *1	-0.5*OCW-3kHz	-13.6	1/1	0	-13.6	0	Pass
Offset *2	-OCW	-38.01	1/10	-10	-48.01	0	Pass
Offset *3	-0.5*OCW-400kHz	-46.71	1/100	-20	-66.71	-27	Pass
Offset *4	-0.5*OCW-1200kHz	-40.27	1/300	-24.8	-65.07	-27	Pass
Offset *5	+0.5*OCW+3kHz	-13.56	1/1	0	-13.56	0	Pass
Offset *6	+OCW	-37.27	1/10	-10	-47.27	0	Pass
Offset *7	+0.5*OCW+400kHz	-46.46	1/100	-20	-66.46	-27	Pass
Offset *8	+0.5*OCW+1200kHz	-39.75	1/300	-24.8	-64.55	-27	Pass

130kHz 868.5MHz

Item	Test Frequency Offset From Centre Frequency	Measure Transient Power (dBm)	$\text{RBW}_{\text{ref}}/\text{RBW}_{\text{meas}}$ (kHz)	Correct Factor (dB)	Transient Power (dBm)	Limit (dBm)	Result
Offset *1	-0.5*OCW-3kHz	-14.63	1/1	0	-14.63	0	Pass
Offset *2	-OCW	-38.18	1/10	-10	-48.18	0	Pass
Offset *3	-0.5*OCW-400kHz	-46.67	1/100	-20	-66.67	-27	Pass
Offset *4	-0.5*OCW-1200kHz	-40.05	1/300	-24.8	-64.85	-27	Pass
Offset *5	+0.5*OCW+3kHz	-15.62	1/1	0	-15.62	0	Pass
Offset *6	+OCW	-37.41	1/10	-10	-47.41	0	Pass
Offset *7	+0.5*OCW+400kHz	-46.37	1/100	-20	-66.37	-27	Pass
Offset *8	+0.5*OCW+1200kHz	-39.76	1/300	-24.8	-64.56	-27	Pass

260kHz 868.3MHz

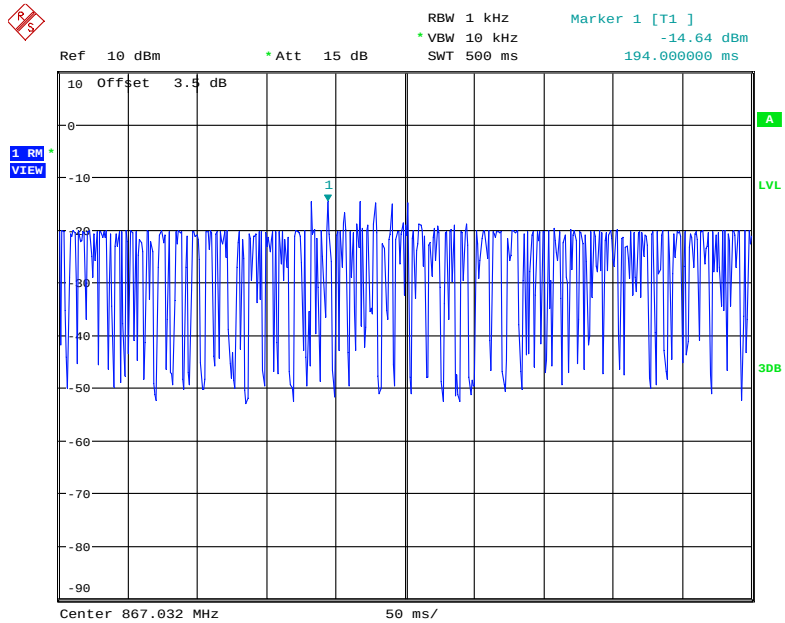
Item	Test Frequency Offset From Centre Frequency	Measure Transient Power (dBm)	$\text{RBW}_{\text{ref}}/\text{RBW}_{\text{meas}}$ (kHz)	Correct Factor (dB)	Transient Power (dBm)	Limit (dBm)	Result
Offset *1	-0.5*OCW-3kHz	-16.19	1/1	0	-16.19	0	Pass
Offset *2	-OCW	-38.10	1/30	-14.8	-52.90	0	Pass
Offset *3	-0.5*OCW-400kHz	-49.03	1/100	-20	-69.03	-27	Pass
Offset *4	-0.5*OCW-1200kHz	-42.83	1/300	-24.8	-67.63	-27	Pass
Offset *5	+0.5*OCW+3kHz	-17.41	1/1	0	-17.41	0	Pass
Offset *6	+OCW	-36.10	1/30	-14.8	-50.90	0	Pass
Offset *7	+0.5*OCW+400kHz	-48.71	1/100	-20	-68.71	-27	Pass
Offset *8	+0.5*OCW+1200kHz	-42.34	1/300	-24.8	-67.14	-27	Pass

Note: Correct factor= $10 \cdot \log(\text{RBW}_{\text{ref}}/\text{RBW}_{\text{meas}})$

Transient power=Absolute level+Correct factor

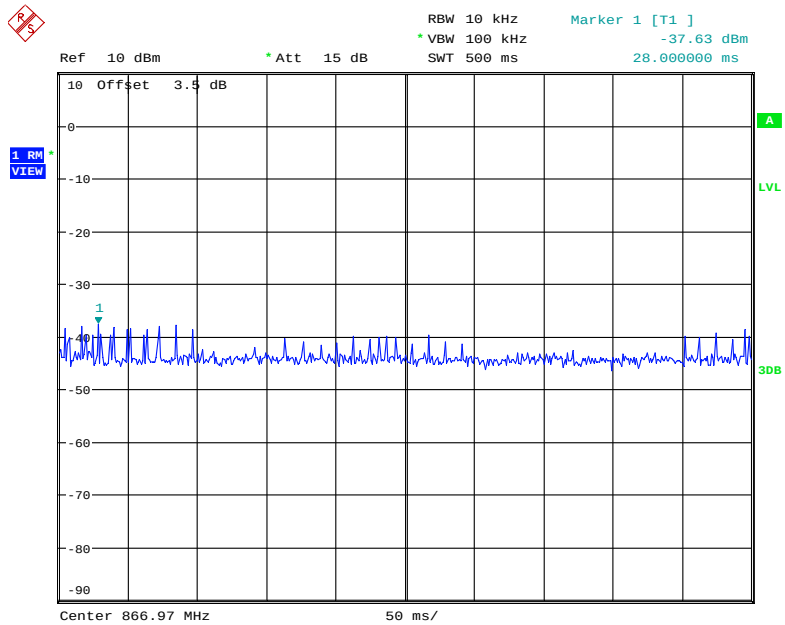
130kHz_867.1MHz:

Offset *1



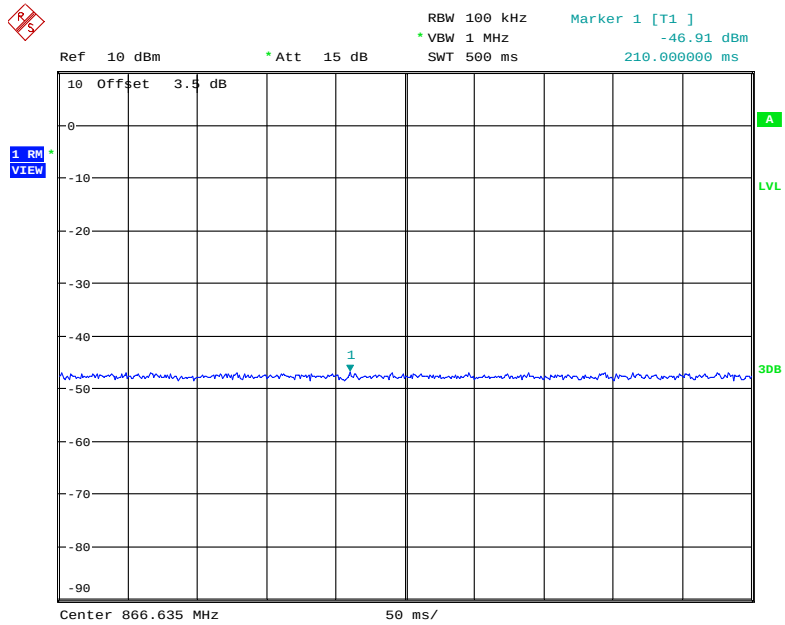
Date: 25.SEP.2023 15:58:13

Offset *2



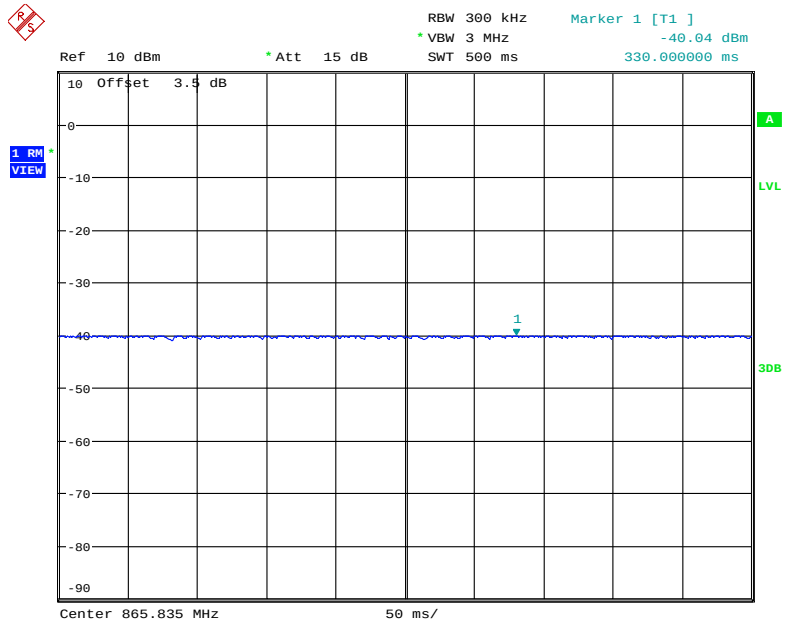
Date: 25.SEP.2023 15:58:48

Offset *3



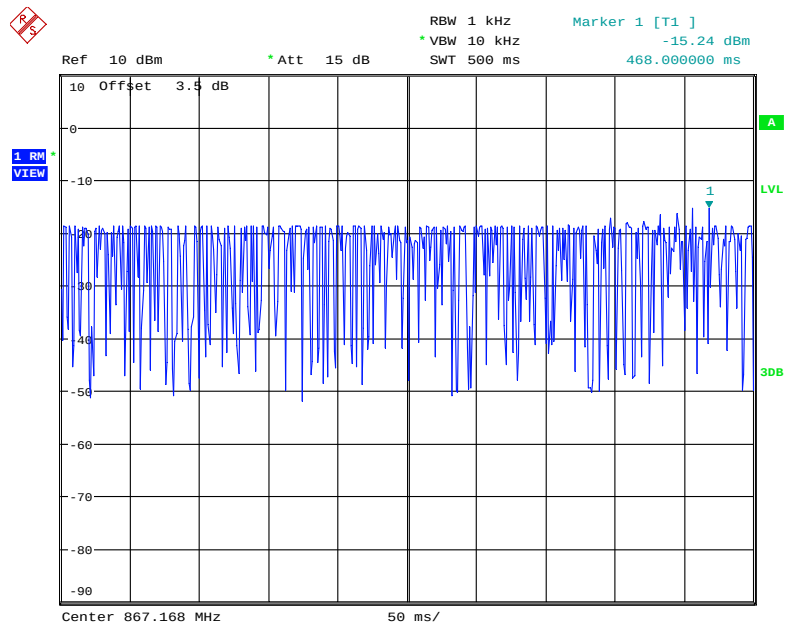
Date: 25.SEP.2023 15:59:24

Offset *4



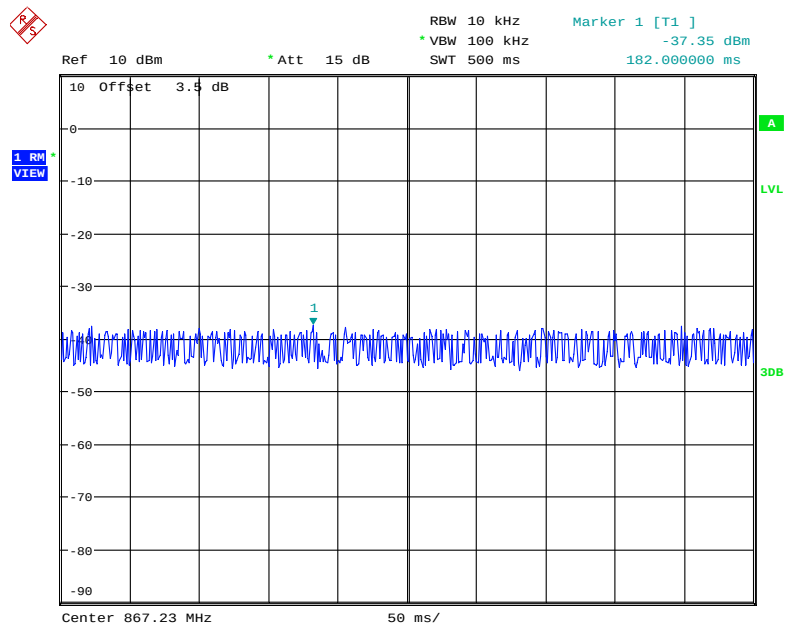
Date: 25.SEP.2023 15:59:59

Offset *5



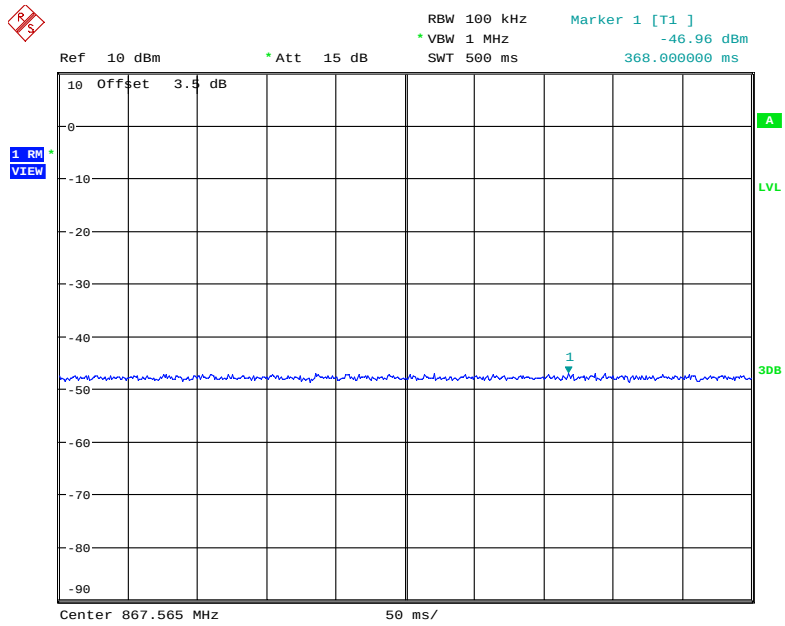
Date: 25.SEP.2023 15:58:31

Offset *6



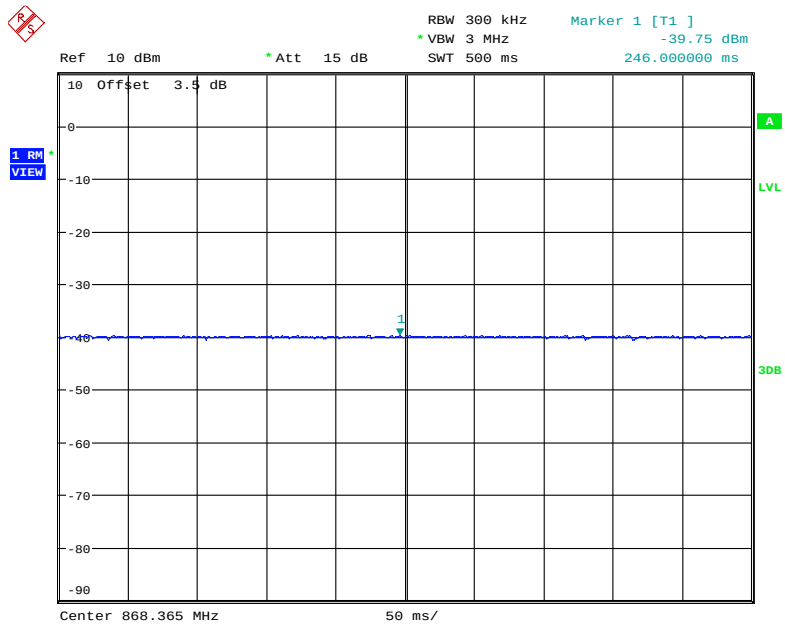
Date: 25.SEP.2023 15:59:06

Offset *7



Date: 25.SEP.2023 15:59:42

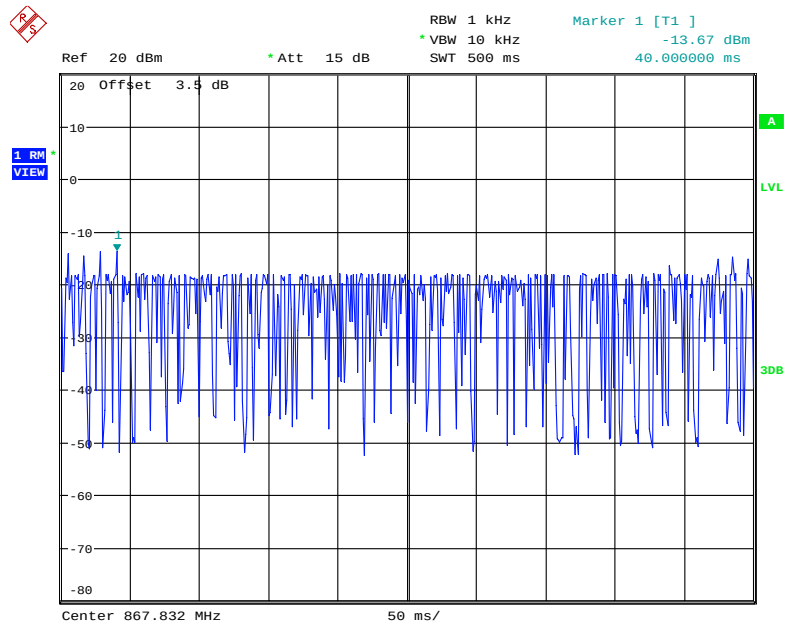
Offset *8



Date: 25.SEP.2023 16:00:17

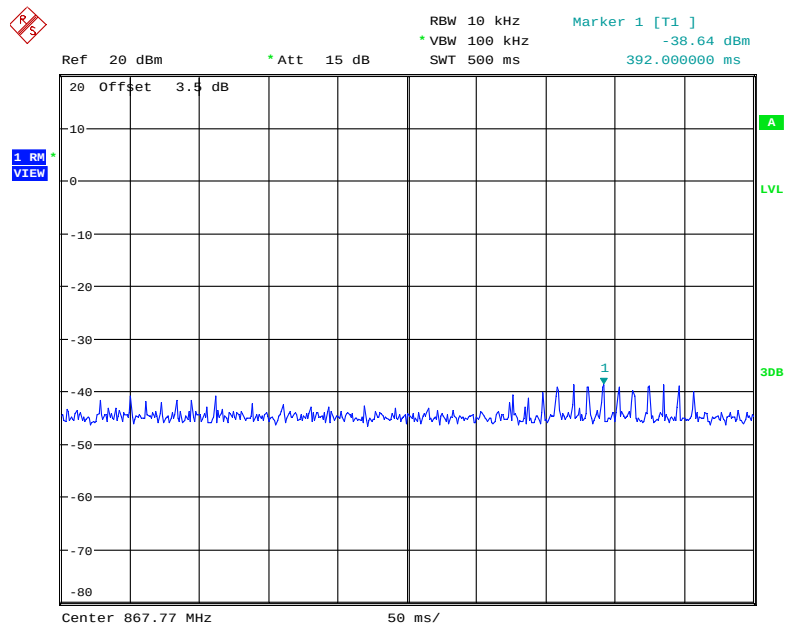
130kHz_867.9MHz:

Offset *1



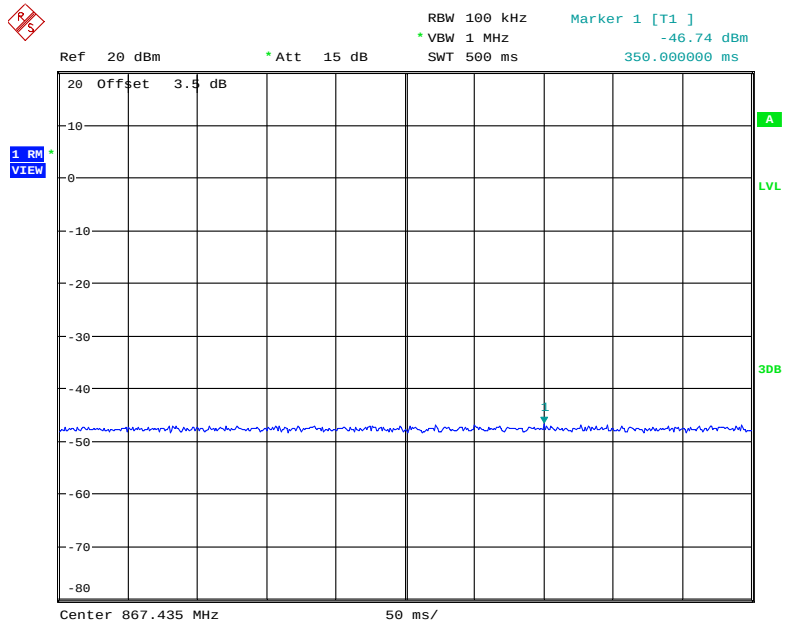
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Offset *2



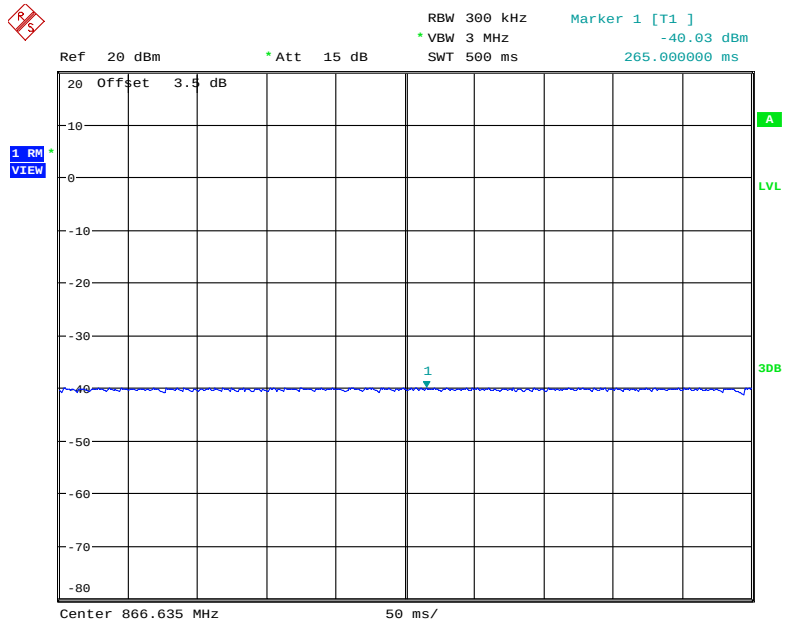
Date: 25.SEP.2023 16:13:11

Offset *3



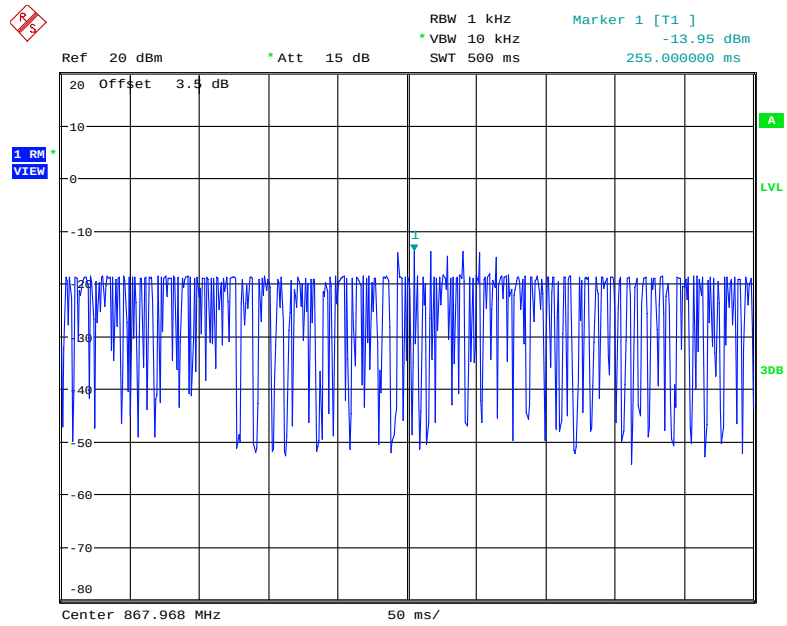
Date: 25.SEP.2023 16:13:47

Offset *4



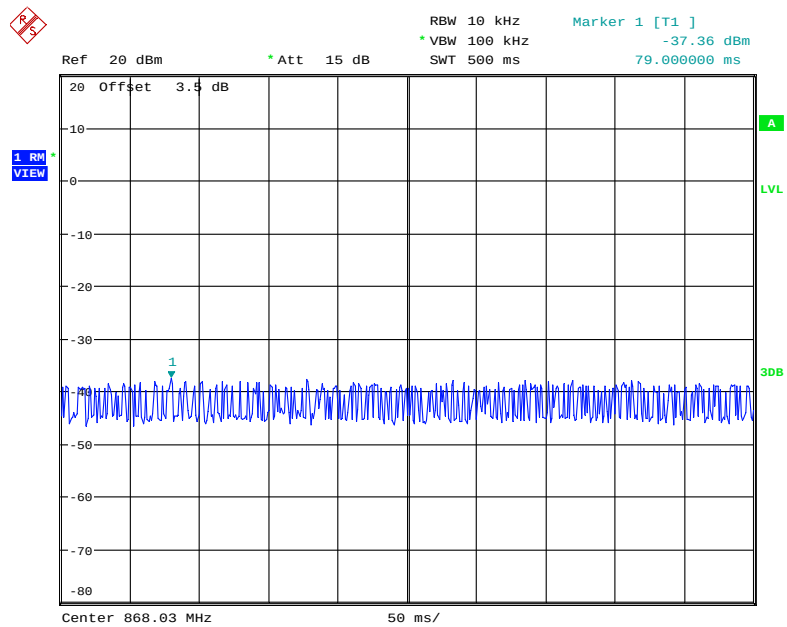
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Offset *5



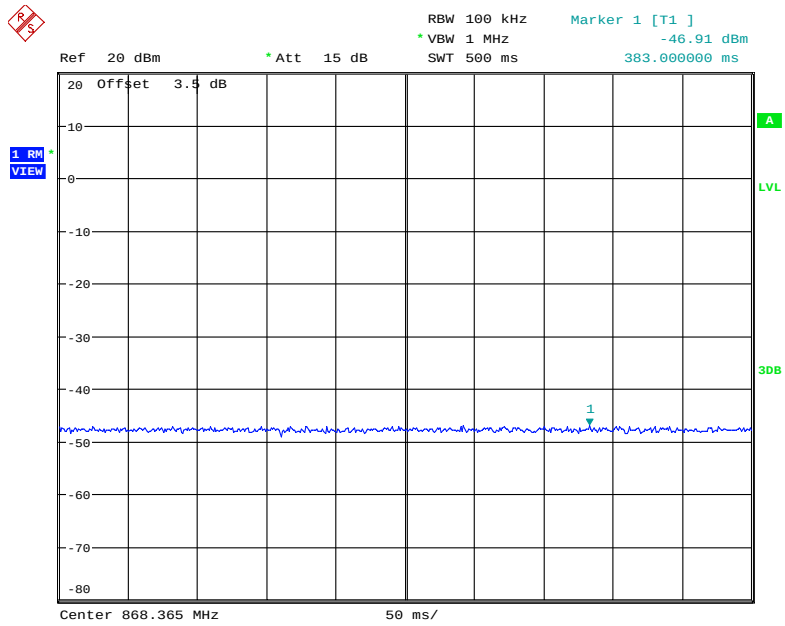
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Offset *6



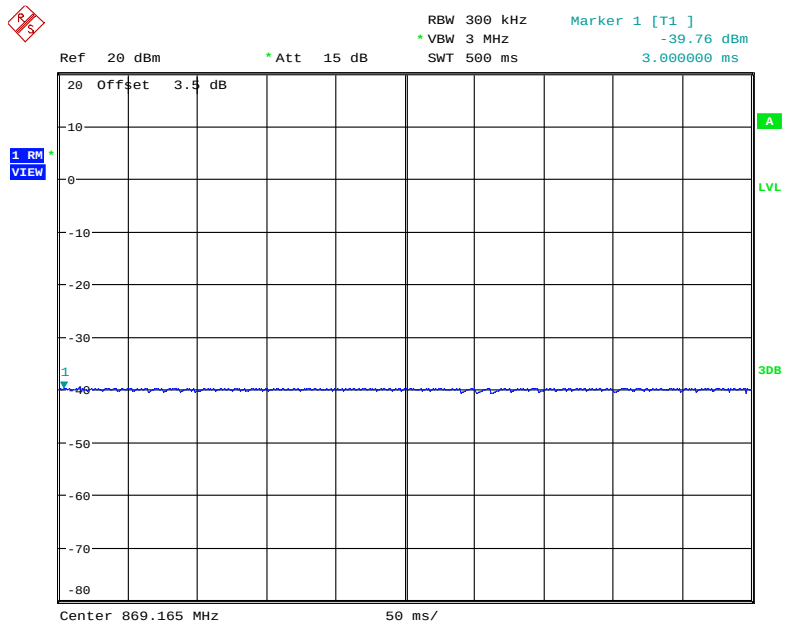
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Offset *7



Date: 25.SEP.2023 16:14:05

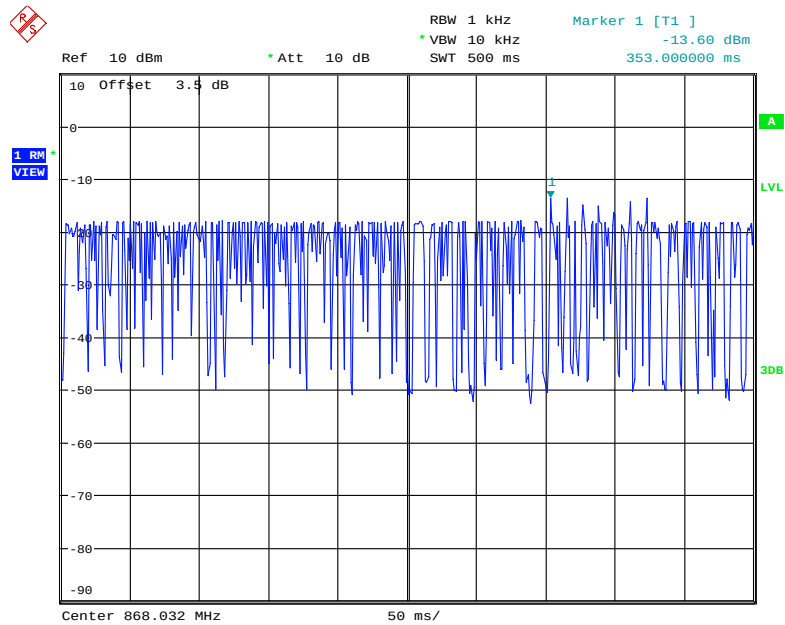
Offset *8



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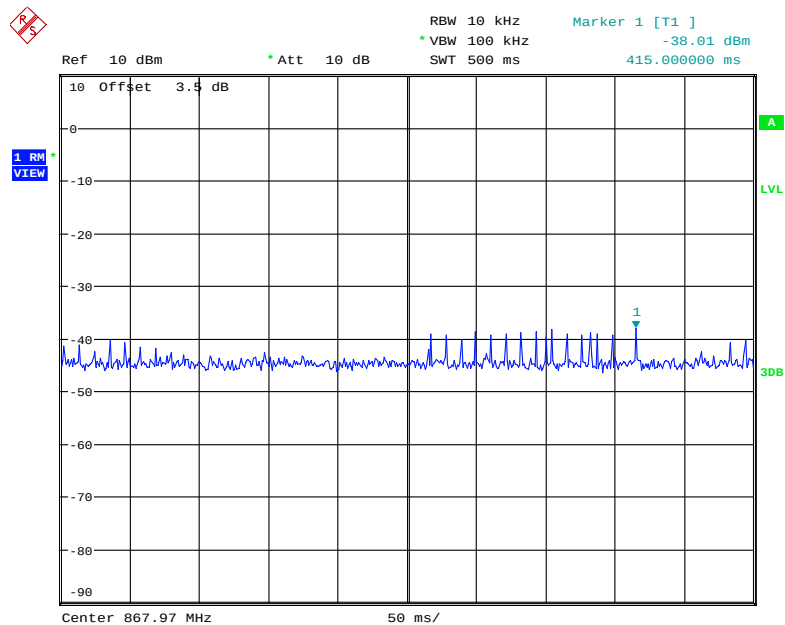
130kHz_868.1MHz:

Offset *1



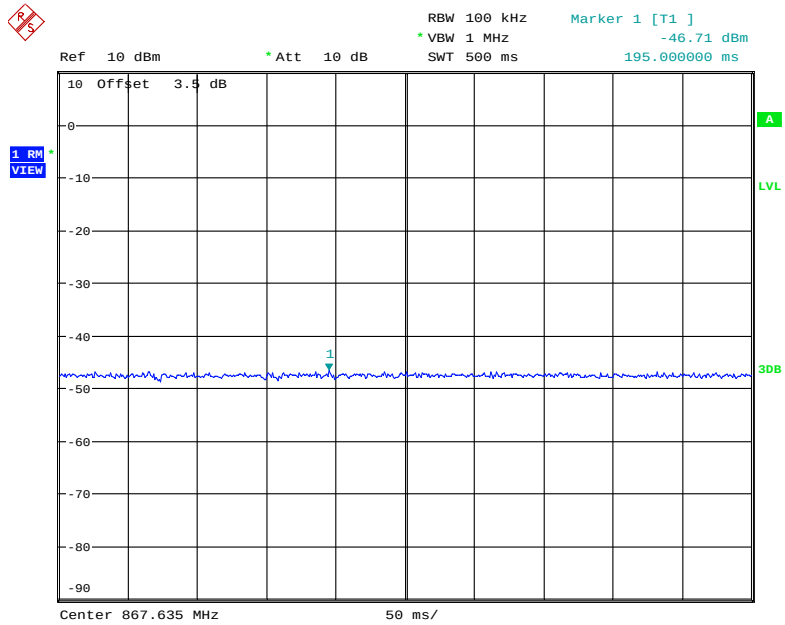
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Offset *2



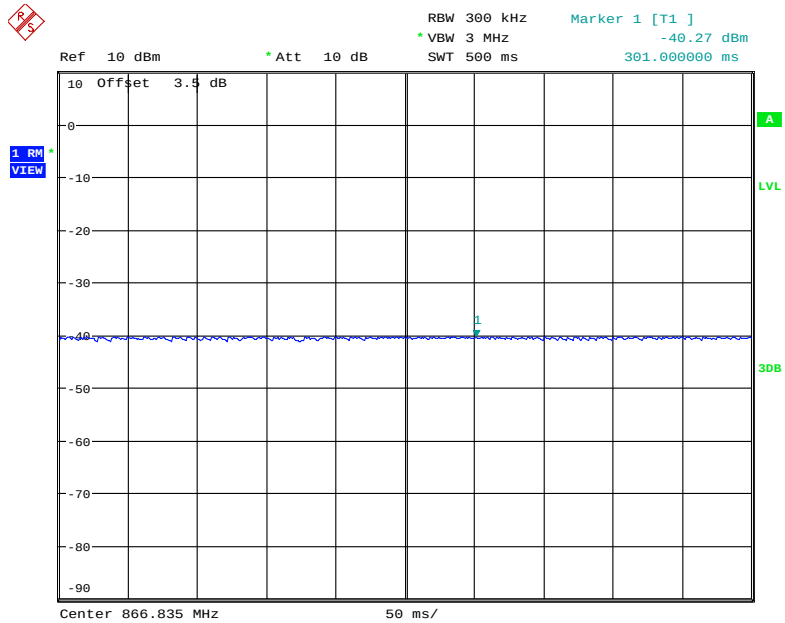
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Offset *3



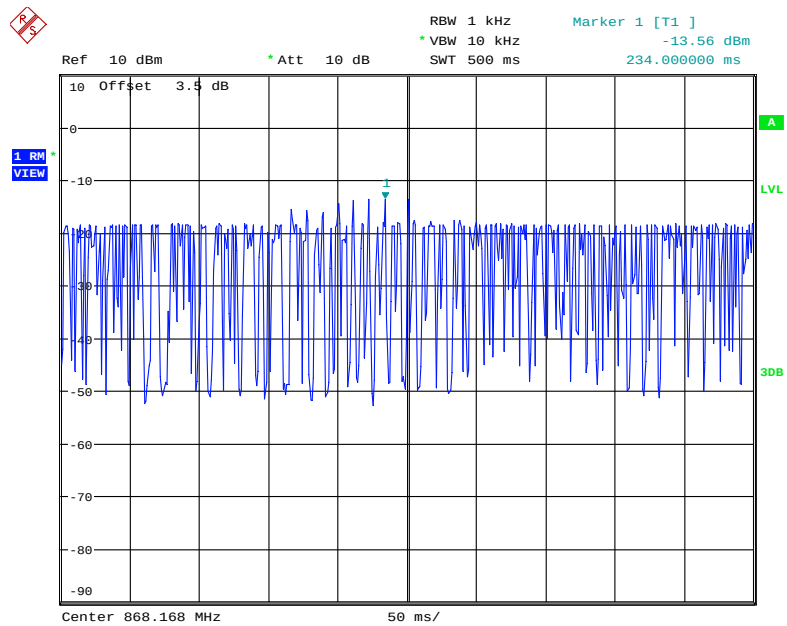
Date: 11.OCT.2023 15:21:48

Offset *4



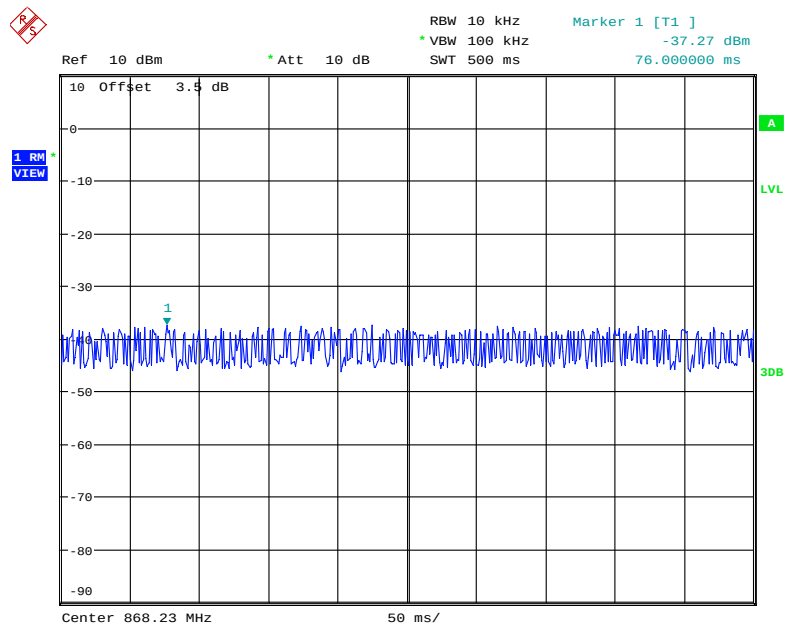
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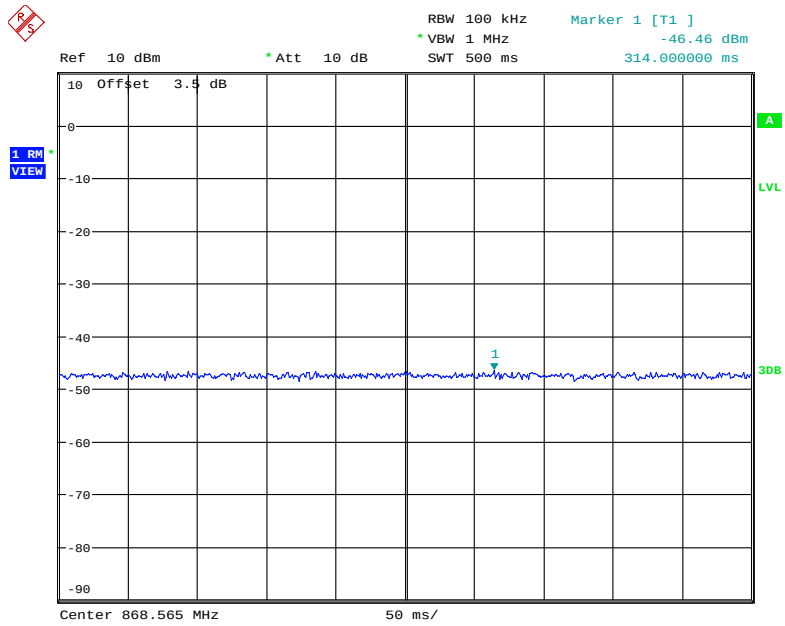
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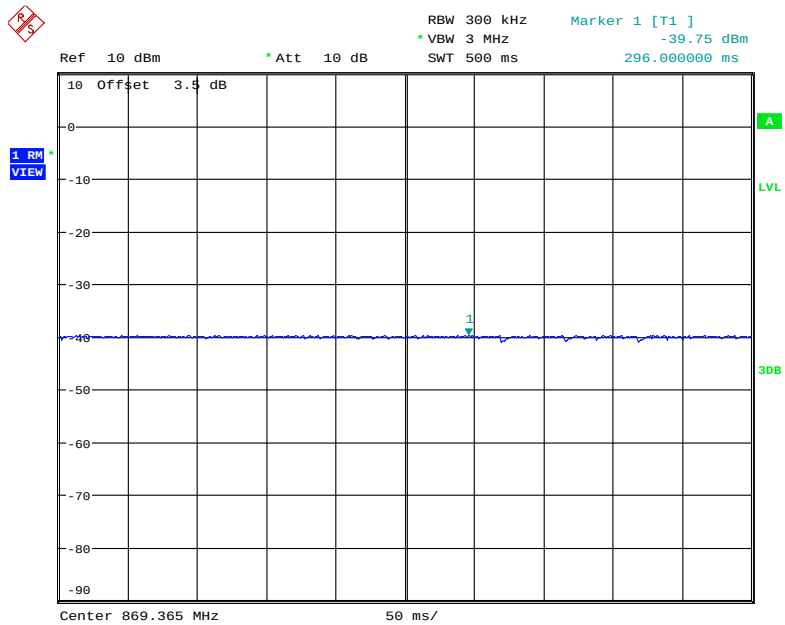
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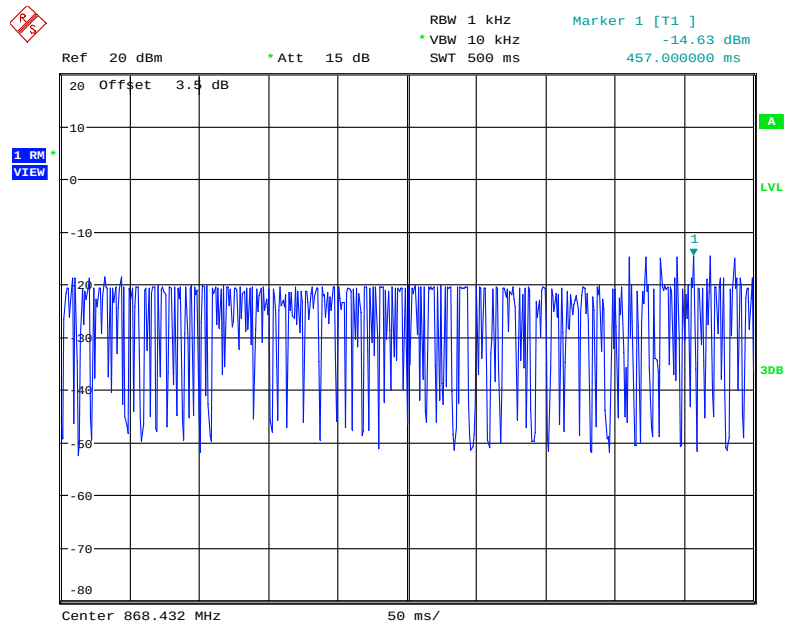
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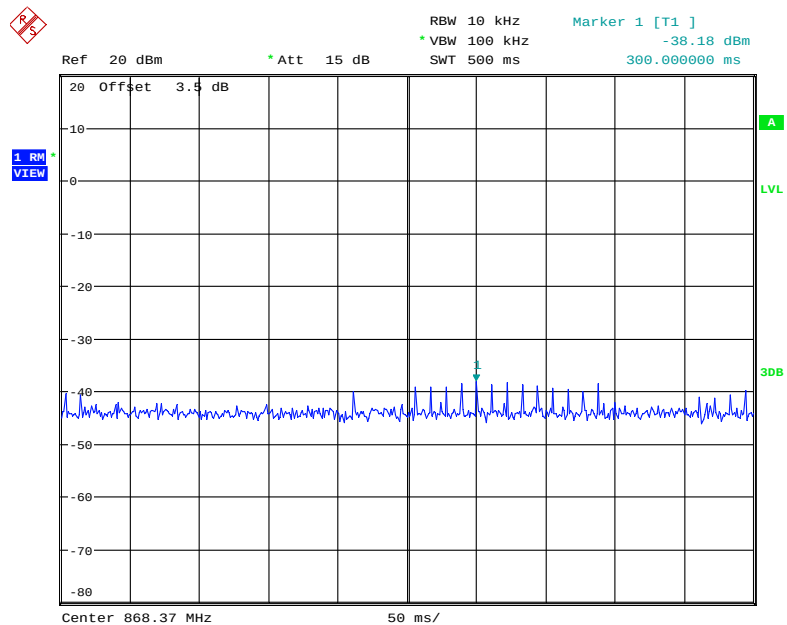
130kHz_868.5MHz

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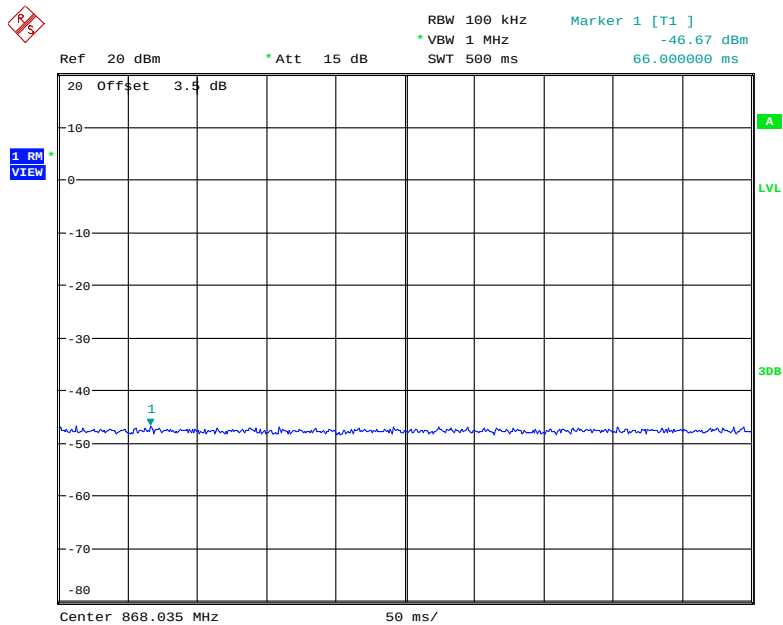
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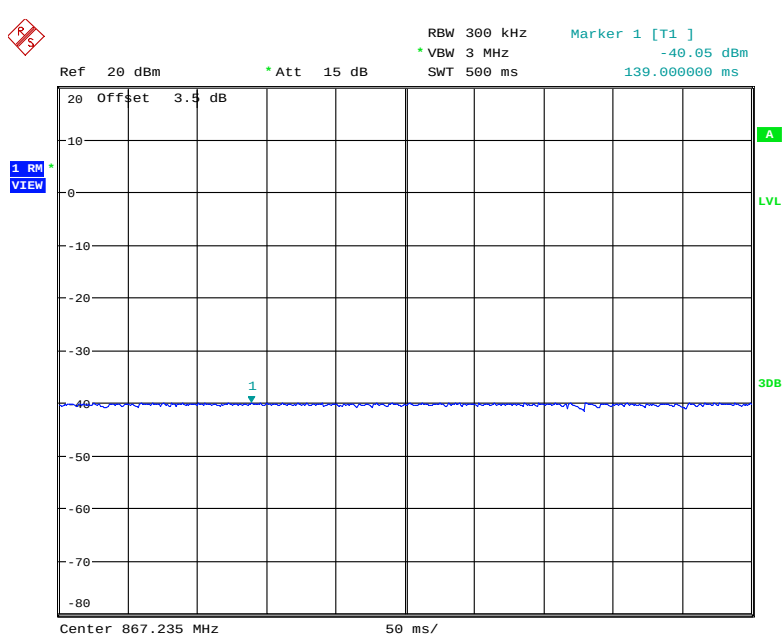
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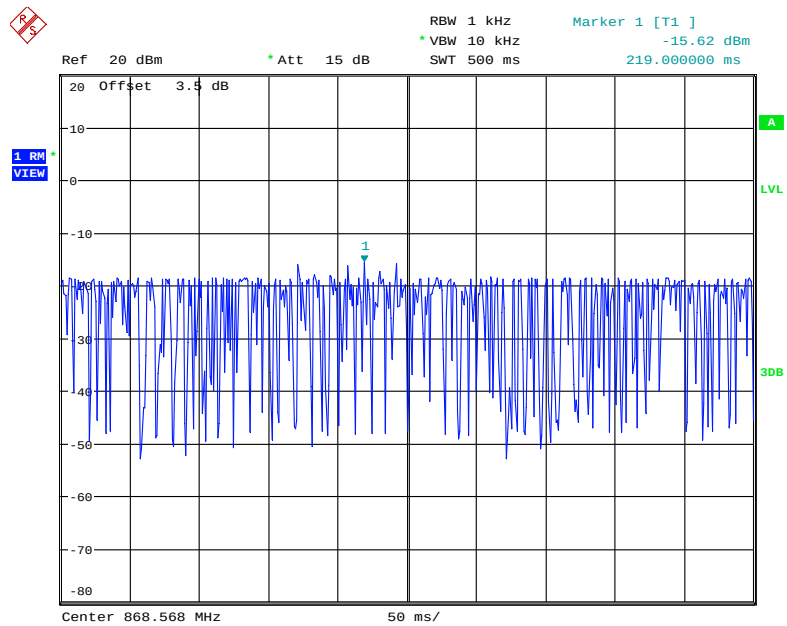
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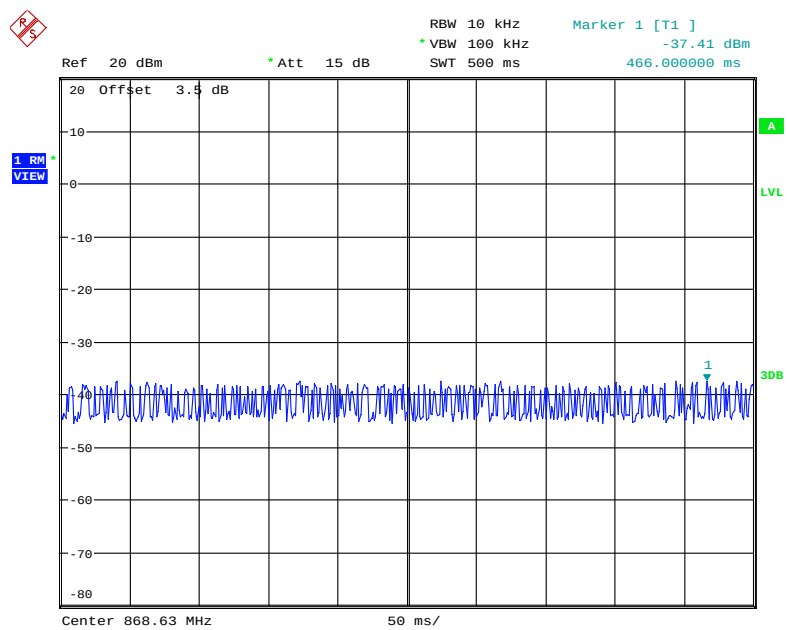
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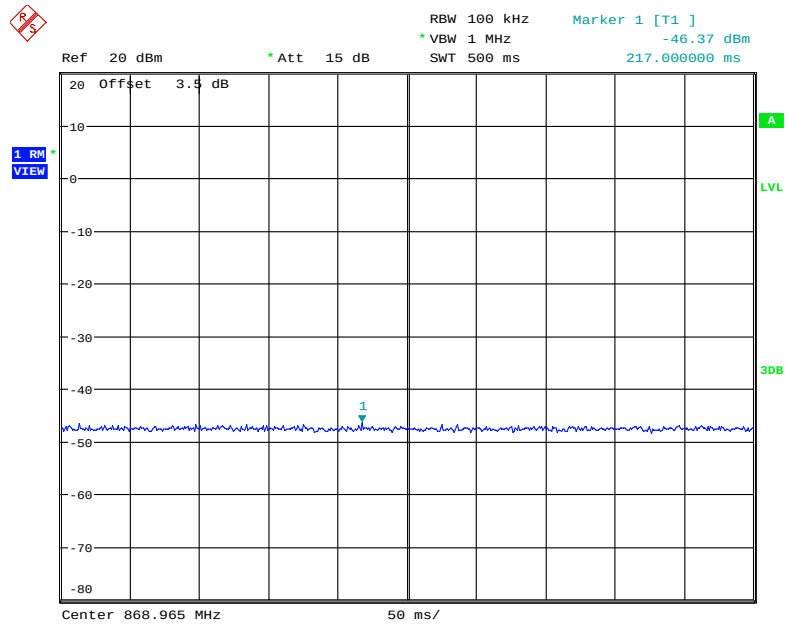
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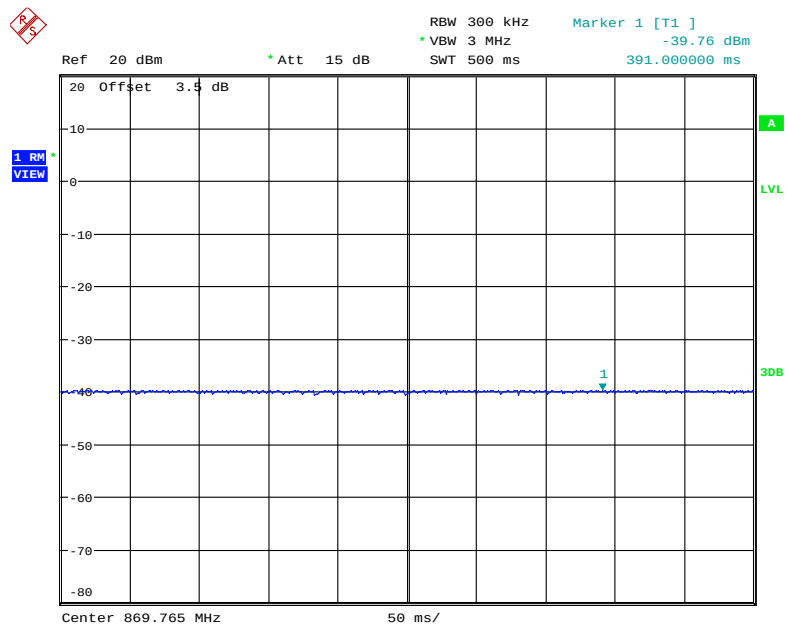
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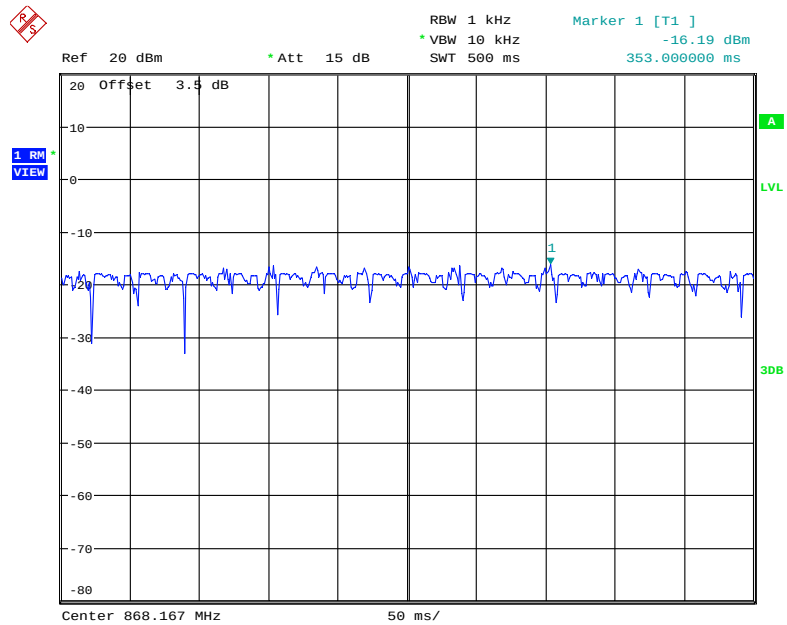
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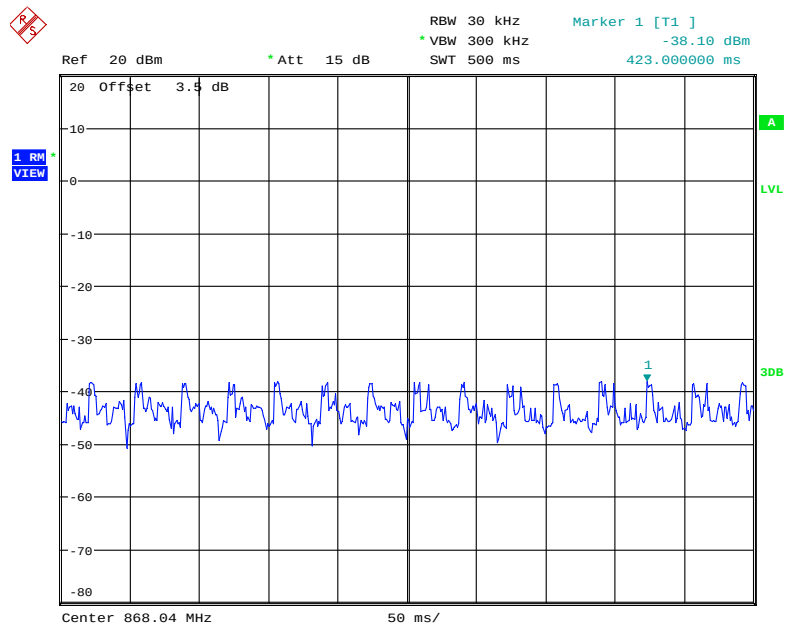
260kHz_868.3MHz

Offset *1



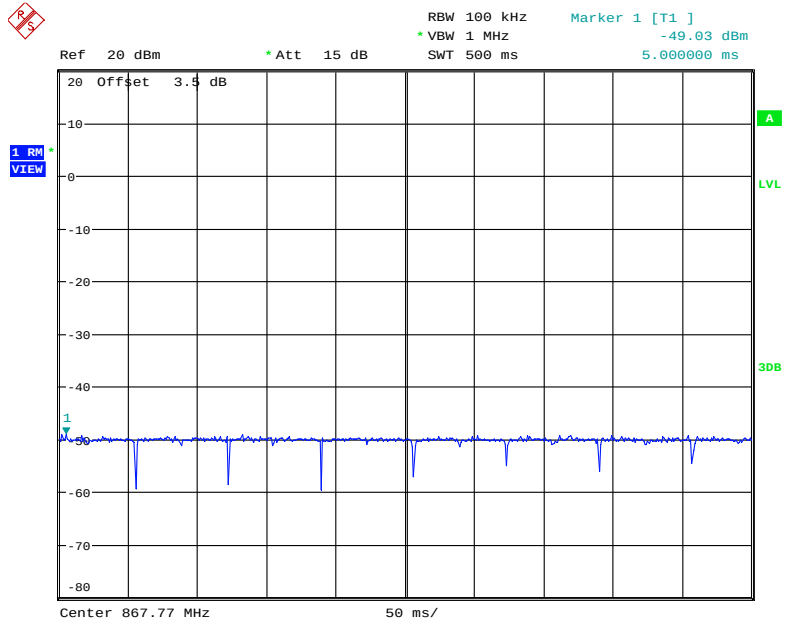
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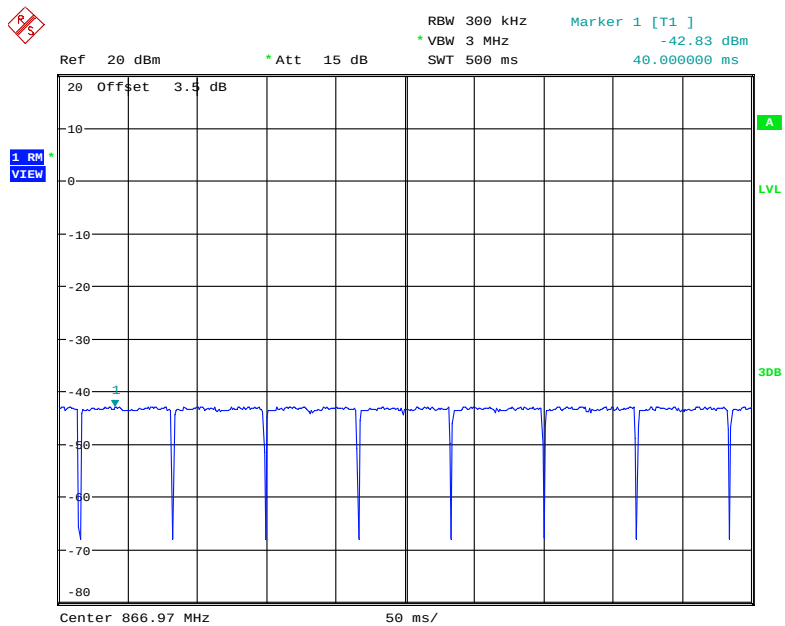
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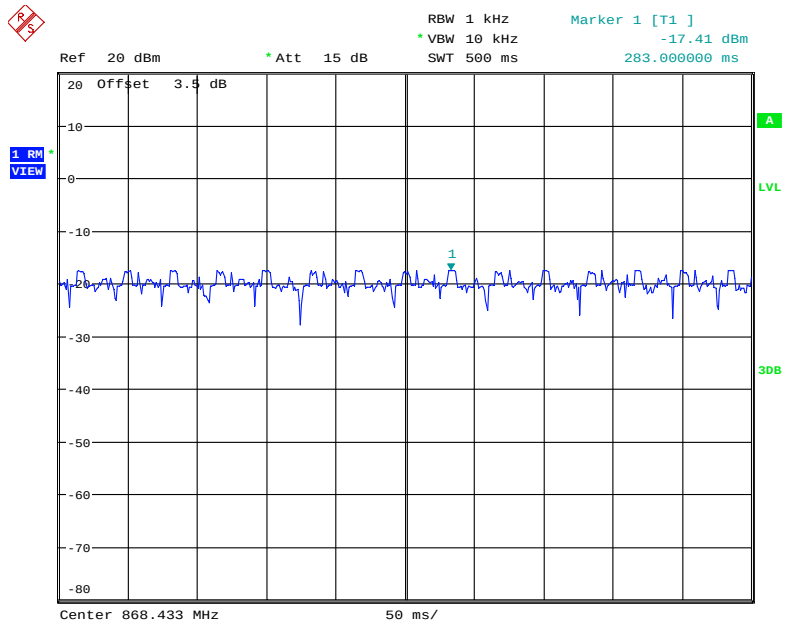
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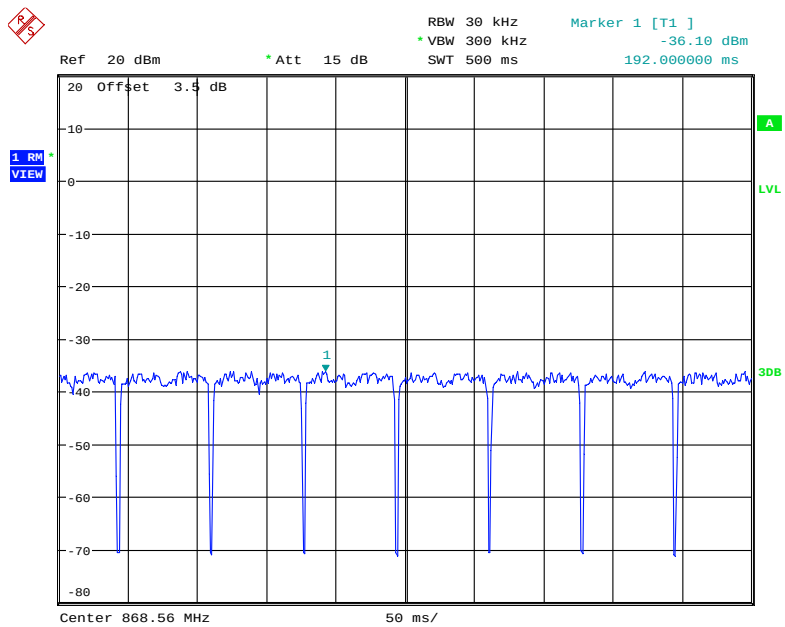
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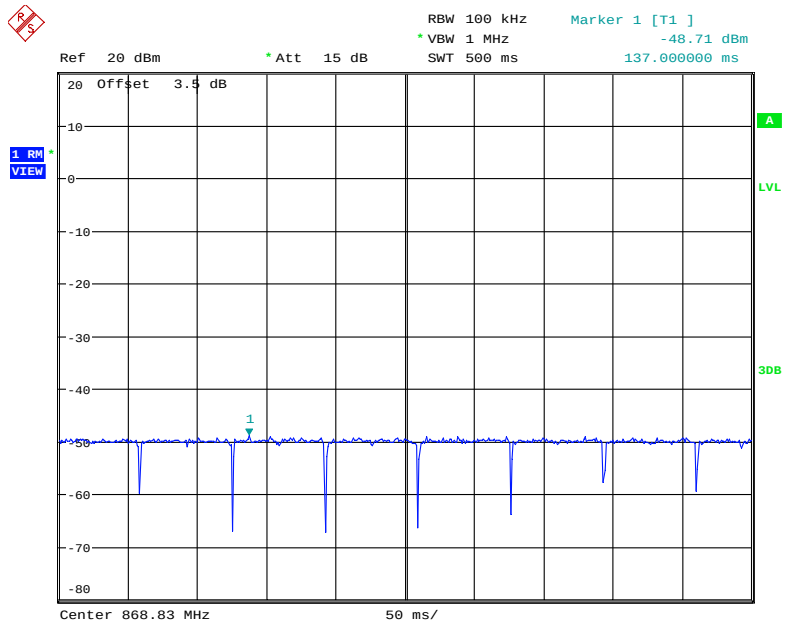
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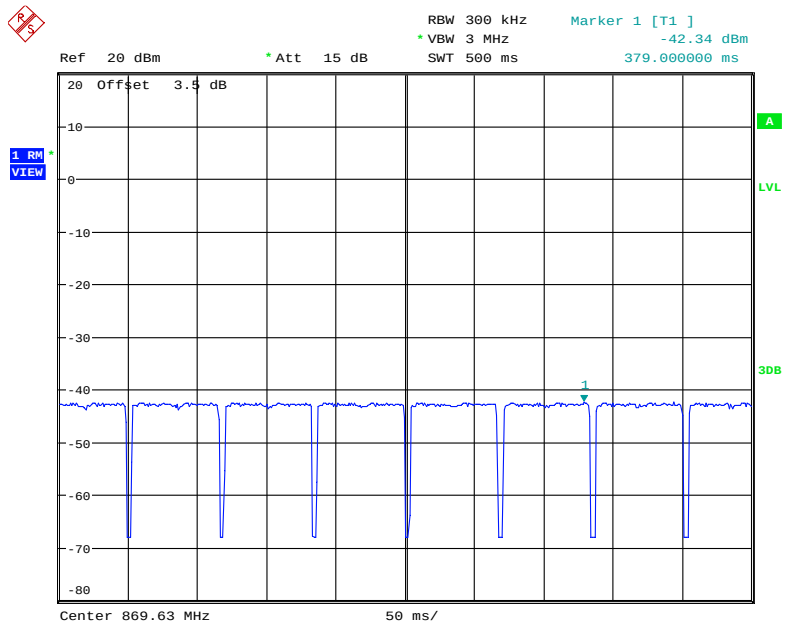
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Offset *7



Date: 25.SEP.2023 16:41:24

Offset *8



Date: 25.SEP.2023 16:42:00

ETSI EN 300 220-2 V3.2.1 (2018-06) §4.3.8 – TX BEHAVIOUR LOW VOLTAGE CONDITIONS

Applicable Standard

According to ETSI EN 300 220-1 V3.1.1 (2017-02) clause 5.12:

The TX behaviour under low voltage condition is the ability of the equipment to maintain its operating frequency and not produce emissions which exceed any relevant limit when the battery voltage falls below the lower extreme voltage level.

Limit:

The equipment shall either:

- a) remain in the Operating Channel OC without exceeding any applicable limits (e.g. Duty Cycle); or
 - b) reduce its effective radiated power below the Spurious Emission limits without exceeding any applicable limits (e.g. Duty Cycle); or
 - c) shut down, (ceasing function);
- as the voltage falls below the manufacturers declared operating voltage.

Method of measurement

Step 1:

Operation of the EUT shall be started, on Operating Frequency as declared by the manufacturer, with the appropriate test signal and with the EUT operating at nominal operating voltage.
The centre frequency of the transmitted signal shall be measured and noted.

Step 2:

The operating voltage shall be reduced by appropriate steps until the voltage reaches zero.
The centre frequency of the transmitted signal shall be measured and noted.
Any abnormal behaviour shall be noted.

Test Data

Environmental Conditions

Temperature:	25.9~26.5 °C
Relative Humidity:	46~48 %
ATM Pressure:	101.0~102.1 kPa

The testing was performed by Hanic Pan from 2023-08-30 to 2023-09-07.

Test Mode: Transmitting

130 kHz_867.1MHz:

Nominal Frequency: 867.1MHz				
Temperature & Humidity (°C & %)	Voltage (V _{DC})	Frequency of Lower point (MHz)	Frequency of Upper point (MHz)	Result (Note)
25.9°C & 46%	3.6	867.033	867.160	Within Operating frequency band and without exceeding any applicable limits
	3.2	867.032	867.159	Within Operating frequency band and without exceeding any applicable limits
	2.8	867.036	867.163	Within Operating frequency band and without exceeding any applicable limits
	2.5	867.035	867.162	Within Operating frequency band and without exceeding any applicable limits
	2.3	/	/	Shut down

130 kHz_867.9MHz

Nominal Frequency: 867.9MHz				
Temperature & Humidity (°C & %)	Voltage (V _{DC})	Frequency of Lower point (MHz)	Frequency of Upper point (MHz)	Result (Note)
25.9°C & 46%	3.6	867.835	867.962	Within Operating frequency band and without exceeding any applicable limits
	3.2	867.834	867.960	Within Operating frequency band and without exceeding any applicable limits
	2.8	867.836	867.963	Within Operating frequency band and without exceeding any applicable limits
	2.5	867.833	867.961	Within Operating frequency band and without exceeding any applicable limits
	2.3	/	/	Shut down

130 kHz_868.1MHz:

Nominal Frequency: 868.1MHz				
Temperature & Humidity (°C & %)	Voltage (V _{DC})	Frequency of Lower point (MHz)	Frequency of Upper point (MHz)	Result (Note)
25.9°C & 46%	3.6	868.035	868.164	Within Operating frequency band and without exceeding any applicable limits
	3.2	868.032	868.162	Within Operating frequency band and without exceeding any applicable limits
	2.8	868.034	868.163	Within Operating frequency band and without exceeding any applicable limits
	2.5	868.033	868.162	Within Operating frequency band and without exceeding any applicable limits
	2.3	/	/	Shut down

130 kHz_868.5MHz

Nominal Frequency: 868.5MHz				
Temperature & Humidity (°C & %)	Voltage (V _{DC})	Frequency of Lower point (MHz)	Frequency of Upper point (MHz)	Result (Note)
25.9°C & 46%	3.6	868.430	868.558	Within Operating frequency band and without exceeding any applicable limits
	3.2	868.438	868.566	Within Operating frequency band and without exceeding any applicable limits
	2.8	868.436	868.564	Within Operating frequency band and without exceeding any applicable limits
	2.5	868.434	868.562	Within Operating frequency band and without exceeding any applicable limits
	2.3	/	/	Shut down

260 kHz_868.3MHz:

Nominal Frequency: 868.3MHz				
Temperature & Humidity (°C & %)	Voltage (V _{DC})	Frequency of Lower point (MHz)	Frequency of Upper point (MHz)	Result (Note)
25.9°C & 46%	3.6	868.165	868.423	Within Operating frequency band and without exceeding any applicable limits
	3.2	868.162	868.420	Within Operating frequency band and without exceeding any applicable limits
	2.8	868.169	868.427	Within Operating frequency band and without exceeding any applicable limits
	2.5	868.170	868.428	Within Operating frequency band and without exceeding any applicable limits
	2.3	/	/	Shut down

Note*: No any other abnormal behaviour was observed.

ETSI EN 300 220-2 V3.2.1 (2018-06) §4.4.1 –RX SENSITIVITY

Applicable Standard

According to ETSI EN 300 220-1 [1], clause 5.14.1 applies:

The receiver sensitivity is the minimum signal power input to the receiver which produces the general performance criterion stated in clause 4.1. The test input signal is generated at the nominal operating frequency and modulated with normal modulation.

Limit: The EUT shall comply with reference limits defined in ETSI EN 300 220-1 [1], clause 5.14.2

The sensitivity for receivers shall be below or equal to Table 32 level.

Table 32: Limits for Receiver sensitivity

$$S = 10 \log RB_{\text{kHz}} - 4 \text{ dB}\mu\text{V emf}; \text{ or}$$

$$S_p = 10 \log RB_{\text{kHz}} - 117 \text{ dBm}$$

where:

- S_p is the sensitivity in dBm.
- RB is the declared receiver bandwidth in kHz.

The receiver bandwidth RB shall be declared by the manufacturer. RB is the usually 3 dB receiver bandwidth selectivity. For example, the sensitivity for a 25 kHz Operating Channel equipment with a 16 kHz bandwidth shall be better than +8 dB μ V emf for a 50 Ω receiver input impedance. This corresponds to a receiver sensitivity of -105 dBm.

Method of measurement

The conformance tests for this requirement shall be as defined in ETSI EN 300 220-1 [1], clause 5.14.3. Conformance shall be established under normal test conditions.

- 1) The measurements shall be performed on the operating frequency as declared by the manufacturer.
- 2) If possible the EUT shall be operated with any FEC or automatic retransmission facility disabled.
- 3) An EUT without a permanent or temporary antenna connector shall be tested according to clause 5.14.3.2.

Radiated measurement A suitable test site shall be selected from those described in clause C.1. The output of the signal generator shall be connected to a transmit test antenna with the same antenna polarization as the EUT. The transmit test antenna shall be placed in the test site.

The EUT shall be placed at the location of the turntable at the orientation of the most sensitive position. The measurement in clause 5.14.3.4 shall be performed using appropriate radiated measurement methods described in clause C.5.4.

- 4) An EUT with a permanent or temporary antenna connector shall be tested according to clause 5.14.3.3.

The EUT shall be connected to the output of a signal generator. The measurements in clause 5.14.3.4 shall be performed.

Measurement procedure The signal generator, modulated with an appropriate test signal, shall be set to Operating Frequency, as declared by the manufacturer.

Step 1: The operation of the EUT shall be started as a receiver on the Operating Frequency, as declared by the manufacturer.

Step 2: The level of the input signal to the EUT shall be adjusted until the wanted criterion as described in clause 4.1 is just exceeded.

Step 3: With the signal generator settings unchanged, the power received by EUT shall be established by appropriate means. The receiver sensitivity shall be noted.

Step 4: Steps 1 to 3 shall be repeated for each data rate at which the EUT is able to operate.

Step 5: The information shown in Table 33 shall be recorded in the test report.

Table 33: Information Recorded in the Test Report

Value	Notes
Test signal	The test signal used
Data rate	EUT data rate
FEC or ARQ state	FEC/ARQ enabled or disabled
Measurement method	BER/message success ratio/other
Measurement description	Description of message success ratio calculation/other measurement method if applicable
Operating frequency	Operating frequency tested
Receiver sensitivity	Measured signal generator power level

Test Data

Environmental Conditions

Temperature:	25.9~26.5 °C
Relative Humidity:	46~48 %
ATM Pressure:	101.0~102.1 kPa

The testing was performed by Hanic Pan from 2023-08-30 to 2023-09-07.

Test mode: *Receiving*

Test Frequency (MHz)	Declared Receiver Bandwidth (kHz)	Measured Sensitivity (dBm)	Limit (dBm)
867.1	130	-104.7	-95.9
867.9		-104.5	-95.9
868.1		-104.6	-95.9
868.5		-104.2	-95.9
868.3	260	-104.9	-92.9

ETSI EN 300 220-2 V3.2.1 (2018-06) §4.4.2 – BLOCKING

Applicable Standard

According to ETSI EN 300 220-1 V3.1.1 (2017-02) clause 5.18.1.

Limit: The blocking level shall be better or equal to category 3 reference limits level defined in ETSI EN 300 220-1 [1], clause 5.18.2.

NOTE: After December 31st, 2018, the receiver category 3 will be withdrawn, therefore receiver category 2 will be the minimum applicable level.

Method of measurement

Signal generator A shall be set to an appropriate modulated test signal at the operating frequency of the EUT receiver.

Signal generator B shall be unmodulated.

Measurements shall be carried out at frequencies of the unwanted signal at approximately the frequency(ies) offset(s) defined in technical requirement avoiding those frequencies at which spurious responses occur. Additional measurement points may be requested by technical requirements clause.

If several operational frequency bands are used by the equipment, at least one blocking measurement by bands has to be performed.

Step 1: Signal generator B shall be powered off. Signal generator A shall be set to the minimum level which gives the wanted performance criterion of EUT or the reference level in Table 32, whichever is the higher. The output level of generator A shall then be increased by 3 dB unless otherwise specified in technical requirement.

Step 2: Signal generator B is powered on and set to operate at the nominal operating frequency - offset frequency.

Signal generator B is then switched on and the signal amplitude is adjusted to the minimum level at which the wanted performance criterion is not achieved.

With signal generator B settings unchanged, the receiver shall be replaced with a suitable RF power measuring equipment. The power into the measuring equipment shall be measured and noted.

The blocking level is then the conducted power received from generator B at the EUT antenna connector. This can either be measured on the antenna connector for conducted test or be calculated for radiated test (see clause C.5.4).

The blocking level shall be higher or equal to the blocking power level requested in the technical requirement clause.

Step 3: The measurement in steps 1 to 3 shall be repeated with signal offsets at required frequencies.

Step 4: The information shown in Table 44 shall be recorded in the test report for each measured signal level and unwanted signal offset.

Table 44: Information Recorded in the Test Report

Value	Notes
Operating Frequency	Nominal centre frequency of the receiver
Signal generator A	Power level of signal generator A
Blocking level	Power level of signal generator B

For equipment using CCA whatever is the receiver category, steps 1 to 4 shall be repeated with signal generator A level adjusted +13 dB higher than in the measurements in clause 5.18.6.4

Test Data

Environmental Conditions

Temperature:	25.9~26.5 °C
Relative Humidity:	46~48 %
ATM Pressure:	101.0~102.1 kPa

The testing was performed by Hanic Pan from 2023-08-30 to 2023-09-07.

Test result: Compliance.

Test mode: Receiving

Receiver Bandwidth: 130 kHz

Frequency	Frequency offset (MHz)	Test result (dBm)	Limit (dBm)	Result
867.1	-2 MHz from OC edge f_{low}	-57.54	-69	Pass
	+2 MHz from OC edge f_{high}	-56.36	-69	Pass
	-10 MHz from OC edge f_{low}	-36.51	-44	Pass
	+10 MHz from OC edge f_{high}	-35.84	-44	Pass
	-5 % of Centre Frequency	-37.56	-44	Pass
	+5 % of Centre Frequency	-36.54	-44	Pass
867.9	-2 MHz from OC edge f_{low}	-57.52	-69	Pass
	+2 MHz from OC edge f_{high}	-57.63	-69	Pass
	-10 MHz from OC edge f_{low}	-37.68	-44	Pass
	+10 MHz from OC edge f_{high}	-37.58	-44	Pass
	-5 % of Centre Frequency	-36.25	-44	Pass
	+5 % of Centre Frequency	-35.26	-44	Pass
868.1	-2 MHz from OC edge f_{low}	-57.84	-69	Pass
	+2 MHz from OC edge f_{high}	-57.63	-69	Pass
	-10 MHz from OC edge f_{low}	-38.69	-44	Pass
	+10 MHz from OC edge f_{high}	-38.06	-44	Pass
	-5 % of Centre Frequency	-36.54	-44	Pass
	+5 % of Centre Frequency	-36.21	-44	Pass
868.5	-2 MHz from OC edge f_{low}	-58.24	-69	Pass
	+2 MHz from OC edge f_{high}	-56.01	-69	Pass
	-10 MHz from OC edge f_{low}	-38.05	-44	Pass
	+10 MHz from OC edge f_{high}	-37.14	-44	Pass
	-5 % of Centre Frequency	-38.53	-44	Pass
	+5 % of Centre Frequency	-35.86	-44	Pass

Receiver Bandwidth: 260kHz

Frequency	Frequency offset (MHz)	Test result (dBm)	Limit (dBm)	Result
868.3	-2 MHz from OC edge f_{low}	-57.74	-69	Pass
	+2 MHz from OC edge f_{high}	-56.56	-69	Pass
	-10 MHz from OC edge f_{low}	-36.21	-44	Pass
	+10 MHz from OC edge f_{high}	-35.56	-44	Pass
	-5 % of Centre Frequency	-37.28	-44	Pass
	+5 % of Centre Frequency	-36.73	-44	Pass

Note: The equipment provider declared that the receiver category for the EUT is 2.

 f_{low} is the low edge of OC, f_{high} is the high edge of OC, f_c is the center frequency.

ETSI EN 300 220-2 V3.2.1 (2018-06) §4.5.2 – CLEAR CHANNEL ASSESSMENT THRESHOLD

Applicable Standard

According to ETSI EN 300 220-1 [1], clause 5.21.2.1 applies.

Limit: The CCA threshold is defined as the received signal level above which the EUT determines that the channel is not available for use.

Reference CCA limits

The CCA threshold shall not exceed the limits given in Table 45.

Table 45: CCA radiated threshold limits

Parameter	Value
CCA threshold for EUT with e.r.p < 100 mW	15 dB above Rx sensitivity level limit as given in Table 32
CCA threshold for EUT with e.r.p from 100 mW to 500 mW	11 dB above Rx sensitivity level limit as given in Table 32
NOTE: The limits are based on an antenna gain of 0 dB relative to a dipole (i.e. +2,15 dBi) maximum. For other antenna gains different to 0 dB the limits shall be adjusted accordingly.	
EXAMPLE: With antenna gain of -5 dB, the CCA threshold shall be corrected and increased by 5 dB at the RF front end.	

Method of measurement

The conformance tests for this requirement shall be as defined in ETSI EN 300 220-1 [1], clause 5.21.2.3. Conformance shall be established under normal test conditions.

1) The measurement is performed on an Operating Frequency declared by the manufacturer. The frequency shall correspond to a nominal channel centre frequency consistent with the highest and lowest frequencies and channel spacing declared by the manufacturer.

2) An EUT without a permanent or temporary antenna connector shall be tested according to clause 5.21.2.3.2.

Radiated measurement

A suitable test site shall be selected from those described in clause C.1. Signal generators together with the combiner/coupler, shown in Figure 11, shall be placed outside the test site.

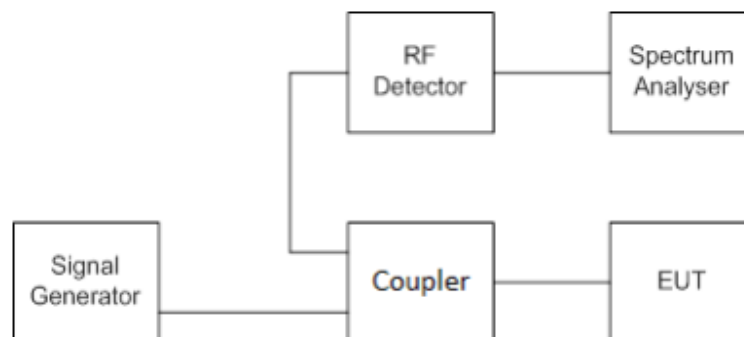


Figure 11: CCA measurement arrangement

The output of the combiner shall be connected to a transmit test antenna with the same antenna polarization as the EUT. The transmit test antenna shall be placed in the test site. The EUT shall be placed at the location of the turntable at the orientation of the most sensitive position. The measurement in clause 5.21.2.3.4 shall be performed using appropriate radiated measurement methods described in clause C.5.

3) An EUT with a permanent or temporary antenna connector shall be tested according to clause 5.21.2.3.3.

Conducted measurement

A signal generator and a power meter or spectrum analyser shall each be combined via appropriate attenuators into the EUT antenna connector as shown in Figure 11. The measurements in clause 5.21.2.3.4 shall be performed.

Table 46: Test Parameters Settings for CCA Threshold Measurement

Setting	Value
Centre frequency	The nominal EUT operating frequency
RBW	Approximately 3 x Operating Channel width OCW
VBW	3 x RBW
Span	Zero span
Detector Mode	RMS
Trace Mode	Max. Hold
NOTE: The nominal operating frequency is agreed between the test laboratory and the manufacturer. The nominal operating frequency shall be consistent with the highest and lowest operating frequencies and channel spacing as declared by the manufacturer. Channel Spacing is declared by the manufacturer.	

The spectrum analyser shall be configured as shown in Table 46.

Step 1: Operation of the EUT as a receiver shall be started with its CCA function active. The signal generator, with normal test modulation, shall be adjusted to the nominal operating frequency. The spectrum analyser levels and RBW shall be adjusted to provide satisfactory display of the signal generator signal.

Step 2: The output power level of the signal generator shall be set to approximately 20 dB above the EUT receiver reference sensitivity. The EUT shall be instructed to transmit. NOTE 1: The means of instructing the EUT to transmit is determined by the manufacturer. The presence of any signal from the EUT detected by the spectrum analyser shall be noted.

Step 3: The level of the signal generator shall be reduced in steps of 1 dB until the equipment starts to transmit. NOTE 2: There may be a delay due to collision avoidance operation before the EUT begins to transmit once the CCA threshold has been reached. Ensure that any such delay is taken into account in the rate at which the signal generator level is reduced. The signal generator level present at the receiver input of the EUT when transmission begins is the CCA threshold and shall be noted.

Step 4: Step 2 and step 3 shall be repeated.

Step 5: The information shown in Table 47 shall be recorded in the test report.

Table 47: Information Recorded in the Test Report

Value	Notes
Test signal	The test signal used
CCA threshold	First measured CCA threshold power level
CCA threshold	Second measured CCA threshold power level
Presence of unexpected EUT signal	Any transmission detected at the spectrum analyser in step 2
NOTE: The presence of unexpected EUT transmission is a test failure.	

All observations and results shall be detailed in the test report.

CCA threshold levels shall not exceed values defined by technical requirements.

Test Data

Environmental Conditions

Temperature:	25.9~26.5 °C
Relative Humidity:	46~48 %
ATM Pressure:	101.0~102.1 kPa

The testing was performed by Hanic Pan from 2023-08-30 to 2023-09-07.

Test result: Compliance.

Test with conducted method:

Declared Receiver Bandwidth (kHz)	Test Frequency (MHz)	Test item	Measured Power Level (dBm)	Limit (dBm)
130	867.1	First CCA Threshold	-96.7	-81.05
		Second CCA Threshold	-97.2	-81.05
130	867.9	First CCA Threshold	-96.6	-81.05
		Second CCA Threshold	-97.2	-81.05
130	868.1	First CCA Threshold	-96.5	-81.05
		Second CCA Threshold	-97.3	-81.05
130	868.5	First CCA Threshold	-97.4	-81.05
		Second CCA Threshold	-96.3	-81.05
260	868.3	First CCA Threshold	-96.5	-78.05
		Second CCA Threshold	-97.2	-78.05

Note: Antenna gain is 2dBi. For antenna gains different to 0 dB the limits shall be adjusted accordingly
2dBi=-0.15dBd

ETSI EN 300 220-2 V3.2.1 (2018-06) §4.5.3 –POLITE SPECTRUM ACCESS TIMING PARAMETERS

Applicable Standard

Polite spectrum access parameters clause applies to EUT with polite spectrum access instead of duty cycle where permitted by table B.1 in annex B, or table C.1 in annex C or any NRI.

Limit:

The EUT shall comply with reference limits defined in ETSI EN 300 220-1 [1], clause 5.21.3.1.

Parameters are shown in Table 48 and their limits shall not be exceeded.

Table 48: Reference limits for Polite Spectrum Access timing parameters

Parameter	Limit	Notes
Minimum CCA interval	160 μ s	Minimum CCA listening period
Minimum deferral period	CCA interval	Minimum value of the deferral interval
Minimum unit of deferral period	See note 1	Smallest interval between two adjacent deferral periods
Dead Time	CCA declared time by the vendor but not exceeding 5 ms	Maximum time between the end of a listening interval and the start of a transmission
Maximum Transmission Duration $T_{on\ Max}$	1 s	For a single Transmission
Maximum Transmission Duration $T_{on\ Max}$	4 s	For a Transmission dialogue or a polling sequence
Max T_{cum_on} over 1 hour	100 s / 1 h per 200 kHz spectrum	Maximum allowed Cumulative On Time over a 200 kHz portion of spectrum per hour
Minimum T_{off_min} on the same Operating Frequency	100 ms	The minimum T-off time period where a specific transmitter shall remain off after a transmission on the same Operating Frequency
NOTE 1: The minimum unit of deferral period is declared by the manufacturer.		
NOTE 2: T_{DIS} is declared by the manufacturer without exceeding 5 ms.		
NOTE 3: Longer accumulated transmission time is possible by implementing more AFA channels.		

Method of measurement

The conformance tests for this requirement shall be as defined in ETSI EN 300 220-1 [1], clause 5.21.3.2. Conformance shall be established under normal test conditions.

The use of polite spectrum access shall be declared by the manufacturer. The manufacturer shall also declare:

- 1) The minimum CCA interval employed. ETSI 59 ETSI EN 300 220-1 V3.1.1 (2017-02)
- 2) The maximum dead time.
- 3) The method employed to randomize timing of re-transmission attempts.

- 4) The units of the deferral period.
- 5) The minimum and maximum values of the deferral random period.
- 6) The maximum transmission duration time, Ton max.
- 7) The minimum transmission off time on the same operating frequency, Toff min. The declared parameters shall not exceed the requested values in technical requirements.

Test Data

Environmental Conditions

Temperature:	25.9~26.5 °C
Relative Humidity:	46~48 %
ATM Pressure:	101.0~102.1 kPa

The testing was performed by Hanic Pan from 2023-08-30 to 2023-09-07.

Test result: Compliance.

Frequency Range (MHz)	Parameter Items	Limit (dBm)	Declared Result
865-868/ 868-868.6	Minimum CCA interval	160 μs	Compliance
	Minimum deferral period	CCA interval	Compliance
	Minimum unit of deferral period	See note 1	Compliance
	Dead Time	CCA declared time by the vendor but not exceeding 5 ms	Compliance
	Maximum Transmission Duration Ton Max	1s	Compliance
	Maximum Transmission Duration Ton Max	4 s	Compliance
	Max Tcum_on over 1 hour	100 s / 1 h per 200 kHz spectrum	Compliance
	Minimum Toff_min on the same Operating Frequency	100 ms	Compliance

NOTE 1: The minimum unit of deferral period is declared by the manufacturer.

NOTE 2: TDIS is declared by the manufacturer without exceeding 5 ms.

NOTE 3: Longer accumulated transmission time is possible by implementing more AFA channels

EXHIBIT A - EUT PHOTOGRAPHS

Please refer to the report number is SZ1230802-44707E-EUT.

EXHIBIT B - TEST SETUP PHOTOGRAPHS

Radiated Spurious Emissions Test View (Below 1GHz)



Radiated Spurious Emissions Test View (Above 1GHz)



EXHIBIT C – PRODUCT SIMILARITY DECLARATION OF LETTER

Dragino Technology Co., Limited
Address: Room 202, BaoChengTai industrial park, No.8 CaiYun LongCheng Street,
LongGang District, Shenzhen China
Tel: 0755-86610821
Email: edwin@dragino.com

Product Similarity Declaration

Date: 2023-08-23

Dear Sir/Madam,

We, Dragino Technology Co., Limited hereby declare that we have a product named as LoRaWAN Sensor Node (model: DDS75-LB), were tested by BACL (Shenzhen). Meanwhile, For our marketing purpose, we would like to list a series model (RS485-LB, SE01-LB, DDS45-LB, DDS20-LB, MDS120-LB, MDS200-LB, LDS12-LB, SPH01-LB, LMS01-LB, DDS04-LB) on reports and certificate. All the models are identical in electrically, the differences are shown in the table below. No other changes were made to them.

Model	Probes
DDS75-LB	Ultrasonic Probe
RS485-LB	N/A
SE01-LB	Soil Moisture Probe
DDS45-LB	Ultrasonic Probe
DDS20-LB	Ultrasonic Probe
MDS120-LB	Microwave Probe
MDS200-LB	Microwave Probe
LDS12-LB	LiDar Probe
SPH01-LB	Soil PH probe
LMS01-LB	Leaf Moisture Probe
DDS04-LB	Ultrasonic Probe

We confirm that all information above is true, and we'll be responsible for all the consequences.
Please contact me if you have any question.

Sincerely Yours,

Signature:

Edwin
Chen
数字签名
者: Edwin Chen
日期: 2023.09.01
15:19:18 +08'00'

Edwin chen
CEO

***** END OF REPORT *****